ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2023)8305270 - 05/12/2023

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101138789
Project name:	A Wood-to-Wood Cascade Upcycling Valorisation Approach
Project acronym:	W2W
Call:	HORIZON-CL4-2023-TWIN-TRANSITION-01
Topic:	HORIZON-CL4-2023-TWIN-TRANSITION-01-42
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 January 2024
Project duration:	48 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

The W2W project will produce and validate a comprehensive framework for multi-dimensional cascade valorisation of wood from C&D and furniture waste, consisting of four core components: 1) Advanced separation & sorting technologies: Adaptive Separation and Sorting system based on cascade valorisation thresholds, making use of humanrobot collaboration and mixed reality technologies
2) Upcycling Processes and Technologies: mechanical, chemical, and bioremediation processes to recover wood or resources from waste and use them in new products which have the same quality as the ones produced by primary resources3) Digital Tools for Improving Circular Flows of Secondary Materials: LCA-based tool to predict and assess the impact of different cascade paths, Supply Chain Management Tool for the optimisation of upcycling of secondary materials, Digital Product Passport, and Digital Platform for circularity/upcycling information exchange.4) Supportive Framework in Policy, Market, and Skills: Framework for CDW Cascade Valorisation, Harmonised Waste Wood Classification Framework, skills development, contribution to standards and certification schemes, business models. W2W will demonstrate efficient and sustainable value chains in 3 Use Cases for the production of: wood without pollutants; biocomposite building materials; biopolymers; polyols; chemical detergents; and the recovery of nutrients, through the use of technologies and tools that allow the selection of optimal cascade paths for further uses of wood products and their materials.The W2W holistic approach will reduce the demand for virgin materials, reduce the amount of waste that ends up in landfills or incineration facilities, allow for the creation of new value-added products from waste materials, and support the transition towards a circular economy by promoting closed-loop systems where materials are continually repurposed and reused, extending their useful life.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	ICCS	EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON	EL	999654356
2	BEN	LERMAB	UNIVERSITE DE LORRAINE	FR	954831626
3	BEN	CIRCE	FUNDACION CIRCE CENTRO DE INVESTIGACION DE RECURSOS Y CONSUMOS ENERGETICOS	ES	999516907
4	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
5	BEN	POLIMI	POLITECNICO DI MILANO	IT	999879881
6	BEN	VTT	TEKNOLOGIAN TUTKIMUSKESKUS VTT OY	FI	932760440
7	BEN	UPV	UNIVERSITAT POLITECNICA DE VALENCIA	ES	999864846
8	BEN	CERTH	ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS	EL	998802502

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC

Associated with document Ref. Ares(2023)8305270 - 05/12/2023

9	BEN	UHE	UNIVERSIDAD DEL PAIS VASCO/ EUSKAL HERRIKO UNIBERTSITATEA	ES	999865234
10	BEN	TUB	TECHNISCHE UNIVERSITAT BERLIN	DE	999986678
11	BEN	UJOH	UNIVERSITY OF JOHANNESBURG	ZA	959044336
12	BEN	ELAK	ELLAKTOR ANONYMOS ETAIREIA	EL	898116890
13	BEN	KIVERDI	KIVERDI NETHERLANDS BV	NL	888472762
14	BEN	P&G	PROCTER & GAMBLE SERVICES COMPANY NV	BE	966022322
15	BEN	FOCCHI	FOCCHI SPA	IT	951961008
16	BEN	DRAXIS	DRAXIS ENVIRONMENTAL SA	EL	996151686
17	BEN	eBOS	EBOS TECHNOLOGIES LIMITED	CY	997848022
18	BEN	QPLAN	Q-PLAN INTERNATIONAL ADVISORS PC	EL	999568608
19	BEN	R2M	R2M SOLUTION SRL	IT	952850110
20	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
21	BEN	CEPS	CENTRE FOR EUROPEAN POLICY STUDIES	BE	999747670
22	BEN	ISWA	INTERNATIONAL SOLID WASTE ASSOCIATION	NL	899789752
23	BEN	RCM	REGION OF CENTRAL MACEDONIA	EL	951347774
24	BEN	UNE	ASOCIACION ESPANOLA DE NORMALIZACION	ES	984232326
25	AP	EcoM	Eco-mobilier	FR	895852425

Associated with document Ref. Ares(2023)8305270 - 05/12/2023

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Project Management and Coordination for RP1	1 - ICCS		1	18	D1.1 – W2W project management handbook D1.2 – Data management plan (version 1)
WP2	Project Management and Coordination for RP2	1 - ICCS		19	36	D2.1 – Data management plan (version 2)
WP3	Project Management and Coordination for RP3	1 - ICCS		37	48	D3.1 – Data management plan (version 3)
WP4	Transformation Framework for Sustainableby-design Construction Demolition Wastes Biorefining	16 - DRAXIS		1	10	D4.1 – Mapping of the actors and key stakeholders D4.2 – Wood characterisation & classification schemes D4.3 – Recycled wood cascade valorisation framework D4.4 – Data sharing and exchange model D4.5 – Impact assessment methodologies and W2W KPIs
WP5	Management of Waste Streams (first release)	1 - ICCS		8	18	D5.1 – CDW flows valorisation and digitalisation (version 1) D5.2 – Wood and glass non-destructive real time techniques analysis & optimal size definition (version 1) D5.3 – Adaptive glass and wood sorting systems (version 1) D5.4 – Better separation and quality control systems (version 1)

WP6	Management of Waste Streams (final release)	1 - ICCS		19	36	D6.1 – CDW flows valorisation and digitalisation (version 2)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D6.2 – Wood and glass non-destructive real time techniques analysis & optimal size definition (version 2) D6.3 – Adaptive glass and wood sorting systems (version 2) D6.4 – Better separation and quality control systems (version 2)
WP7	Cascade Refinement Technologies for Wood Waste Upcycling (TRL5)	4 - NTUA		8	18	D7.1 – Cascade to function, mixed wood treatment (version 1) D7.2 – New mix between wood, glass and composites for wallboards (version 1) D7.3 – Final products and initial validation (version 1)
WP8	Cascade Refinement Technologies for Wood Waste Upcycling (TRL6)	4 - NTUA		19	36	D8.1 – Cascade to function, mixed wood treatment (version 2) D8.2 – New mix between wood, glass and composites for wallboards (version 2) D8.3 – Final products and initial validation (version 2)

WP9	Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL5)	2 - LERMAB		8	18	D9.1 – Glue separation from wood and bioremediation wood treatment (version 1) D9.2 – Liquefaction process of mixed wood waste (version 1) D9.3 – Final products and initial validation (version 3)
WP10	Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL6)	2 - LERMAB		19	36	D10.1 – Glue separation from wood and bioremediation wood treatment (version 2) D10.2 – Liquefaction process of mixed wood waste (version 2)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D10.3 – Final products and initial validation (version 4)
WP11	Energy, Gas and Ashes Valorisation (TRL5)	3 - CIRCE		8	18	D11.1 – Hydrothermal carbonisation and gasification process of wood (version 1) D11.2 – Valorisation of ashes from CDW and thermal gasification (version 1) D11.3 – Optimised engineered metabolic pathways enhancing dodecanol (version 1) D11.4 – Assessment of dodecanol and validation of detergent formulations (version 1)

WP12	Energy, Gas and Ashes Valorisation (TRL6)	3 - CIRCE		19	36	D12.1 – Hydrothermal carbonisation and gasification process of wood (version 2) D12.2 – Valorisation of ashes from CDW and thermal gasification (version 2) D12.3 – Optimised engineered metabolic pathways enhancing dodecanol (version 2) D12.4 – Assessment of dodecanol and validation of detergent formulations (version 2)
WP13	Digital Tools for Improving Circular Flows of Secondary Materials (first release)	17 - eBOS		8	18	D13.1 – End-to-end tracing module and DPP (version 1) D13.2 – Material volume estimation tool (version 1) D13.3 – Supply chain optimisation tool (version 1) D13.4 – Life cycle analysis computation tool (version 1)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D13.5 – Integrated Circular Wood Upcycling Platform (version 1)

WP14	Digital Tools for Improving Circular Flows of Secondary Materials (final release)	17 - eBOS		19	36	D14.1 – End-to-end tracing module and DPP (version 2) D14.2 – Material volume estimation tool (version 2) D14.3 – Supply chain optimisation tool (version 2) D14.4 – Life cycle analysis computation tool (version 2) D14.5 – Integrated Circular Wood Upcycling Platform (version 2)
WP15	Material Flows and Pilot Activities	4 - NTUA		36	48	D15.1 – CDW pilot installation and validation D15.2 – Assessment of the cascade valorisation scheme
WP16	Skills Development, and Standardisation, Policy & Certification Influencing for RP1	6 - VTT		1	18	D16.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 1) D16.2 – Skills activities, trainings and lifelong learning programmes (version 1) D16.3 – Standardisation activities (version 1) D16.4 – Policy recommendations for developing a circular and sustainable process industry (version 1)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables

WP17	Skills Development, and Standardisation, Policy & Certification Influencing for RP2	6 - VTT		19	36	D17.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 2) D17.2 – Skills activities, trainings and lifelong learning programmes (version 2) D17.3 – Standardisation activities (version 2) D17.4 – Policy recommendations for developing a circular and sustainable process industry (version 2)
WP18	Skills Development, and Standardisation, Policy & Certification Influencing for RP3	6 - VTT		37	48	D18.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 3) D18.2 – Skills activities, trainings and lifelong learning programmes (version 3) D18.3 – Standardisation activities (version 3) D18.4 – Policy recommendations for developing a circular and sustainable process industry (version 3)
WP19	Impact Maximisation and Increasing Competitiveness for RP1	19 - R2M		1	18	D19.1 – Communication and Dissemination plan and activities (version 1) D19.2 – Exploitation and Business Innovation Plan and activities (version 1) D19.3 – Synergies with relevant initiatives and best practises exchange (version 1)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D19.4 – Exploitation and Business Innovation Plan and activities (version 2) D19.5 – Market analysis, business models and revenue streams (version 1)
WP20	Impact Maximisation and Increasing Competitiveness for RP2	19 - R2M		19	36	D20.1 – Communication and Dissemination plan and activities (version 2) D20.2 – Synergies with relevant initiatives and best practises exchange (version 2) D20.3 – Exploitation and Business Innovation Plan and activities (version 3) D20.4 – Market analysis, business models and revenue streams (version 2)
WP21	Impact Maximisation and Increasing Competitiveness for RP3	19 - R2M		37	48	D21.1 – Communication and Dissemination plan and activities (version 3) D21.2 – Synergies with relevant initiatives and best practises exchange (version 3) D21.3 – Exploitation and Business Innovation Plan and activities (version 4) D21.4 – Market analysis, business models and revenue streams (version 3)

Work package WP1 – Project Management and Coordination for RP1

Work Package Number	WP1	**Lead Beneficiary**	1. ICCS
Work Package Name	Project Management and Coordination for RP1		
Start Month	1	**End Month**	18
Objectives			
● Project management according to EU Grant Agreement provisions and conformingly to the respective Project Board organisation; ● Consortium coordination of the activities of the project, and ensure that partners work together effectively, by monitoring the overall technical coordination and progress of the project; ● Controlling and refining the objectives of the project by Quality Assurance (QA), performance management and conflict resolution to ensure scientific excellence for all outputs and internal processes; ● Establish Executive Board and assign managers (Technical, QA and Risk, Ethics); ● Coordination of the involvement of an External Expert Advisory Board in the project; ● Develop an efficient data management plan and open science guidelines towards maximisation of the exploitation potential; ● Set out project's GDPR and gender balance requirements and effectively manage legal and ethics matters.			
Description			

T1.1 Project coordination and management (including quality assurance and risk management) [M01-M18] (ICCS, All partners): will provide the overall project coordination, management and administration by developing and implementing management processes for an efficient and successful project execution, including the QA and risk management. The following tasks will be performed: i) preparation of technical and financial interim reports in a 6-month basis so that deliverables and milestones of each period are closely followed and technical activities follow resource consumption; ii) preparation of the external EC reports, monitoring technical and financial activities as well as audit reports in applicable cases; iii) organisation of consortium meetings; iv) distribution of the EC financial contribution and costs coordination and controlling; v) communication with EC and submission of deliverables to the EC; vi) ensuring the communication between partners at all levels and the proper information exchange; vii) preparation and maintenance of an appropriate communication and collaborative environment for the consortium members (e.g., mailing lists, project repository, internal wikis, reporting tools); viii) support of the evolution of the work plan; ix) implementation of QA procedures with particular focus on monitoring review activities and submitting quality review reports –to undertake this task; x) ensuring that proactive measures are taken to mitigate risks which could jeopardise the planned project outcomes, as well as monitoring the potential critical implementation risks; and xi) management of the external advisory board to provide advice, and validation of the vision, global impact, and outreach of the project. Outputs: D1.1

T1.2 Technological management and scientific connection (including data management) [M01-M18] (NTUA, All partners): will monitor the overall technical progress and work quality to support Project Coordinator and follow up the technical part of the review meetings. It will guarantee the smooth technical cooperation between partners and interconnections between WPs and tasks, participate in the consideration of, and responding to, the opinions and contributions of the relevant working groups, and will suggest alternative technical routes in case of stumbling blocks. Additionally, it will track the milestones and coordinate the technical meetings and will ensure that the scientific and technological outcomes comply with the project's work plan, and the results fulfil the technical requirements for effective progress towards the achievement of the project's goals. In accordance with EC Guidelines regarding data management, this task will also set up a Research Data Management Plan defining how research data collected and/or generated throughout the project will be handled during (and after) the project duration. Following FAIR principles, T1.2 will produce an overview of how research data will be made Findable, Accessible, Interoperable and Reusable elaborating, among other, on the data types, the accessibility to datasets, the use of standards and the publication methodology. In this context, special emphasis will be put on making the research data accessible through open repositories, to maximise the visibility of the project's data results, aiming also, to assist other research activities to find and utilise the knowledge produced by W2W. The task will deliver the project's open science guidelines (as part of D1.1) and Data Management Plan (D1.2) by M06, capturing the envisioned practices at that early stage. Outputs: D1.1, D1.2

T1.3 GDPR, gender balance, legal and ethical issues management [M01-M18] (EBOS, All partners): will design and monitor the procedures and protocols necessary for handling ethical issues during the project. A framework of ethical requirements will be constructed to ensure W2W carrying out of responsible research practice with specific reference to the European Code of Conduct for Research Integrity (ALLEA) and the Data Protection legislation and regulation. It will also set out the ethics requirements (e.g. procedures, criteria to identify/recruit research participants, templates for informed consent, etc.) and manage the ethical issues concerning the design of the information and consent forms to be used in case of involvement of humans through the use cases demonstrations, while ensuring GDPR and Privacy Compliance Policy by producing and properly disseminate informed consent forms for all project participatory activities (demonstration activities, online surveys, evaluation forms, workshops, etc.) according to the EU GDPR regulations and national data protection legislation. It will also ensure that SSH is embedded in the W2W as an enabler of RRI. Outputs: D1.1

Work package WP2 – Project Management and Coordination for RP2

Work Package Number	WP2	**Lead Beneficiary**	1. ICCS
Work Package Name	Project Management and Coordination for RP2		
Start Month	19	**End Month**	36
Objectives			

● Project management according to EU Grant Agreement provisions and conformingly to the respective Project Board organisation; ● Consortium coordination of the activities of the project, and ensure that partners work together effectively, by monitoring the overall technical coordination and progress of the project; ● Controlling and refining the objectives of the project by Quality Assurance (QA), performance management and conflict resolution to ensure scientific excellence for all outputs and internal processes; ● Establish Executive Board and assign managers (Technical, QA and Risk, Ethics); ● Coordination of the involvement of an External Expert Advisory Board in the project; ● Develop an efficient data management plan and open science guidelines towards maximisation of the exploitation potential; ● Set out project's GDPR and gender balance requirements and effectively manage legal and ethics matters.
Description
T2.1 Project coordination and management (including quality assurance and risk management) [M19-36] (ICCS, All partners): will provide the overall project coordination, management and administration by developing and implementing management processes for an efficient and successful project execution, including the QA and risk management. The following tasks will be performed: i) preparation of technical and financial interim reports in a 6-month basis so that deliverables and milestones of each period are closely followed and technical activities follow resource consumption; ii) preparation of the external EC reports, monitoring technical and financial activities as well as audit reports in applicable cases; iii) organisation of consortium meetings; iv) distribution of the EC financial contribution and costs coordination and controlling; v) communication with EC and submission of deliverables to the EC; vi) ensuring the communication between partners at all levels and the proper information exchange; vii) preparation and maintenance of an appropriate communication and collaborative environment for the consortium members (e.g., mailing lists, project repository, internal wikis, reporting tools); viii) support of the evolution of the work plan; ix) implementation of QA procedures with particular focus on monitoring review activities and submitting quality review reports –to undertake this task; x) ensuring that proactive measures are taken to mitigate risks which could jeopardise the planned project outcomes, as well as monitoring the potential critical implementation risks; and xi) management of the external advisory board to provide advice, and validation of the vision, global impact, and outreach of the project. T2.2 Technological management and scientific connection (including data management) [M19-36] (NTUA, All partners): will monitor the overall technical progress and work quality to support Project Coordinator and follow up the technical part of the review meetings. It will guarantee the smooth technical cooperation between partners and interconnections between WPs and tasks, participate in the consideration of, and responding to, the opinions and contributions of the relevant working groups, and will suggest alternative technical routes in case of stumbling blocks. Additionally, it will track the milestones and coordinate the technical meetings and will ensure that the scientific and technological outcomes comply with the project's work plan, and the results fulfil the technical requirements for effective progress towards the achievement of the project's goals. In accordance with EC Guidelines regarding data management, this task will also set up a Research Data Management Plan defining how research data collected and/or generated throughout the project will be handled during (and after) the project duration. Following FAIR principles, T2.2 will produce an overview of how research data will be made Findable, Accessible, Interoperable and Reusable elaborating, among other, on the data types, the accessibility to datasets, the use of standards and the publication methodology. In this context, special emphasis will be put on making the research data accessible through open repositories, to maximise the visibility of the project's data results, aiming also, to assist other research activities to find and utilise the knowledge produced by W2W. Outputs: D2.1
T2.3 GDPR, gender balance, legal and ethical issues management [M19-36] (EBOS, All partners): will design and monitor the procedures and protocols necessary for handling ethical issues during the project. A framework of ethical requirements will be constructed to ensure W2W carrying out of responsible research practice with specific reference to the European Code of Conduct for Research Integrity (ALLEA) and the Data Protection legislation and regulation. It will also set out the ethics requirements (e.g. procedures, criteria to identify/recruit research participants, templates for informed consent, etc.) and manage the ethical issues concerning the design of the information and consent forms to be used in case of involvement of humans through the use cases demonstrations, while ensuring GDPR and Privacy Compliance Policy by producing and properly disseminate informed consent forms for all project participatory activities (demonstration activities, online surveys, evaluation forms, workshops, etc.) according to the EU GDPR regulations and national data protection legislation. It will also ensure that SSH is embedded in the W2W as an enabler of RRI.

Work package WP3 – Project Management and Coordination for RP3

Work Package Number	WP3	**Lead Beneficiary**	1. ICCS

Work Package Name	Project Management and Coordination for RP3		
Start Month	37	**End Month**	48

Objectives

● Project management according to EU Grant Agreement provisions and conformingly to the respective Project Board organisation; ● Consortium coordination of the activities of the project, and ensure that partners work together effectively, by monitoring the overall technical coordination and progress of the project; ● Controlling and refining the objectives of the project by Quality Assurance (QA), performance management and conflict resolution to ensure scientific excellence for all outputs and internal processes; ● Establish Executive Board and assign managers (Technical, QA and Risk, Ethics); ● Coordination of the involvement of an External Expert Advisory Board in the project; ● Develop an efficient data management plan and open science guidelines towards maximisation of the exploitation potential; ● Set out project's GDPR and gender balance requirements and effectively manage legal and ethics matters.

Description

T3.1 Project coordination and management (including quality assurance and risk management) [M37-48] (ICCS, All partners): will provide the overall project coordination, management and administration by developing and implementing management processes for an efficient and successful project execution, including the QA and risk management. The following tasks will be performed: i) preparation of technical and financial interim reports in a 6-month basis so that deliverables and milestones of each period are closely followed and technical activities follow resource consumption; ii) preparation of the external EC reports, monitoring technical and financial activities as well as audit reports in applicable cases; iii) organisation of consortium meetings; iv) distribution of the EC financial contribution and costs coordination and controlling; v) communication with EC and submission of deliverables to the EC; vi) ensuring the communication between partners at all levels and the proper information exchange; vii) preparation and maintenance of an appropriate communication and collaborative environment for the consortium members (e.g., mailing lists, project repository, internal wikis, reporting tools); viii) support of the evolution of the work plan; ix) implementation of QA procedures with particular focus on monitoring review activities and submitting quality review reports –to undertake this task; x) ensuring that proactive measures are taken to mitigate risks which could jeopardise the planned project outcomes, as well as monitoring the potential critical implementation risks; and xi) management of the external advisory board to provide advice, and validation of the vision, global impact, and outreach of the project.

T3.2 Technological management and scientific connection (including data management) [M37-48] (NTUA, All partners): will monitor the overall technical progress and work quality to support Project Coordinator and follow up the technical part of the review meetings. It will guarantee the smooth technical cooperation between partners and interconnections between WPs and tasks, participate in the consideration of, and responding to, the opinions and contributions of the relevant working groups, and will suggest alternative technical routes in case of stumbling blocks. Additionally, it will track the milestones and coordinate the technical meetings and will ensure that the scientific and technological outcomes comply with the project's work plan, and the results fulfil the technical requirements for effective progress towards the achievement of the project's goals. In accordance with EC Guidelines regarding data management, this task will also set up a Research Data Management Plan defining how research data collected and/or generated throughout the project will be handled during (and after) the project duration. Following FAIR principles, T3.2 will

produce an overview of how research data will be made Findable, Accessible, Interoperable and Reusable elaborating, among other, on the data types, the accessibility to datasets, the use of standards and the publication methodology. In this context, special emphasis will be put on making the research data accessible through open repositories, to maximise the visibility of the project's data results, aiming also, to assist other research activities to find and utilise the knowledge produced by W2W. Outputs: D3.1

T3.3 GDPR, gender balance, legal and ethical issues management [M37-48] (EBOS, All partners): will design and monitor the procedures and protocols necessary for handling ethical issues during the project. A framework of ethical requirements will be constructed to ensure W2W carrying out of responsible research practice with specific reference to the European Code of Conduct for Research Integrity (ALLEA) and the Data Protection legislation and regulation. It will also set out the ethics requirements (e.g. procedures, criteria to identify/recruit research participants, templates for informed consent, etc.) and manage the ethical issues concerning the design of the information and consent forms to be used in case of involvement of humans through the use cases demonstrations, while ensuring GDPR and Privacy Compliance Policy by producing and properly disseminate informed consent forms for all project participatory activities

(demonstration activities, online surveys, evaluation forms, workshops, etc.) according to the EU GDPR regulations and national data protection legislation. It will also ensure that SSH is embedded in the W2W as an enabler of RRI.

Work package WP4 – Transformation Framework for Sustainable-by-design Construction Demolition Wastes Biorefining

Work Package Number	WP4	**Lead Beneficiary**	16. DRAXIS
Work Package Name	Transformation Framework for Sustainable-by-design Construction Demolition Wastes Biorefining		
Start Month	1	**End Month**	10
Objectives			
Description			

T4.1 Actors mapping and key stakeholders' interactions [M01-M10] (DRAXIS, ICCS, NTUA, UPV, CERTH, RCM, QPLAN): will undertake an in-depth analysis to define the stakeholders involved in each one of the value chains falling under the scope of the project. More specifically, a detailed mapping will take place aiming to provide information regarding the stakeholders and their key interactions taking into consideration that the general actor categories of a value chain are wood waste owners/generators, wood waste sorting and upcycling technology providers, recyclers, products end users, public authorities, and standardization actors. Outputs: D4.1

T4.2 Wood characterisation and classification schemes [M01-M10] (CEPS, NTUA, ICCS, DRAXIS): will perform a study on parameters linked to wood characterisation, as well as on classification schemes at country level, combining inputs from T4.1, T4.3 and perspectives from different stakeholders. Based on this work the task will provide a proposal for a common classification scheme. Given that various characterisation and classification schemes are applied per country, this task aims to create a common framework for recycled wood management. That is essential for developing appropriate standards and regulations for recycled wood products to ensure their safety and sustainability, since recycled wood products may contain hazardous chemicals that can leach into the environment or pose a health risk to workers and consumers. Thus, this task will list the developing innovative processes that utilise clean, waste, and polluted wood, to lead to the development of high-value products and contribute to a more sustainable and circular economy. Outputs: D4.2 T4.3 CDW recycled wood cascade valorisation framework for pure and mixed treatment [M01-M10] (NTUA, ICCS, ELAK, DRAXIS, R2M): will create a first analysis of all the available routes and valorisation schemes from the provided feedstock. A list of available technologies accompanied with the developed ones from W2W shall be considered for the development of the most appropriate valorisation routes. The characterisation of the available feedstock as provided in T5-6.2 as well as the dynamic LCA (T13-14.4) calculation shall also be provisionally considered for the selection of each potential route. Additional elements related to offer vs demand, logistics optimisations, supply chain management etc. will set the base towards a dynamic valorisation framework aiming to maximise the resource recovery potential, based on the principles of circular management. The developed framework will be validated in the context of W2W and set the principles for replication towards the additional identified sectors. Outputs: D4.3

T4.4 Data sharing model, FAIR digital assets and digital passports principles [M01-M10] (TUB, UPV, DRAXIS, EBOS):

will develop the concept for the W2W data sharing model based on FAIR principles; the model will support the development of the Digital Product Passport (DPP), including the DPP principles, as well as the definition of the digital data required to FAIR use of the DPP in the project context. Therefore, existing solutions will be analysed, and based on the outcomes of this background analysis, a model with a suitable ontology will be developed, focusing on the increased confidence in the materials' lifecycle. The model will also support the FAIR data sharing and communication among stakeholders, while the data quality is ensured. This is intended to contribute to the achievement of high-quality, lowrisk project outcomes and to ensure their long-term sustainability. Outputs: D4.4

T4.5 Definition of impact assessment methodologies and KPIs [M01-M10] (DRAXIS, NTUA): will set up the most suitable methodologies and Key Performance Indicators (KPIs) required to quantitatively evaluate the results obtained

by implementing the W2W framework in the project use cases. The definition will start with an in-depth analysis of all the current systems to establish the starting point (KPIs baseline values) and to identify the detailed requirements of the W2W use cases. A detailed specification for the development of all composing elements of the experiments will be laid down, establishing the boundaries for the integration of the W2W framework. Outputs: D4.5

Work package WP5 – Management of Waste Streams (first release)

Work Package Number	WP5	**Lead Beneficiary**	1. ICCS
Work Package Name	Management of Waste Streams (first release)		
Start Month	8	**End Month**	18

Objectives

● Develop the overall methodology for the CDW separation and sorting systems; ● Put in place the most adequate optical and spectroscopic characterisation techniques for the most efficient particle size of wood and glass coming from CDW to further processing these materials; ● Deploy an adaptive system for sorting glass waste; ● Develop an adaptive cyber-physical system for sorting wood waste extracted from CDW; ● Create a station where a collaborative robot will cooperate with workforce to jointly process the wood stream with existing assembled elements separated; ● Develop an MR system detecting the material and state of the recycled products.

Description

T5.1 CDW flows valorisation and digitalisation [M08-M18] (ELAK, ICCS, RCM): will develop an overall methodology for the CDW separation and sorting systems. The task will evaluate three district levels: (1) at each construction site where CDW is primarily disposed of, (2) at a regional level that consists of two or more urban core areas, and (3) at the city/urban level of the pilots' countries. This methodology will contribute to the broader goal of promoting a circular economy and sustainable waste management practices towards the aims of the Hubs for Circularity. The relevant streams for optimising the cascade valorisation of wood particles will be analysed. The total investment costs for the development of private recycling centres handling various volumes of materials, in correlation to logistics optimisation shall be considered for the overall system. The overall analysis shall result towards a digital tool supporting the decision and investment needs for each of the analysed scenarios. Any permission needed for the installation of the developed sorting system will also be covered in this task. Outputs: D5.1

T5.2 Wood and glass non-destructive real time techniques analysis and definition of optimal size [M08-M18] (IRIS, ICCS): will put in place the most adequate optical and spectroscopic characterisation techniques to determinate which is the most efficient particle size of wood and glass coming from CDW to further process these materials. Through this task, ICCS will run trials within the wood and glass recycling processes by pre-selecting different sizes of shredding in the processes to collect samples with different sizes of sorted wood and glass. These referenced samples will be sent to IRIS to be analysed through analytical techniques such as NIR/SWIR-HSI or XRF, and the optimal particle size will be defined according to factors such as the presence of impurities in the sorted wood and glass. Moreover, IRIS will design and develop an offline handheld NIR device, as part of the offline monitoring techniques, which will allow to obtain materials' information in real-time (moisture, type of wood, among other parameters) at the end of the sorting process. Chemometric models will be built, which will allow to predict and assess the Critical Quality Attributes (CQA) of wood and glass final products developed through the valorisation processes (i.e., wood panels, among others.). The device will be shifted and validated by ICCS to assess relevant product parameters. Outputs: D5.2

T5.3 Adaptive glass sorting system based on recycling thresholds [M08-M18] (ICCS): will deploy an adaptive system for sorting glass waste, by gradually separating the grades of glass according to quality, purity and existence of foreign materials. After separating glass from other CDW via a first robotic conveyor, glass wastes will enter a crushing mechanism to be reduced to the most efficient particle size for further processing. The position and composition of each particle will be defined, using a combination of a multi-sensor system (as advanced non-destructive characterisation techniques) and a dynamic criteria adaptation, for the particles to be sorted in each glass waste stream. After separating the pure glass from glass with foreign materials, pure glass will be further separated into four categories depending on the quality and valorisation potential. Glass including other materials will proceed in a different conveyor where the

combination of characterisation techniques will be used again to direct it into flows, and to be separated by either mechanical or chemical means. Outputs: D5.3

T5.4 Adaptive wood sorting system based on cascade valorisation thresholds [M08-M18] (ICCS): will develop an adaptive cyber-physical system for sorting wood waste extracted from CDW by applying cascade valorisation, to separate different grades of wood depending on their potential for recycling. Initially, wood will be separated via a robotic arm from other CDW, and separated into two flows depending on the presence of assembly elements. Wood waste assembled with metallic or other components will be directed to a separate conveyor line incorporating Human-Robot Collaboration, which is described in T5.5. Waste comprised purely of wood will enter a shredding system to reach optimal particle size. A combination of a multi-sensor system and dynamic adaptation based on LCA will be used to define the position and composition of each particle, so as to sort it in the proper stream. Wood containing no foreign materials will be categorised into four divisions depending on quality and valorisation potential. Wood including contaminants will be directed to a separate conveyor to be characterised again with the combination of techniques, and subsequently separated by either mechanical or chemical means. Outputs: D5.3

T5.5 Robotics and Human Robot Collaboration for better separation [M08-M18] (ICCS): will create a station where a collaborative robot (cobot) will cooperate with workforce to jointly process the wood stream with existing assembled elements separated in T5.4. A single workpiece will be moved from the infeed conveyor using a robot or another appropriate machine and will be fixtured. Computer vision techniques will assist in locating the assembly elements and their material will be classified with hyperspectral imaging employed in this or one of the previous sorting stages. Depending on the type of the assembly element e.g., screw, nail, and bolt, the time required to remove it from the workpiece and other key factors, a plan for the required dismantling tasks will be created aiming to optimise processing time, while reducing fatigue for the human worker. Each task will be assigned to either the human worker (T5.6) or the robot and an appropriately designed human robot interface will be guiding the human worker and the overall process. Finally, the now pure wooden workpiece will be released from its fixture and directed to the conveyors described in T5.4 to be sorted anew. The system will be developed in ICCS premises, before assembled to the wood sorting system and tested in pilot area. Outputs: D5.4

T5.6 MR system for better separation and quality control [M08-M18] (ICCS): will program an application for an MRHMD to interact with the cobot to i) stream data from the cobot to monitor its state and operating modes, and ii) guide the operator to the needed task. A user-friendly interface will deliver fast and clear information about the current state of the wood material and the actions to be performed. Upon performance of the requested action (including, separation of metallic bits or other components from the wood material), the MR system will communicate to the cobot to resume in action. Object detection algorithms will be embodied to the MR system, detecting the material and state of the recycled products to control the outcome of the tasks performed not only by the worker, but the cobot itself to ensure the accuracy of actions is high. Assessment through pre-determined evaluation criteria will ensure that a structured quality control is performed. The task will follow an iterative approach and is organised in three main phases, as follows: 1) design and development of a configuration layer enabling the interface between cobot/MR system; 2) design and development of a user-friendly MR interface for operator guidance/data visualisation; 3) development of the quality control component. Outputs: D5.4

Work package WP6 – Management of Waste Streams (final release)

Work Package Number	WP6	**Lead Beneficiary**	1. ICCS
Work Package Name	Management of Waste Streams (final release)		
Start Month	19	**End Month**	36
Objectives			
● Develop the overall methodology for the CDW separation and sorting systems; ● Put in place the most adequate optical and spectroscopic characterisation techniques for the most efficient particle size of wood and glass coming from CDW to further processing these materials; ● Deploy an adaptive system for sorting glass waste; ● Develop an adaptive cyber-physical system for sorting wood waste extracted from CDW; ● Create a station where a collaborative robot will			

cooperate with workforce to jointly process the wood stream with existing assembled elements separated; ● Develop an MR system detecting the material and state of the recycled products

Description
T6.1 CDW flows valorisation and digitalisation [M19-M36] (ELAK, ICCS, RCM): will develop an overall methodology for the CDW separation and sorting systems. The task will evaluate three district levels: (1) at each construction site where CDW is primarily disposed of, (2) at a regional level that consists of two or more urban core areas, and (3) at the city/urban level of the pilots' countries. This methodology will contribute to the broader goal of promoting a circular economy and sustainable waste management practices towards the aims of the Hubs for Circularity. The relevant streams for optimising the cascade valorisation of wood particles will be analysed. The total investment costs for the development of private recycling centres handling various volumes of materials, in correlation to logistics optimisation shall be considered for the overall system. The overall analysis shall result towards a digital tool supporting the decision and investment needs for each of the analysed scenarios. Any permission needed for the installation of the developed sorting system will also be covered in this task. Outputs: D6.1 T6.2 Wood and glass non-destructive real time techniques analysis and definition of optimal size [M19-M36] (IRIS, ICCS): will put in place the most adequate optical and spectroscopic characterisation techniques to determinate which is the most efficient particle size of wood and glass coming from CDW to further process these materials. Through this task, ICCS will run trials within the wood and glass recycling processes by pre-selecting different sizes of shredding in the processes to collect samples with different sizes of sorted wood and glass. These referenced samples will be sent to IRIS to be analysed through analytical techniques such as NIR/SWIR-HSI or XRF, and the optimal particle size will be defined according to factors such as the presence of impurities in the sorted wood and glass. Moreover, IRIS will design and develop an offline handheld NIR device, as part of the offline monitoring techniques, which will allow to obtain materials' information in real-time (moisture, type of wood, among other parameters) at the end of the sorting process. Chemometric models will be built, which will allow to predict and assess the Critical Quality Attributes (CQA) of wood and glass final products developed through the valorisation processes (i.e., wood panels, among others.). The device will be shifted and validated by ICCS to assess relevant product parameters. Outputs: D6.2 T6.3 Adaptive glass sorting system based on recycling thresholds [M19-M36] (ICCS): will deploy an adaptive system for sorting glass waste, by gradually separating the grades of glass according to quality, purity and existence of foreign materials. After separating glass from other CDW via a first robotic conveyor, glass wastes will enter a crushing mechanism to be reduced to the most efficient particle size for further processing. The position and composition of each particle will be defined, using a combination of a multi-sensor system (as advanced non-destructive characterisation techniques) and a dynamic criteria adaptation, for the particles to be sorted in each glass waste stream. After separating the pure glass from glass with foreign materials, pure glass will be further separated into four categories depending on the quality and valorisation potential. Glass including other materials will proceed in a different conveyor where the combination of characterisation techniques will be used again to direct it into flows, and to be separated by either mechanical or chemical means. Outputs: D6.3 T6.4 Adaptive wood sorting system based on cascade valorisation thresholds [M19-M36] (ICCS): will develop an adaptive cyber-physical system for sorting wood waste extracted from CDW by applying cascade valorisation, to separate different grades of wood depending on their potential for recycling. Initially, wood will be separated via a robotic arm from other CDW, and separated into two flows depending on the presence of assembly elements. Wood waste assembled with metallic or other components will be directed to a separate conveyor line incorporating Human-Robot Collaboration, which is described in T6.5. Waste comprised purely of wood will enter a shredding system to reach optimal particle size. A combination of a multi-sensor system and dynamic adaptation based on LCA will be used to define the position and composition of each particle, so as to sort it in the proper stream. Wood containing no foreign materials will be categorised into four divisions depending on quality and valorisation potential. Wood including contaminants will be directed to a separate conveyor to be characterised again with the combination of techniques, and subsequently separated by either mechanical or chemical means. Outputs: D6.3 T6.5 Robotics and Human Robot Collaboration for better separation [M19-M36] (ICCS): will create a station where a collaborative robot (cobot) will cooperate with workforce to jointly process the wood stream with existing assembled elements separated in T6.4. A single workpiece will be moved from the infeed conveyor using a robot or another appropriate machine and will be fixtured. Computer vision techniques will assist in locating the assembly elements and their material will be classified with hyperspectral imaging employed in this or one of the previous sorting stages.

Depending on the type of the assembly element e.g., screw, nail, and bolt, the time required to remove it from the workpiece and other key factors, a plan for the required dismantling tasks will be created aiming to optimise processing time, while reducing fatigue for the human worker. Each task will be assigned to either the human worker (T6.6) or the

robot and an appropriately designed human robot interface will be guiding the human worker and the overall process. Finally, the now pure wooden workpiece will be released from its fixture and directed to the conveyors described in T6.4 to be sorted anew. The system will be developed in ICCS premises, before assembled to the wood sorting system and tested in pilot area. Outputs: D6.4

T6.6 MR system for better separation and quality control [M19-M36] (ICCS): will program an application for an MRHMD to interact with the cobot to i) stream data from the cobot to monitor its state and operating modes, and ii) guide the operator to the needed task. A user-friendly interface will deliver fast and clear information about the current state of the wood material and the actions to be performed. Upon performance of the requested action (including, separation of metallic bits or other components from the wood material), the MR system will communicate to the cobot to resume in action. Object detection algorithms will be embodied to the MR system, detecting the material and state of the recycled products to control the outcome of the tasks performed not only by the worker, but the cobot itself to ensure the accuracy of actions is high. Assessment through pre-determined evaluation criteria will ensure that a structured quality control is performed. The task will follow an iterative approach and is organised in three main phases, as follows: 1) design and development of a configuration layer enabling the interface between cobot/MR system; 2) design and development of a user-friendly MR interface for operator guidance/data visualisation; 3) development of the quality control component. Outputs: D6.4

Work package WP7 – Cascade Refinement Technologies for Wood Waste Upcycling (TRL5)

Work Package Number	WP7	**Lead Beneficiary**	4. NTUA
Work Package Name	Cascade Refinement Technologies for Wood Waste Upcycling (TRL5)		
Start Month	8	**End Month**	18

Objectives

● Develop and optimise the extraction of cellulose nanofibers form waste wood; ● Improve the compatibility of the CNF via functionalisation with organic compounds, and the compatibility of the lignin via functionalisation with fatty acids; ● Deliver composite compounds that will serve as low-cost sustainable recycled materials; ● Create a study on glass recycling thresholds on post-consumer flow; ● Validate the final products through the integration of the composite components into a developed prefabricated façade system.

Description

T7.1 Cascade to function, mixed wood treatment [M08-M18] (UHE): will be executed through 3 subtasks. Subtask 1: Isolation of cellulose nanofillers: will prepare and purify cellulose nanofillers from wood waste using chemical and mechanical treatments. Organosolv pulping based on mixture of formic and acetic acids will used to fractionate the wood waste into solid fraction (cellulose pulp) rich in cellulose and liquid fraction (liquor) rich in lignin. The process variable such as chemical concentration, treatment time at different extraction conditions and particle size will be analysed. Isolation will be carried out at lab scale. Mechanical treatment will be performed to achieve cellulose nanofibers. CNF will be isolated by using mechanical processing routes (ultra fine grinding using a special grinder or high-pressure homogenization). The CNF will be characterised via X-Ray, AFM, TGA, DSC, particle size analysis and other techniques to completely define their physicochemical and thermal properties. Subtask 2: Functionalisation / Modification of cellulose: the final properties achieved by composites containing nanofillers are highly dependent on how well the fillers are dispersed and interact within the PLA matrix. For this reason, it is important that specific functionalisation methods based on these nanoparticles are to be developed following the PLA particularities. Thus, taking advantage of the hydroxyl functionalities available at the surface of cellulose nanofillers, functionalization via salinisation and via esterification with fatty acids will be carried out. The modified CNF will be characterised via FTIR, 13C NMR, TGA, DSC, contact angle analysis and other techniques to completely define their physicochemical and thermal properties. Subtask 3 Lignin isolation and modification: lignin will be precipitated from the liquor by adjusting the pH of the solution. Recovered lignin will be washed and dried. Lignin will be subjected to a chemical modification to enhance its compatibility with PLA. Esterification is one of the easiest reactions to perform considering the reaction parameters and used reactants. Moreover, when lignin is modified by esterification, hydroxyl groups are replaced by ester

substituent and thus, reduce the number of hydrogen bonding and generate an increased free volume in the molecule, providing mobility of the chains reducing the glass transition point of lignin and increasing its thermoplasticity. These gained thermal properties are highly desirable for current industrial polymeric processing technology. Lignin will be modified by fatty acids. The reaction condition (time, temperature, chain length of the fatty acid) will be optimised. The un-modified and modified lignins will be characterised by different analytical techniques (GPC, FT-IR, 31P NMR, DSC, TGA and contact angle). Outputs: D7.1

T7.2 New mix between wood, glass and composites for wallboards [M08-M18] (NTUA, ICCS): will combine the recovered lignocellulose-based materials from wood waste (prepared in T7.1) with thermoplastic matrices, to deliver composite compounds that will serve as low-cost sustainable recycled material aiming at construction applications. Different grades of biodegradable matrices (e.g., PLA grades of different optical purity, PLA copolymers, PLA blends) will be examined with different ratios (<30 %) of materials from T7.1 to meet the targeted specifications in terms of thermal and mechanical performance. The biocomposites processing will be performed in mixers and twin-screw extruders and the melted mass or extruded strands will be grinded or cut in pellets, which will serve as a compound for the subsequent transformation process. Compounding conditions, such as feedstock preheating for drying, temperature profile (including melt temperature), shearing rate (screw rotation speed), degassing via vacuum, screw configuration through the addition of mixing elements, will be thoroughly studied and optimised to achieve good compatibility and avoid filler treatment and/or addition of compatibilizers which are costly. Compounds will be characterised with SotA techniques (SEM, DSC, TGA, FTIR-ATR, GPC, tensile, DMA, rheometry) to assess filler dispersion and distribution in the matrix, extent of thermomechanical degradation of both fillers and matrix, thermal stability, mechanical performance and visco-elastic behavior which is a critical parameter for processing. A feedback loop will be established with the partners involved in T7.1 to construct correlations between cascade conditions and biocomposites processability. Then, on selected biocomposite compounds, a proper additivation strategy will be developed to prepare hydrolytically stabilized PLA for long-term applications, such as in construction field. Anti-hydrolysis additives (such as carbodiimides, polycarbodiimides) will be added the polymeric matrix in various concentrations via extrusion, and moulded in thin films to be submitted in accelerated ageing. The molecular weight decrease will be monitored and associated to mechanical performance deterioration. The target is to minimize the MW change compared to the non-stabilized PLA compounds. The optimised stabilizer formulation and processing conditions (set in T7.2) will be applied for biocomposite processing at higher amounts. The compounds will be used to manufacture composite elements in collaboration with end user via compression molding in heated plates press. Outputs: D7.2

T7.3 Final products and initial validation [M08-M18] (FOCCHI, NTUA, UHE): will validate the final products through the integration of the composite components into a developed prefabricated façade system. The physical prototype will allow testing the possibility of introducing innovative composite materials into the façade industry. Through the manufacturing process, technical aspects (e.g., dimension, fixing mode, aesthetic finishes, etc.) will be evaluated to clarify their potential use and the compliance with regulatory and market requirements. Specifically, the validation process aims to investigate acoustic performance of composite components. Acoustic test activities will be carried out to compare integrated façade with the “traditional” one. The benefits in terms of noise reduction will be evaluated to clarify the potential applicability in the construction market. Outputs: D7.3

Work package WP8 – Cascade Refinement Technologies for Wood Waste Upcycling (TRL6)

Work Package Number	WP8	**Lead Beneficiary**	4. NTUA
Work Package Name	Cascade Refinement Technologies for Wood Waste Upcycling (TRL6)		
Start Month	19	**End Month**	36
Objectives			
● Develop and optimise the extraction of cellulose nanofibers form waste wood; ● Improve the compatibility of the CNF via functionalisation with organic compounds, and the compatibility of the lignin via functionalisation with fatty acids; ● Deliver composite compounds that will serve as low-cost sustainable recycled materials; ● Create a study on glass recycling thresholds on post-consumer flow; ● Validate the final products through the integration of the composite components into a developed prefabricated façade system.			

Description
T8.1 Cascade to function, mixed wood treatment [M19-M36] (UHE): will be executed through 3 subtasks. Subtask

1: Isolation of cellulose nanofillers: will prepare and purify cellulose nanofillers from wood waste using chemical and mechanical treatments. Organosolv pulping based on mixture of formic and acetic acids will used to fractionate the wood waste into solid fraction (cellulose pulp) rich in cellulose and liquid fraction (liquor) rich in lignin. The process variable such as chemical concentration, treatment time at different extraction conditions and particle size will be analysed. Isolation will be carried out at lab scale. Mechanical treatment will be performed to achieve cellulose nanofibers. CNF will be isolated by using mechanical processing routes (ultra fine grinding using a special grinder or high-pressure homogenization). The CNF will be characterised via X-Ray, AFM, TGA, DSC, particle size analysis and other techniques to completely define their physicochemical and thermal properties. Subtask 2: Functionalisation / Modification of cellulose: the final properties achieved by composites containing nanofillers are highly dependent on how well the fillers are dispersed and interact within the PLA matrix. For this reason, it is important that specific functionalisation methods based on these nanoparticles are to be developed following the PLA particularities. Thus, taking advantage of the hydroxyl functionalities available at the surface of cellulose nanofillers, functionalization via salinisation and via esterification with fatty acids will be carried out. The modified CNF will be characterised via FTIR, 13C NMR, TGA, DSC, contact angle analysis and other techniques to completely define their physicochemical and thermal properties. Subtask 3 Lignin isolation and modification: lignin will be precipitated from the liquor by adjusting the pH of the solution. Recovered lignin will be washed and dried. Lignin will be subjected to a chemical modification to enhance its compatibility with PLA. Esterification is one of the easiest reactions to perform considering the reaction parameters and used reactants. Moreover, when lignin is modified by esterification, hydroxyl groups are replaced by ester substituent and thus, reduce the number of hydrogen bonding and generate an increased free volume in the molecule, providing mobility of the chains reducing the glass transition point of lignin and increasing its thermoplasticity. These gained thermal properties are highly desirable for current industrial polymeric processing technology. Lignin will be modified by fatty acids. The reaction condition (time, temperature, chain length of the fatty acid) will be optimised. The un-modified and modified lignins will be characterised by different analytical techniques (GPC, FT-IR, 31P NMR, DSC, TGA and contact angle). Outputs: D8.1

T8.2 New mix between wood, glass and composites for wallboards [M19-M36] (NTUA, ICCS): will combine the recovered lignocellulose-based materials from wood waste (prepared in T8.1) with thermoplastic matrices, to deliver composite compounds that will serve as low-cost sustainable recycled material aiming at construction applications. Different grades of biodegradable matrices (e.g., PLA grades of different optical purity, PLA copolymers, PLA blends) will be examined with different ratios (<30 %) of materials from T8.1 to meet the targeted specifications in terms of thermal and mechanical performance. The biocomposites processing will be performed in mixers and twin-screw extruders and the melted mass or extruded strands will be grinded or cut in pellets, which will serve as a compound for the subsequent transformation process. Compounding conditions, such as feedstock preheating for drying, temperature profile (including melt temperature), shearing rate (screw rotation speed), degassing via vacuum, screw configuration through the addition of mixing elements, will be thoroughly studied and optimised to achieve good compatibility and avoid filler treatment and/or addition of compatibilizers which are costly. Compounds will be characterised with SotA techniques (SEM, DSC, TGA, FTIR-ATR, GPC, tensile, DMA, rheometry) to assess filler dispersion and distribution in the matrix, extent of thermomechanical degradation of both fillers and matrix, thermal stability, mechanical performance and visco-elastic behavior which is a critical parameter for processing. A feedback loop will be established with the partners involved in T8.1 to construct correlations between cascade conditions and biocomposites processability. Then, on selected biocomposite compounds, a proper additivation strategy will be developed to prepare hydrolytically stabilized PLA for long-term applications, such as in construction field. Anti-hydrolysis additives (such as carbodiimides, polycarbodiimides) will be added the polymeric matrix in various concentrations via extrusion, and moulded in thin films to be submitted in accelerated ageing. The molecular weight decrease will be monitored and associated to mechanical performance deterioration. The target is to minimize the MW change compared to the non-stabilized PLA compounds. The optimised stabilizer formulation and processing conditions (set in T8.2) will be applied for biocomposite processing at higher amounts. The compounds will be used to manufacture composite elements in collaboration with end user via compression molding in heated plates press. Outputs: D8.2

T8.3 Final products and initial validation [M19-M36] (FOCCHI, NTUA, UHE): will validate the final products through the integration of the composite components into a developed prefabricated façade system. The physical prototype will allow testing the possibility of introducing innovative composite materials into the façade industry. Through the manufacturing process, technical aspects (e.g., dimension, fixing mode, aesthetic finishes, etc.) will be evaluated to clarify their potential use and the compliance with regulatory and market requirements. Specifically, the validation process aims to investigate acoustic performance of composite components. Acoustic test activities will be carried out to compare integrated façade with the “traditional” one. The benefits in terms of noise reduction will be evaluated to clarify the potential applicability in the construction market. Outputs: D8.3

Work package WP9 – Chemical and Bioremediation Technologies for Wood Waste Upcycling

(TRL5)

Work Package Number	WP9	**Lead Beneficiary**	2. LERMAB
Work Package Name	Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL5)		
Start Month	8	**End Month**	18

Objectives

● Defibration and pre-treatment of CDW by steam explosion; ● Removal of adhesives, treatment and coating products, such as glue; ● Bioremediation of the effluent using fungi; ● Production of advanced wood panel; ● Validate the final products produced.

Description

T9.1 Glue separation from wood [M08-M18] (LERMAB): will perform a chemical and morphological characterisation of fibres. As steam explosion combines a hydrolytic step, a physical degradation allows both the defibration of the wood waste and the removal of glue from the fibres (> 90%). During this task, a chemical-free steam explosion process will be optimised for fibre generation at lab scale (1 Kg fibres/ day). The upscaling of the process optimised will be then carried out using a pilot-scale facility. A unique batch prototype equipped with a gas scrubber will be used. Then, an effort will be made on the wood fibers/particles characterisation but also on the gaseous effluent potentially rich in formaldehyde and nitrogen gases produced from the depolymerisation of resin. Gaseous contaminant will be recovered by gas washing and treated as done for the TRL5 version. Outputs: D9.1

T9.2 Bioremediation wood treatment [M08-M18] (LERMAB): will perform a mycoremediation by solid state and submerged fermentation. Heavy metals desorption will be addressed by fungal produced oxalate chelation. Fungal tolerance and assimilation of organic preservatives in association with heavy metals will be investigated. For bioremediation of the SE effluent, submerged fermentation with formaldehyde tolerant fungi will be used. Removal of formaldehyde and characterisation of environmental toxicity of aquaeous effluent will be monitored. Water recycling in the process will be also investigated. Characterisation of contaminant removal will be performed by spectrometric techniques that can be used at the industrial scale. Finally, a screening of added value molecules (exopolysaccarides, nano polymers) resulting from the fermentation and SE combination will be implemented. The upscaling of the process optimised will be then carried out using a pilot-scale facility. The mycoremediation of C&D wood containing wood preservatives will be performed at the pilot scale. Optimisation of the fungal growth on wood wafers and particles will be performed at the lowest possible cost. Bioremediation of the aqueous effluents will be done at higher scale TRL6 using statistical representative sampling. Outputs: D9.1

T9.3 Liquefaction Process of mixed wood waste and green adhesives based on polyols [M08-M18] (UHE, LERMAB): will be executed in two subtasks: Subtask 1: Liquefaction: the reaction condition of the liquefaction process (temperature, time, ratio biomass/solvent) will be optimised using experimental design. The optimum conditions of the process in term of yield and functionalities of the bio-polyols will be determined. The liquefaction yield and properties of the bio-polyol (chemical structure, hydroxyl number, acid number, viscosity) will be determined. Subtask 2: Cationic ringopening polymerisation: is the synthesis of a family bio-polyols by cationic ROP using acid catalysts. This process allows to synthesise bio-polyols of controlled structure with a low homopolymer content at low temperatures (20-80ºC) and atmospheric pressure from lignin produced in T7-8.1. Different types of catalysis and reaction conditions will be tested to optimise the properties of the bio-polyols. The reaction yield and properties of the bio-polyol (chemical structure, hydroxyl number, acid number, viscosity) will be determined. Obtained polyols will be used for the synthesis of nonisocyanate polyurethanes (NIPUs) adhesives through polycondensation reactions. The NIPUs properties will be also determined. Outputs: D9.2

T9.4 Final products and initial validation [M08-M18] (LERMAB, UHE, ICCS): will validate the final products produced in this WP. Particleboard (one layer and 3 layers) will be produced to TRL5 scale from T9.1 and fully characterised (Internal bond strength, MOE, MOR, Swelling, density profiles). Innovative lightweight panels with no VOC emissions with mycocomposites incorporated will be produced at TRL 4-5. Particleboard (one layer and 3 layers) will be produced then to TRL6 scale and characterised (Internal bond strength, MOE, MOR, Swelling, density profiles). Innovative lightweight panels with no VOC emissions with mycocomposites incorporated will also be produced and validated.
Outputs: D9.3

Work package WP10 – Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL6)

Work Package Number	WP10	**Lead Beneficiary**	2. LERMAB
Work Package Name	Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL6)		
Start Month	19	**End Month**	36
Objectives			
● Defibration and pre-treatment of CDW by steam explosion; ● Removal of adhesives, treatment and coating products, such as glue; ● Bioremediation of the effluent using fungi; ● Production of advanced wood panel; ● Validate the final products produced.			
Description			

T10.1 Glue separation from wood [M19-M36] (LERMAB): will perform a chemical and morphological characterisation of fibres. As steam explosion combines a hydrolytic step, a physical degradation allows both the defibration of the wood waste and the removal of glue from the fibres (> 90%). During this task, a chemical-free steam explosion process will be optimised for fibre generation at lab scale (1 Kg fibres/ day). The upscaling of the process optimised will be then carried out using a pilot-scale facility. A unique batch prototype equipped with a gas scrubber will be used. Then, an effort will be made on the wood fibers/particles characterisation but also on the gaseous effluent potentially rich in formaldehyde and nitrogen gases produced from the depolymerisation of resin. Gaseous contaminant will be recovered by gas washing and treated as done for the TRL5 version. Outputs: D10.1

T10.2 Bioremediation wood treatment [M19-M36] (LERMAB): will perform a mycoremediation by solid state and submerged fermentation. Heavy metals desorption will be addressed by fungal produced oxalate chelation. Fungal tolerance and assimilation of organic preservatives in association with heavy metals will be investigated. For bioremediation of the SE effluent, submerged fermentation with formaldehyde tolerant fungi will be used. Removal of formaldehyde and characterisation of environmental toxicity of aquaeous effluent will be monitored. Water recycling in the process will be also investigated. Characterisation of contaminant removal will be performed by spectrometric techniques that can be used at the industrial scale. Finally, a screening of added value molecules (exopolysaccarides, nano polymers) resulting from the fermentation and SE combination will be implemented. The upscaling of the process optimised will be then carried out using a pilot-scale facility. The mycoremediation of C&D wood containing wood preservatives will be performed at the pilot scale. Optimisation of the fungal growth on wood wafers and particles will be performed at the lowest possible cost. Bioremediation of the aqueous effluents will be done at higher scale TRL6 using statistical representative sampling. Outputs: D10.1

T10.3 Liquefaction Process of mixed wood waste and green adhesives based on polyols [M19-M36] (UHE, LERMAB): will be executed in two subtasks: Subtask 1: Liquefaction: the reaction condition of the liquefaction process (temperature, time, ratio biomass/solvent) will be optimised using experimental design. The optimum conditions of the process in term of yield and functionalities of the bio-polyols will be determined. The liquefaction yield and properties of the bio-polyol (chemical structure, hydroxyl number, acid number, viscosity) will be determined. Subtask 2: Cationic ringopening polymerisation: is the synthesis of a family bio-polyols by cationic ROP using acid catalysts. This process allows to synthesise bio-polyols of controlled structure with a low homopolymer content at low temperatures (20-80ºC) and atmospheric pressure from lignin produced in T7-8.1. Different types of catalysis and reaction conditions will be tested to optimise the properties of the bio-polyols. The reaction yield and properties of the bio-polyol (chemical structure, hydroxyl number, acid number, viscosity) will be determined. Obtained polyols will be used for the synthesis of nonisocyanate polyurethanes (NIPUs) adhesives through polycondensation reactions. The NIPUs properties will be also determined. Outputs: D10.2

T10.4 Final products and initial validation [M19-M36] (LERMAB, UHE, ICCS): will validate the final products produced in this WP. Particleboard (one layer and 3 layers) will be produced to TRL5 scale from T10.1 and fully characterised (Internal bond strength, MOE, MOR, Swelling, density profiles). Innovative lightweight panels with no VOC emissions with mycocomposites incorporated will be produced at TRL 4-5. Particleboard (one layer and 3 layers) will be produced then to TRL6 scale and characterised (Internal bond strength, MOE, MOR, Swelling, density profiles). Innovative lightweight panels with no VOC emissions with mycocomposites incorporated will also be produced and validated. Outputs: D10.3

Work package WP11 – Energy, Gas and Ashes Valorisation (TRL5)

Work Package Number	WP11	**Lead Beneficiary**	3. CIRCE
Work Package Name	Energy, Gas and Ashes Valorisation (TRL5)		
Start Month	8	**End Month**	18
Objectives			
●Develop compilation of experimental conditions and conduct hydrothermal carbonisation and fluidised bed gasification experimentations; ● Perform microwave hydrothermal treatment of CDW-FA taking as input material incineration CDW flying ashes; ● Create optimised engineered metabolic pathways enhancing dodecanol productivity; ●Preliminary assessment of compatibility of dodecanol and exhausted cells from fermentation with surfactants formulation; ● Perform small-scale consumer exposure tests for final validations.			

Description
T11.1 Hydrothermal carbonisation [M08-M18] (CIRCE): will involve the compilation of experimental conditions reported for similar systems in high impact journal papers. Conditioning of existing microwave-assisted batch system setup for hydrothermal carbonisation experimentations will be also conducted, while chemical and physical characterisation of selected feedstocks to stablish byproduct yield forecast will be developed. Experimental test of operational conditions to enhance the chemo-selectivity towards hydrochar yield and its energy gain value will be conducted as the final stage of this task. Outputs: D11.1 T11.2 Fluidised bed gasification process of wood [M08-M18] (CIRCE): will involve larger scale operation (2kg/h) of carbonaceous material similar to hydrochar to T11.1. More specifically, preliminary prediction analysis for the production of H2 and CO2 through gasification of the selected waste streams based on their reported characteristics will be executed through this task. Adaptation of existing pilot gasification plant to feedstock and operational conditions will also be implemented. Optimisation of experimental conditions (temperature, feeding rate, carrier gas flow) to maximise hydrogen yield in syngas composition will be developed, as well as online characterisation of product gas streams by using micro-GC techniques. Outputs: D11.1 T11.3 Valorisation of ashes from CDW and thermal gasification [M08-M18] (CIRCE): will perform microwave hydrothermal treatment of CDW-FA taking as input material incineration CDW flying ashes containing high heavy inorganics concentrations and toxic volatile organics, such as PCBs and dioxines with different concentrations of alkaline. As an output, there will be produced Silicoaluminate mineral structures of stabilised and fixed heavy inorganic compounds (Pb, Zn, Cd, Cu, Ca) that must be quantified by analytic techniques such as ICP spectroscopy, - Alkali/ acid residual solutions containing contaminants degradation products that could be further characterised and ashes with reduced inorganic and organic compositions. The results will be compared with conventional methods of treatment, evaluated based on i) Contaminant degradation indicated as the variation on dioxin/PCBs concentration of byproducts, and ii) Toxicity tests to evaluate the leakage potential of zeolitic and solid products of the process and will be used for the replication studies in other sectors and the complete valorisation framework that will be proposed. Outputs: D11.2 T11.4 Upcycling of H2 and CO2 [M08-M18] (KIV, CIRCE): will employ high-throughput genetic and analytical tools to improve dodecanol production by reengineering fatty acid biosynthesis and fatty acid beta-oxidation through two phases. Phase 1: metabolic modelling and 13oC metabolic flux analysis will be applied to supporting the enhancement of the microorganism strain (C.necator) towards the desired methabolic pathway. This task will also identify conditions for high expression of heterologous genes for dodecanol, including judicious choice of thioesterases for chain-length specificity. Fermentation at 20Lt gas fermentation scale (TRL4) using syngas simulating detailed composition provided by CIRCE from T11.2 will be leveraged; the improved strains will be used during fermentation trials to to achieve targeted productivity and titer. Downstream protocol based on tricanter separation will be applied by optimising process parameters based on fermentation broth composition and maximising water recycling. The fermentation protocol will be adjusted to the engineered microorganisms tackling the optimisation and control of different parameters pH, temperature, operating pressure, mineral nutrients composition and flow rate, (syn)gas flow rate and composition, dissolved oxygen, etc. Characterisation of the product using UHPLC liquid chromatography coupled to mass spectrometry (UHPLC-ESI/ QTOF-MS) will also take place in this task. The chemical-physical analysis will be also used to evaluate the batchto-batch consistency of the production process. First samples (100-500 gr) of pure dodecanol for validation in T11.5. Preparation of demo scale (100 Lt) facility for trials: first trials using standard CO2 and H2 bottles and with nonoptimised strain for ensuring the correct functioning of the demo scale fermenter (100 Lt) addressing all safety and health requirements. Phase 2: the overall process (upstream and downstream) will be scaled up at 100 Lt using as feedstock bottle of syngas from CIRCE process and leveraging the access from KIV R&D team to upstream and downstream open facilities located in Spain (KIV external service provider). The fermentation protocol will be adjusted by KIV staff to the engineered microorganisms tackling the optimisation and control of different parameters pH, temperature, operating pressure, mineral nutrients composition and flow rate, syngas flow rate and composition, dissolved oxygen, etc. Characterisation of the product using UHPLC liquid chromatography coupled to mass spectrometry (UHPLC-ESI/ QTOF-MS). Maximisation of water and exhausted biomass recycling; digitalisation of process through sensors and automatic controls both for process optimisation and for health and safety aspects fulfilment. The different analysis will be also used to evaluate the batch-to-batch consistency of the production process achieving a reliable and stable productivity. Running of the demo continuously for at least 1 month proving process reliability and productivity; 30 kg in different batches of purified dodecanol and exhausted biomass for validation in T11.5. Outputs: D11.3 T11.5 Final products and initial validation [M08-M18] (P&G, KIV,CIRCE, ICCS): will perform the lab scale trials for a) the preliminary assessment of compatibility of dodecanol and exhausted cells from fermentation with surfactants

formulation by performing different stress tests (such as Stability, Colour, Odour, Safety, etc..), to reveal eventual technical characteristics that could not be identified in the product assessment in T11.4 to provide feedback to KIV for process fine tuning D11.5, and b) samples from TRL6 tests (kg scale) will be then progressively validated into formulation of low impact detergents leveraging a micro falling film pilot reactor (TRL6) owned by P&G in Brussels centre that allows to convert fatty alcohols and aminoacids from exhausted cell biomass in ionic-surfactants for specific application in detergency (e.g., laundry detergents, fabric enhancers and dishwashing liquid). Small scale consumer exposure tests will be carried out for final validation of technical and safety performances. Outputs: D11.4

Work package WP12 – Energy, Gas and Ashes Valorisation (TRL6)

Work Package Number	WP12	**Lead Beneficiary**	3. CIRCE
Work Package Name	Energy, Gas and Ashes Valorisation (TRL6)		
Start Month	19	**End Month**	36

Objectives

●Develop compilation of experimental conditions and conduct hydrothermal carbonisation and fluidised bed gasification experimentations; ● Perform microwave hydrothermal treatment of CDW-FA taking as input material incineration CDW flying ashes; ● Create optimised engineered metabolic pathways enhancing dodecanol productivity; ●Preliminary assessment of compatibility of dodecanol and exhausted cells from fermentation with surfactants formulation; ● Perform small-scale consumer exposure tests for final validations.

Description

T12.1 Hydrothermal carbonisation [M19-M36] (CIRCE): will involve the compilation of experimental conditions reported for similar systems in high impact journal papers. Conditioning of existing microwave-assisted batch system setup for hydrothermal carbonisation experimentations will be also conducted, while chemical and physical characterisation of selected feedstocks to stablish byproduct yield forecast will be developed. Experimental test of operational conditions to enhance the chemo-selectivity towards hydrochar yield and its energy gain value will be conducted as the final stage of this task. Outputs: D12.1

T12.2 Fluidised bed gasification process of wood [M19-M36] (CIRCE): will involve larger scale operation (2kg/h) of carbonaceous material similar to hydrochar to T12.1. More specifically, preliminary prediction analysis for the production of H2 and CO2 through gasification of the selected waste streams based on their reported characteristics will be executed through this task. Adaptation of existing pilot gasification plant to feedstock and operational conditions will also be implemented. Optimisation of experimental conditions (temperature, feeding rate, carrier gas flow) to maximise hydrogen yield in syngas composition will be developed, as well as online characterisation of product gas streams by using micro-GC techniques. Outputs: D12.1

T12.3 Valorisation of ashes from CDW and thermal gasification [M19-M36] (CIRCE): will perform microwave hydrothermal treatment of CDW-FA taking as input material incineration CDW flying ashes containing high heavy inorganics concentrations and toxic volatile organics, such as PCBs and dioxines with different concentrations of alkaline. As an output, there will be produced Silicoaluminate mineral structures of stabilised and fixed heavy inorganic compounds (Pb, Zn, Cd, Cu, Ca) that must be quantified by analytic techniques such as ICP spectroscopy, - Alkali/ acid residual solutions containing contaminants degradation products that could be further characterised and ashes with reduced inorganic and organic compositions. The results will be compared with conventional methods of treatment, evaluated based on i) Contaminant degradation indicated as the variation on dioxin/PCBs concentration of byproducts, and ii) Toxicity tests to evaluate the leakage potential of zeolitic and solid products of the process and will be used for the replication studies in other sectors and the complete valorisation framework that will be proposed. Outputs: D12.2

T12.4 Upcycling of H2 and CO2 [M19-M36] (KIV, CIRCE): will employ high-throughput genetic and analytical tools to improve dodecanol production by reengineering fatty acid biosynthesis and fatty acid beta-oxidation through two phases. Phase 1: metabolic modelling and 13oC metabolic flux analysis will be applied to supporting the enhancement of the microorganism strain (C.necator) towards the desired methabolic pathway. This task will also identify conditions for high expression of heterologous genes for dodecanol, including judicious choice of thioesterases for chain-length

specificity. Fermentation at 20Lt gas fermentation scale (TRL4) using syngas simulating detailed composition provided by CIRCE from T12.2 will be leveraged; the improved strains will be used during fermentation trials to to achieve targeted productivity and titer. Downstream protocol based on tricanter separation will be applied by optimising process parameters based on fermentation broth composition and maximising water recycling. The fermentation protocol will be adjusted to the engineered microorganisms tackling the optimisation and control of different parameters pH, temperature, operating pressure, mineral nutrients composition and flow rate, (syn)gas flow rate and composition, dissolved oxygen, etc. Characterisation of the product using UHPLC liquid chromatography coupled to mass spectrometry (UHPLC-ESI/ QTOF-MS) will also take place in this task. The chemical-physical analysis will be also used to evaluate the batchto-batch consistency of the production process. First samples (100-500 gr) of pure dodecanol for validation in T12.5. Preparation of demo scale (100 Lt) facility for trials: first trials using standard CO2 and H2 bottles and with nonoptimised strain for ensuring the correct functioning of the demo scale fermenter (100 Lt) addressing all safety and health requirements. Phase 2: the overall process (upstream and downstream) will be scaled up at 100 Lt using as feedstock bottle of syngas from CIRCE process and leveraging the access from KIV R&D team to upstream and downstream open facilities located in Spain (KIV external service provider). The fermentation protocol will be adjusted by KIV staff to the engineered microorganisms tackling the optimisation and control of different parameters pH, temperature, operating pressure, mineral nutrients composition and flow rate, syngas flow rate and composition, dissolved oxygen, etc. Characterisation of the product using UHPLC liquid chromatography coupled to mass spectrometry (UHPLC-ESI/ QTOF-MS). Maximisation of water and exhausted biomass recycling; digitalisation of process through sensors and automatic controls both for process optimisation and for health and safety aspects fulfilment. The different analysis will be also used to evaluate the batch-to-batch consistency of the production process achieving a reliable and stable productivity. Running of the demo continuously for at least 1 month proving process reliability and productivity; 30 kg in different batches of purified dodecanol and exhausted biomass for validation in T12.5. Outputs: D12.3

T12.5 Final products and initial validation [M19-M36] (P&G, KIV,CIRCE, ICCS): will perform the lab scale trials for a) the preliminary assessment of compatibility of dodecanol and exhausted cells from fermentation with surfactants formulation by performing different stress tests (such as Stability, Colour, Odour, Safety, etc..), to reveal eventual technical characteristics that could not be identified in the product assessment in T11.4 to provide feedback to KIV for process fine tuning D11.5, and b) samples from TRL6 tests (kg scale) will be then progressively validated into formulation of low impact detergents leveraging a micro falling film pilot reactor (TRL6) owned by P&G in Brussels centre that allows to convert fatty alcohols and aminoacids from exhausted cell biomass in ionic-surfactants for specific application in detergency (e.g., laundry detergents, fabric enhancers and dishwashing liquid). Small scale consumer exposure tests will be carried out for final validation of technical and safety performances. Outputs: D12.4

Work package WP13 – Digital Tools for Improving Circular Flows of Secondary Materials (first release)

Work Package Number	WP13	**Lead Beneficiary**	17. eBOS
Work Package Name	Digital Tools for Improving Circular Flows of Secondary Materials (first release)		
Start Month	8	**End Month**	18

Objectives

● Develop the end-to-end module and implement the dynamic digital product passport; ● Develop the models and algorithms for material volume estimation through multi-source data collection; ● Develop methodologies and algorithms for the supply chain optimisation of the secondary materials; ● Develop an LCA, LCCA and S-LCA performance analysis tool; ● Develop the Integrated Circular Wood Upcycling Platform.

Description

T13.1 End-to-end tracing module and digital product passport [M08-M18] (CERTH, UPV, EBOS): will produced the end-to-end tracing module and W2W Digital Product Passport (DPP) to facilitate the storage and sharing of manufacturing data related to wood waste materials. The primary objective of this task is to create interfaces, interface standards, and interface protocols that will allow for reliable and effective data exchange between various computer-aided manufacturing systems and stages. The collection, management, and sharing of multi-source, large-scale manufacturing data across various devices and tasks will be handled by a smart platform. The developed DPP will direct knowledge management and information flow for upcycling wood waste materials by utilising Explainable AI algorithms. To help with decision-making and direct the best configuration strategies for the recycling and reuse of wood waste materials, Machine Learning knowledge will be used to provide data that is interpretable, consistent, and compatible (in close collaboration with T13.4). Outputs: D13.1

T13.2 Material volume estimation [M08-M18] (ICCS, CERTH, EBOS): will conduct the estimation of the materials' volume. Multi-source data collection will also take place in this task. Images will be collected using both aerial (drone) and ground images (assistive apps for site workers) - images will come from conventional RGB cameras. The collected material will be used for i) developing 3D models for volume estimation and ii) feeding them in a deep learning algorithm based on CNNs for material detection. For the latter, tools for annotations will be developed to collect the correct label of each waste material. To this end, initial deep learning models will be developed and validated on the collected data. The final models will be created, validated and tested in real conditions, while continuous monitoring on the performance of the developed methods and further parameter adjustment to the deep learning algorithms will take place if needed.
Outputs: D13.2

T13.3 Supply chain optimisation (including primary materials and zero-defect management) [M08-M18] (UPV, CERTH, EBOS): will develop the supply chain optimisation digital tool for secondary materials for managing the process of recycling wood along the supply chain. The developed tool will allow companies to design, simulate, and optimise their supply chain networks for secondary materials by creating virtual representations of their supply chains for the secondary materials providers, recyclers, transporters, formulators and users. The tool will use advanced algorithms and data analysis techniques to simulate various scenarios to help companies identify the most efficient and cost-effective supply chain design and operation for optimising the flow of secondary materials throughout the recycling process; from collection to processing and distribution. By determining the optimal levels of secondary materials inventory to hold at different points in the supply chain, users will meet demand while minimising costs, and optimising the routing and scheduling of the secondary materials transportation, including modes of transportation and carrier selection. The final tool will be connected to the supply chain companies' information systems, to track and monitor the supply chain, providing real-time information on the location and status of secondary materials at all stages of the process. The final tool will also include a range of reporting and analytics features, allowing users to generate customised reports and track performance metrics over time. This can help identify areas for improvement and optimise recycling process further.
Outputs: D13.3

T13.4 Life cycle sustainability assessment for cascade pathways [M08-M18] (DRAXIS, ICCS, EBOS): will develop a tool to predict and assess the impact of different cascade paths or sequences for possible further uses of wood products and their materials. Comprehensive studies on Life Cycle Assessment (LCA), Life Cycle Cost Analysis (LCCA), and Social Life Cycle Assessment (S-LCA) will be conducted for the project use cases, and for different possible scenarios that make use of the project technologies. Based on the output of these scenarios, and other data (regarding product properties and technologies), an interactive tool will be developed. The LCA will be conducted based on the guidelines of ISO 14040 and 14044 using the software EASETECH or SimaPro and the Ecoinvent database. Among the environmental impacts to be analysed and quantified will be Climate change, Human toxicity, Ecotoxicity, etc. The LCA will serve both as a characterisation tool for the performance of the use cases developed in the project and as a means of identifying areas where process optimisation can make the largest environmental savings (hot spots). The LCCA will assess the economic performance of the solutions developed within project, presenting a similar structure and will be conducted in parallel to the LCA based on the same system boundaries, data inventory and scenarios. The social-LCA will be focusing on assessing the social impacts of the suggested solutions compared to the reference baseline scenario and in focusing on defining potential sources of critical social aspects for the new processes. Input and output data will be gathered from partners involved in the use cases. Specialised databases and publicly available references will be used to define potential sources of critical social aspects for the new processes. Machine Learning will be used to fill gaps in available data, and in cleaning data for life cycle inventories, estimating flow data for unit processes, improving the quality and quantity of data used to determine impact characterisation factors, and generating inventory data for scenario analyses.

Machine learning techniques will also be used to assess uncertainties in LCA data and models and help improve the reliability and robustness of LCA results, leading to more informed decision-making and better environmental outcomes. Outputs: D13.4

T13.5 Integrated Circular Wood Upcycling Platform [M08-M18] (EBOS, DRAXIS, UPV, ICCS, CERTH): will integrate all the tools developed in WP13 (and updated in WP14) into a common multifunctional and interoperable Integrated Circular Wood Upcycling platform for the complete monitoring, tracking and tracing of materials and products through dynamically updated digital product passport in the whole value chains by the end-users. The platform will include visualisation, controlling features, AI-based optimisation and decision-making functionalities and will facilitate modelling, simulation and assessments designed. It will support the tasks of end-uses (operators, formulators, recyclers, public authorities, standardisation actors, etc.); required interactions, communication and coordination of stakeholders along the supply chain; exploitation of the KERs to create communities of practice for resource exchange and matchmaking; documentation of results, statistics, best practices, barriers; support the skills activities of WP16-18; dissemination and knowledge exchange in WP19-21. The lead pilot partners will contribute to the data acquisition and integration providing SCADA and multiprotocol experience, software modules for data aggregation and remote communication. These resources can be applied on gateways or devices with Linux-based distributed developments, using docker modules or the encapsulation needed for a better portability of these components. Outputs: D13.5

Work package WP14 – Digital Tools for Improving Circular Flows of Secondary Materials (final release)

Work Package Number	WP14	**Lead Beneficiary**	17. eBOS
Work Package Name	Digital Tools for Improving Circular Flows of Secondary Materials (final release)		
Start Month	19	**End Month**	36
Objectives			
● Develop the end-to-end module and implement the dynamic digital product passport; ● Develop the models and algorithms for material volume estimation through multi-source data collection; ● Develop methodologies and algorithms for the supply chain optimisation of the secondary materials; ● Develop an LCA, LCCA and S-LCA performance analysis tool; ● Develop the Integrated Circular Wood Upcycling Platform.			
Description			

T14.1 End-to-end tracing module and digital product passport [M19-M36] (CERTH, UPV, EBOS): will produced the end-to-end tracing module and W2W Digital Product Passport (DPP) to facilitate the storage and sharing of manufacturing data related to wood waste materials. The primary objective of this task is to create interfaces, interface standards, and interface protocols that will allow for reliable and effective data exchange between various computer-aided manufacturing systems and stages. The collection, management, and sharing of multi-source, large-scale manufacturing data across various devices and tasks will be handled by a smart platform. The developed DPP will direct knowledge management and information flow for upcycling wood waste materials by utilising Explainable AI algorithms. To help with decision-making and direct the best configuration strategies for the recycling and reuse of wood waste materials, Machine Learning knowledge will be used to provide data that is interpretable, consistent, and compatible (in close collaboration with T14.4). Outputs: D14.1

T14.2 Material volume estimation [M19-M36] (ICCS, CERTH, EBOS): will conduct the estimation of the materials' volume. Multi-source data collection will also take place in this task. Images will be collected using both aerial (drone) and ground images (assistive apps for site workers) - images will come from conventional RGB cameras. The collected material will be used for i) developing 3D models for volume estimation and ii) feeding them in a deep learning algorithm based on CNNs for material detection. For the latter, tools for annotations will be developed to collect the correct label of each waste material. To this end, initial deep learning models will be developed and validated on the collected data. The final models will be created, validated and tested in real conditions, while continuous monitoring on the performance of the developed methods and further parameter adjustment to the deep learning algorithms will take place if needed. Outputs: D14.2

T14.3 Supply chain optimisation (including primary materials and zero-defect management) [M19-M36] (UPV, CERTH, EBOS): will develop the supply chain optimisation digital tool for secondary materials for managing the process of recycling wood along the supply chain. The developed tool will allow companies to design, simulate, and optimise their supply chain networks for secondary materials by creating virtual representations of their supply chains for the secondary materials providers, recyclers, transporters, formulators and users. The tool will use advanced algorithms and data analysis techniques to simulate various scenarios to help companies identify the most efficient and cost-effective supply chain design and operation for optimising the flow of secondary materials throughout the recycling process; from collection to processing and distribution. By determining the optimal levels of secondary materials inventory to hold at different points in the supply chain, users will meet demand while minimising costs, and optimising the routing and scheduling of the secondary materials transportation, including modes of transportation and carrier selection. The final tool will be connected to the supply chain companies' information systems, to track and monitor the supply chain, providing real-time information on the location and status of secondary materials at all stages of the process. The final tool will also include a range of reporting and analytics features, allowing users to generate customised reports and track performance metrics over time. This can help identify areas for improvement and optimise recycling process further. Outputs: D14.3

T14.4 Life cycle sustainability assessment for cascade pathways [M19-M36] (DRAXIS, ICCS, EBOS): will develop a tool to predict and assess the impact of different cascade paths or sequences for possible further uses of wood products and their materials. Comprehensive studies on Life Cycle Assessment (LCA), Life Cycle Cost Analysis (LCCA), and Social Life Cycle Assessment (S-LCA) will be conducted for the project use cases, and for different possible scenarios that make use of the project technologies. Based on the output of these scenarios, and other data (regarding product properties and technologies), an interactive tool will be developed. The LCA will be conducted based on the guidelines of ISO 14040 and 14044 using the software EASETECH or SimaPro and the Ecoinvent database. Among the environmental impacts to be analysed and quantified will be Climate change, Human toxicity, Ecotoxicity, etc. The LCA will serve both as a characterisation tool for the performance of the use cases developed in the project and as a means of identifying areas where process optimisation can make the largest environmental savings (hot spots). The LCCA will assess the economic performance of the solutions developed within project, presenting a similar structure and will be conducted in parallel to the LCA based on the same system boundaries, data inventory and scenarios. The social-LCA will be focusing on assessing the social impacts of the suggested solutions compared to the reference baseline scenario and in focusing on defining potential sources of critical social aspects for the new processes. Input and output data will be gathered from partners involved in the use cases. Specialised databases and publicly available references will be used to define potential sources of critical social aspects for the new processes. Machine Learning will be used to fill gaps in available data, and in cleaning data for life cycle inventories, estimating flow data for unit processes, improving the quality and quantity of data used to determine impact characterisation factors, and generating inventory data for scenario analyses. Machine learning techniques will also be used to assess uncertainties in LCA data and models and help improve the reliability and robustness of LCA results, leading to more informed decision-making and better environmental outcomes. Outputs: D14.4

T14.5 Integrated Circular Wood Upcycling Platform [M19-M36] (EBOS, DRAXIS, UPV, ICCS, CERTH): will integrate all the tools updated in WP14 into a common multifunctional and interoperable Integrated Circular Wood Upcycling platform for the complete monitoring, tracking and tracing of materials and products through dynamically updated digital product passport in the whole value chains by the end-users. The platform will include visualisation, controlling features, AI-based optimisation and decision-making functionalities and will facilitate modelling, simulation and assessments designed. It will support the tasks of end-uses (operators, formulators, recyclers, public authorities, standardisation actors, etc.); required interactions, communication and coordination of stakeholders along the supply chain; exploitation of the KERs to create communities of practice for resource exchange and matchmaking; documentation of results, statistics, best practices, barriers; support the skills activities of WP16-18; dissemination and knowledge exchange in WP19-21. The lead pilot partners will contribute to the data acquisition and integration providing SCADA and multiprotocol experience, software modules for data aggregation and remote communication. These resources can be applied on gateways or devices with Linux-based distributed developments, using docker modules or the encapsulation needed for a better portability of these components. Outputs: D14.5

Work package WP15 – Material Flows and Pilot Activities

Work Package Number	WP15	**Lead Beneficiary**	4. NTUA
Work Package Name	Material Flows and Pilot Activities		
Start Month	36	**End Month**	48

Objectives

● Install, operate and validate the CDW sorting system; ● Record/register the material exchange network; ● Introduce the basic steps and approaches to optimise the logistics; ● Detect critical points in the logistics supply chain; ● Support the development of more sustainable supply chains; ● Validate the sustainability of the overall cascade valorisation scheme.

Description

T15.1 CDW pilot installation and validation [M36-M48] (ICCS, ELAK): will implement the installation, operation and validation activities of the CDW sorting system. In line with the analysis performed in T5-6.1 and the ongoing contruction schedule of ELAK, the most appropriate location for the installation of the system shall be identified. The system is developed in a modular format, and we directly shipped assembled in the final location by M37. The operation of the system will be performed few hours per day, allowing the processing of the required volume of materials as per the needs of each value chain. Characterisation of the recycled streams will also be performed to identify links between the type of construction sites and the separated wood classifications. Outputs: D15.1

T15.2 Material exchanges and logistics optimisation between the pilot sites [M36-M48] (NTUA, CIRCE, UHE): will have as its main objective the optimisation of material flows and logistics, considering the transportation of raw materials, intermediate and final products, storage, and management, along with their availability (supply) and demand. For this purpose, a thorough logging of material and products flows across the pilot sites will be conducted, thus identifying the critical points and significant sustainability impacts of the supply chain and logistics. Subsequently, the gaps in the material exchanges and logistics supply chain will be analysed and preventive measures, by adapting internal structures and processes, will be implemented, to improve the overall environmental and economic impact of all the involved processes. Outputs: D15.2

T15.3 Final validation and sustainability assessment of the overall cascade valorisation scheme [M36-M48] (NTUA, R2M, QPLAN, RCM): will focus on verifying the sustainability of the proposed CDW valorisation schemes across the overall cascade scheme. Results obtained from T15.2 will be utilised to validate that the optimisation of materials exchanges and logistics is viable, promoting circular economy and upcycling of waste. The overall sustainability will be assessed using necessary control mechanisms, including a step-by-step evaluation system in the logistics supply chain, combined with a weighed sustainability assessment performance indicator. The aforementioned indicator will focus on the environmental, social and economic attributes of the overall cascade valorisation scheme, incorporating the six key dimensions of the supply chain: customs performance, infrastructure quality, ease of arranging shipments, logistics services quality, consignments tracking and tracing, and tracing and timeliness of shipments. Outputs: D15.2

Work package WP16 – Skills Development, and Standardisation, Policy & Certification Influencing for RP1

Work Package Number	WP16	**Lead Beneficiary**	6. VTT
Work Package Name	Skills Development, and Standardisation, Policy & Certification Influencing for RP1		
Start Month	1	**End Month**	18
Objectives			
● Develop and provide skills foresight to European policy-makers, workers, educators, and firms with actionable and granular information on demand for circular economy occupations in the process industries by 2035; ● Create synergies with existing learning initiatives to serve as a continuous point of reference for the establishment of collaborations; ● Co-design training activities and lifelong learning programmes; ● Launch capacity building activities among the wider public; ● Contribute to standardisation and certification of secondary materials; ● Identify and study the best practices, and provide policy recommendations towards a truly circular and sustainable process industry.			
Description			
T16.1 Identification of trends and scenario development [M01-M18] (VTT, UPV, POLIMI) will firstly focused on identifying broad trends with the potential to impact the European labour market in circular economy in the process industries, by 2035. The trends will be identified using the horizon scanning technique, which will help identify weak signals of change and emerging opportunities for prospective policy development, as well as to assess future policy risks,			

threats, and vulnerabilities. The STEEPV (societal, technological, economic, environmental, political, or values-based) conceptual framework will be used to identify and categorise signals of change and assess their degree of maturity. Data will consist of machine-learning processed secondary data emerging from job adverts, academic journals, media and patents, and from a qualitative survey of 300 multidisciplinary experts (labour market, industrial technology, design, education, and socio-technical transition experts) to identify possible signals of change. The signals of change will be synthesised into key megatrends, for which a maturity index will be developed, capturing an in-depth description of the projected change and a detailed explanation of the labour market implications associated with the trends identified. Subsequently, the task moves into the design and execution of regional experiential foresight workshops (two workshops in each of the project partner countries) with a diverse mix of experts (government, consulting firms, manufacturing firms, unions, service delivery organisations, and post-secondary education institutions) to develop scenarios on how selected process industries occupations might change, as well as to explore emerging occupations based on the circular economy in process industry trends identified and on participants' own knowledge of trends impacting employment. Outputs: D16.1

T16.2 Navigating the future of skill development [M01-M18] (POLIMI, UPV, VTT, ICCS, KIV): will be executed in four subtasks: Subtask 1: Investigating emerging roles and skills in the applied context of the pilots: examine and introduce the necessary skills, professions, and roles needed to successfully implement a Circular Economy, including Industrial and Urban Symbiosis, as well as virtuous circular value chains. Digital Skills and competencies, such as Data Analytics and AI, using the SPIRE-06 cluster of projects, such as CAPRI, and relevant resources such as the annual reports of WMF and WEF will be explored and identified. Subtask 2: Cooperation with existing learning initiatives for skills activities: by using the output of T16.2, skill gaps will be identified and analysed to find potential solutions. Workshops and training programs will then be implemented to address these gaps, complete and validate the findings of the initial iteration. Interactive workshops approach will be taken to ensure that the introduced jobs and skills are suitable for the organisational conditions of the experiments. Subtask 3: Jobs, competences and training action plan: focusing on people dimension, a thorough analysis of current jobs and professions involved in this project context will be conducted to identify possible skills gaps derived by Circular Economy, data and digital adoption. The main objective is to design and develop a structured methodology (Survey, and interview based) able to assess the current level of circularity, and digital maturity of experiments (AS-IS). The final iteration of surveys and interviews will be implemented to analyse the TO-BE situation of the experiments, as well as to measure the level of improvement that has been achieved throughout the project's lifetime. Subtask 4: Training activities and lifelong learning programmes on upcycling, replication and cooperation: an investigation into various training sources will be conducted to identify relevant courses that can help bridge the identified skill gaps. These training courses will be analysed at different levels (such as awareness, foundation, extended know-how) to ensure that they are appropriate for experiments with varying backgrounds and levels of knowledge. Along with identifying relevant training courses various potential pathways will be explored to address skill gaps, including reskilling and upskilling programs, partnerships, and others. The final subtask will involve analysing the responses received from the final iteration of surveys and interviews. Valuable insights into the effectiveness of the project and areas where further improvements may be necessary can be gained by reviewing this data. Additionally, a thorough analysis of the lessons learned throughout the project's lifetime will be conducted. This will allow for the identification of best practices, areas for improvement, and potential opportunities for future development. Based on these insights, recommendations will be provided for further action to ensure that the project's objectives are fully realized and that all stakeholders have been adequately supported. The analysis aims to maximize the impact of the project and ensure that it delivers the desired outcomes. Outputs: D16.2

T16.3 Standardisation activities [M01-M18] (UNE): will firstly analyse the applicable standardisation landscape and applicable certification programmes and report the relevant Technical Committee(s) together with applicable standards, and where relevant, existing certification schemes. Based on the applicable landscape, a strategy will be developed to allow W2W to effectively contribute to standardisation of the secondary materials. Further, the possibility and the extent to which a contribution to standardisation can be made will be investigated through interacting with the relevant committees, disseminating the project and getting feedback from their members. The existing landscape, the interaction with the technical committees as well as the work progress around the midpoint of the project will allow for the identification of candidate topics for a contribution. This will be done considering the deliverables from other WPs as well as gaps or barriers in existing standards or topics not yet addressed by standards and existing certifications. At the final stage, W2W's contribution to the development of new standards in topics related to the project's objectives and results will be carried out. This may involve providing information, participating in ongoing works, submitting proposals for the modification of existing standards or promoting the elaboration of new standards like technical specifications (TS). The development of fast-track standards like CEN-CENELEC Workshop Agreements (CWA) is usually the bestsuited option in the frame of a Horizon Europe project. Outputs: D16.3

T16.4 Policy influencing towards a truly circular and sustainable process industry [M01-M18] (CEPS, ICCS): will

perform a consolidation on the regulation, legislation, and certification barriers that affect the implementation of the proposed framework. As achieving efficient and effective use of wood resources requires considering multidimensional valorisation of wood, including time, value, and function as to be defined in T4.3 and further validated in T15.3, this task will be continuously updated through the implementation of the project, resulting in the creation of a policy report summarising the actions needed for a EU Scale implementation of the W2W proposed cascade valorisation scheme. Outputs: D16.4

Work package WP17 – Skills Development, and Standardisation, Policy & Certification Influencing for RP2

Work Package Number	WP17	**Lead Beneficiary**	6. VTT
Work Package Name	Skills Development, and Standardisation, Policy & Certification Influencing for RP2		
Start Month	19	**End Month**	36
Objectives			
● Develop and provide skills foresight to European policy-makers, workers, educators, and firms with actionable and granular information on demand for circular economy occupations in the process industries by 2035; ● Create synergies with existing learning initiatives to serve as a continuous point of reference for the establishment of collaborations; ● Co-design training activities and lifelong learning programmes; ● Launch capacity building activities among the wider public; ● Contribute to standardisation and certification of secondary materials; ● Identify and study the best practices, and provide policy recommendations towards a truly circular and sustainable process industry.			
Description			

T17.1 Identification of trends and scenario development [M19-36] (VTT, UPV, POLIMI) will firstly focused on identifying broad trends with the potential to impact the European labour market in circular economy in the process industries, by 2035. The trends will be identified using the horizon scanning technique, which will help identify weak signals of change and emerging opportunities for prospective policy development, as well as to assess future policy risks, threats, and vulnerabilities. The STEEPV (societal, technological, economic, environmental, political, or values-based) conceptual framework will be used to identify and categorise signals of change and assess their degree of maturity. Data will consist of machine-learning processed secondary data emerging from job adverts, academic journals, media and patents, and from a qualitative survey of 300 multidisciplinary experts (labour market, industrial technology, design, education, and socio-technical transition experts) to identify possible signals of change. The signals of change will be synthesised into key megatrends, for which a maturity index will be developed, capturing an in-depth description of the projected change and a detailed explanation of the labour market implications associated with the trends identified. Subsequently, the task moves into the design and execution of regional experiential foresight workshops (two workshops in each of the project partner countries) with a diverse mix of experts (government, consulting firms, manufacturing firms, unions, service delivery organisations, and post-secondary education institutions) to develop scenarios on how selected process industries occupations might change, as well as to explore emerging occupations based on the circular economy in process industry trends identified and on participants' own knowledge of trends impacting employment.
Outputs: D17.1

T17.2 Navigating the future of skill development [M19-36] (POLIMI, UPV, VTT, ICCS, KIV): will be executed in four subtasks: Subtask 1: Investigating emerging roles and skills in the applied context of the pilots: examine and introduce the necessary skills, professions, and roles needed to successfully implement a Circular Economy, including Industrial and Urban Symbiosis, as well as virtuous circular value chains. Digital Skills and competencies, such as Data Analytics and AI, using the SPIRE-06 cluster of projects, such as CAPRI, and relevant resources such as the annual reports of WMF and WEF will be explored and identified. Subtask 2: Cooperation with existing learning initiatives for skills activities: by using the output of T17.2, skill gaps will be identified and analysed to find potential solutions. Workshops and training programs will then be implemented to address these gaps, complete and validate the findings of the initial iteration. Interactive workshops approach will be taken to ensure that the introduced jobs and skills are suitable for the organisational conditions of the experiments. Subtask 3: Jobs, competences and training action plan: focusing on people dimension, a thorough analysis of current jobs and professions involved in this project context will be conducted to identify possible skills gaps derived by Circular Economy, data and digital adoption. The main objective is to design and develop a structured methodology (Survey, and interview based) able to assess the current level of circularity, and digital maturity of experiments (AS-IS). The final iteration of surveys and interviews will be implemented to analyse the TO-BE situation of the experiments, as well as to measure the level of improvement that has been achieved throughout the project's lifetime. Subtask 4: Training activities and lifelong learning programmes on upcycling, replication and cooperation: an investigation into various training sources will be conducted to identify relevant courses that can help bridge the identified skill gaps. These training courses will be analysed at different levels (such as awareness, foundation, extended know-how) to ensure that they are appropriate for experiments with varying backgrounds and levels of knowledge. Along with identifying relevant training courses various potential pathways will be explored to address skill gaps, including reskilling and upskilling programs, partnerships, and others. The final subtask will involve analysing the responses received from the final iteration of surveys and interviews. Valuable insights into the effectiveness of the project and areas where further improvements may be necessary can be gained by reviewing this data. Additionally, a thorough analysis of the lessons learned throughout the project's lifetime will be conducted. This will allow for the identification of best practices, areas for improvement, and potential opportunities for future development. Based on these insights, recommendations will be provided for further action to ensure that the project's objectives are fully realized and that all stakeholders have been adequately supported. The analysis aims to maximize the impact of the project and ensure that it delivers the desired outcomes. Outputs: D17.2

T17.3 Standardisation activities [M19-36] (UNE): will firstly analyse the applicable standardisation landscape and applicable certification programmes and report the relevant Technical Committee(s) together with applicable standards, and where relevant, existing certification schemes. Based on the applicable landscape, a strategy will be developed to allow W2W to effectively contribute to standardisation of the secondary materials. Further, the possibility and the extent to which a contribution to standardisation can be made will be investigated through interacting with the relevant committees, disseminating the project and getting feedback from their members. The existing landscape, the interaction with the technical committees as well as the work progress around the midpoint of the project will allow for the identification of candidate topics for a contribution. This will be done considering the deliverables from other WPs as well as gaps or barriers in existing standards or topics not yet addressed by standards and existing certifications. At the final stage, W2W's contribution to the development of new standards in topics related to the project's objectives and results will be carried out. This may involve providing information, participating in ongoing works, submitting proposals

for the modification of existing standards or promoting the elaboration of new standards like technical specifications (TS). The development of fast-track standards like CEN-CENELEC Workshop Agreements (CWA) is usually the bestsuited option in the frame of a Horizon Europe project. Outputs: D17.3

T17.4 Policy influencing towards a truly circular and sustainable process industry [M19-36] (CEPS, ICCS): will perform a consolidation on the regulation, legislation, and certification barriers that affect the implementation of the proposed framework. As achieving efficient and effective use of wood resources requires considering multidimensional valorisation of wood, including time, value, and function as to be defined in T4.3 and further validated in T15.3, this task will be continuously updated through the implementation of the project, resulting in the creation of a policy report summarising the actions needed for a EU Scale implementation of the W2W proposed cascade valorisation scheme. Outputs: D17.4

Work package WP18 – Skills Development, and Standardisation, Policy & Certification Influencing for RP3

Work Package Number	WP18	**Lead Beneficiary**	6. VTT
Work Package Name	Skills Development, and Standardisation, Policy & Certification Influencing for RP3		
Start Month	37	**End Month**	48
Objectives			
● Develop and provide skills foresight to European policy-makers, workers, educators, and firms with actionable and granular information on demand for circular economy occupations in the process industries by 2035; ● Create synergies with existing learning initiatives to serve as a continuous point of reference for the establishment of collaborations; ● Co-design training activities and lifelong learning programmes; ● Launch capacity building activities among the wider public; ● Contribute to standardisation and certification of secondary materials; ● Identify and study the best practices, and provide policy recommendations towards a truly circular and sustainable process industry.			
Description			

T18.1 Identification of trends and scenario development [M37-48] (VTT, UPV, POLIMI) will firstly focused on identifying broad trends with the potential to impact the European labour market in circular economy in the process industries, by 2035. The trends will be identified using the horizon scanning technique, which will help identify weak signals of change and emerging opportunities for prospective policy development, as well as to assess future policy risks, threats, and vulnerabilities. The STEEPV (societal, technological, economic, environmental, political, or values-based) conceptual framework will be used to identify and categorise signals of change and assess their degree of maturity. Data will consist of machine-learning processed secondary data emerging from job adverts, academic journals, media and patents, and from a qualitative survey of 300 multidisciplinary experts (labour market, industrial technology, design, education, and socio-technical transition experts) to identify possible signals of change. The signals of change will be synthesised into key megatrends, for which a maturity index will be developed, capturing an in-depth description of the projected change and a detailed explanation of the labour market implications associated with the trends identified. Subsequently, the task moves into the design and execution of regional experiential foresight workshops (two workshops in each of the project partner countries) with a diverse mix of experts (government, consulting firms, manufacturing firms, unions, service delivery organisations, and post-secondary education institutions) to develop scenarios on how selected process industries occupations might change, as well as to explore emerging occupations based on the circular economy in process industry trends identified and on participants' own knowledge of trends impacting employment.
Outputs: D18.1

T18.2 Navigating the future of skill development [M37-48] (POLIMI, UPV, VTT, ICCS, KIV): will be executed in four subtasks: Subtask 1: Investigating emerging roles and skills in the applied context of the pilots: examine and introduce the necessary skills, professions, and roles needed to successfully implement a Circular Economy, including Industrial and Urban Symbiosis, as well as virtuous circular value chains. Digital Skills and competencies, such as Data Analytics and AI, using the SPIRE-06 cluster of projects, such as CAPRI, and relevant resources such as the annual reports of WMF and WEF will be explored and identified. Subtask 2: Cooperation with existing learning initiatives for skills activities: by using the output of T18.2, skill gaps will be identified and analysed to find potential solutions. Workshops and training programs will then be implemented to address these gaps, complete and validate the findings of the initial iteration. Interactive workshops approach will be taken to ensure that the introduced jobs and skills are suitable for the organisational conditions of the experiments. Subtask 3: Jobs, competences and training action plan: focusing on people dimension, a thorough analysis of current jobs and professions involved in this project context will be conducted to identify possible skills gaps derived by Circular Economy, data and digital adoption. The main objective is to design and develop a structured methodology (Survey, and interview based) able to assess the current level of circularity, and digital maturity of experiments (AS-IS). The final iteration of surveys and interviews will be implemented to analyse the TO-BE situation of the experiments, as well as to measure the level of improvement that has been achieved throughout the project's lifetime. Subtask 4: Training activities and lifelong learning programmes on upcycling, replication and cooperation: an investigation into various training sources will be conducted to identify relevant courses that can help bridge the identified skill gaps. These training courses will be analysed at different levels (such as awareness, foundation, extended know-how) to ensure that they are appropriate for experiments with varying backgrounds and levels of knowledge. Along with identifying relevant training courses various potential pathways will be explored to address skill gaps, including reskilling and upskilling programs, partnerships, and others. The final subtask will involve analysing the responses received from the final iteration of surveys and interviews. Valuable insights into the effectiveness of the project and areas where further improvements may be necessary can be gained by reviewing this data. Additionally, a thorough analysis of the lessons learned throughout the project's lifetime will be conducted. This will allow for the identification of best practices, areas for improvement, and potential opportunities for future development. Based on these insights, recommendations will be provided for further action to ensure that the project's objectives are fully realized and that all stakeholders have been adequately supported. The analysis aims to maximize the impact of the project and ensure that it delivers the desired outcomes. Outputs: D18.2

T18.3 Standardisation activities [M37-48] (UNE): will firstly analyse the applicable standardisation landscape and applicable certification programmes and report the relevant Technical Committee(s) together with applicable standards, and where relevant, existing certification schemes. Based on the applicable landscape, a strategy will be developed to allow W2W to effectively contribute to standardisation of the secondary materials. Further, the possibility and the extent to which a contribution to standardisation can be made will be investigated through interacting with the relevant committees, disseminating the project and getting feedback from their members. The existing landscape, the interaction with the technical committees as well as the work progress around the midpoint of the project will allow for the identification of candidate topics for a contribution. This will be done considering the deliverables from other WPs as well as gaps or barriers in existing standards or topics not yet addressed by standards and existing certifications. At the final stage, W2W's contribution to the development of new standards in topics related to the project's objectives and results will be carried out. This may involve providing information, participating in ongoing works, submitting proposals for the modification of existing standards or promoting the elaboration of new standards like technical specifications

(TS). The development of fast-track standards like CEN-CENELEC Workshop Agreements (CWA) is usually the bestsuited option in the frame of a Horizon Europe project. Outputs: D18.3

T18.4 Policy influencing towards a truly circular and sustainable process industry [M37-48] (CEPS, ICCS): will perform a consolidation on the regulation, legislation, and certification barriers that affect the implementation of the proposed framework. As achieving efficient and effective use of wood resources requires considering multidimensional valorisation of wood, including time, value, and function as to be defined in T4.3 and further validated in T15.3, this task will be continuously updated through the implementation of the project, resulting in the creation of a policy report summarising the actions needed for a EU Scale implementation of the W2W proposed cascade valorisation scheme. Outputs:
D18.4

Work package WP19 – Impact Maximisation and Increasing Competitiveness for RP1

Work Package Number	WP19	**Lead Beneficiary**	19. R2M
Work Package Name	Impact Maximisation and Increasing Competitiveness for RP1		
Start Month	1	**End Month**	18
Objectives			
● Shape a tailored dissemination and communication plan, ensuring continuous stakeholder engagement and collaboration throughout the project; ● Create synergies with relevant associations, projects and initiatives, and provide the methodological framework to serve as a continuous point of reference for the establishment of collaborations; ● Develop replication guidelines and methodology with technical and practical information to replicate a successful case internationally; ● Map the market landscape, identifying opportunities and obstacles; ● Develop an efficient exploitation and IPR management plan towards sustainability; ● Provide customised revenue streams to improve the economy scale in the upcycling of the new materials.			
Description			

T19.1 Dissemination and communication activities for promoting resources consciously [M01-M18] (ISWA, All Partners): will build the foundation for and prepares W2W's dissemination, communication and outreach of the project achievements and research results to the priority target audience groups mapped under T4.1. At the very beginning of the project, the visual identity (logo, a set of graphic elements and images, templates for presentations and publications) based on the material of the proposal documents, and the project public website containing a portfolio of materials and contents, will be established – the project website will be constantly updated with the project's latest activities and outcomes, including the communication objectives, its main messages, the underlying communication principles, the communication channels and activities and the respective consortium partners' roles and responsibilities. This task will also coordinate the efforts of W2W partners in the dissemination of the project's achievements by: (a) submitting high quality papers at high impact, peer-reviewed journals and magazines, (b) presenting and publishing results in prestigious conferences, (c) organising workshops, trainings, conference session, and tutorials, and (d) link the dissemination and communication activities with related initiatives and ongoing projects. Apart from the aforementioned dissemination activities, the consortium will additionally be engaged in dedicated stakeholders clustering and decision makers engagement and awareness raising, target mainly core initiatives and decision-making communities that hold more potential in commercially exploiting the results and applying them in daily practice and in regional or even national extent, as well as best practices for social acceptance. Capitalising on already formulated ecosystems, and enhancing them with additional stakeholder groups (such as policy formulators, decisions makers, policy makers, organisations and initiatives focused on entrepreneurship, business support services, such as (but not limited to) incubators or accelerators programmes), the consortium will raise awareness with regards to its objectives and results, and through them try to promote the project results and bring W2W closer to market, ensuring dialogue with potential funders and customers. Outputs: D19.1, D19.3

T19.2 Cross-sectoral cooperation and link existing initiatives [M01-M18] (ICCS, ISWA, CIRCE): will set up synergies between different projects ,funding sources and technology roadmaps along the innovation chain proposed by W2W. Collaboration will be established with European-wide initiatives, as well as networking activities with related SPIRE projects, in previous and future calls at HE or other relevant programmes to link the project results and conclusions with available or under development roadmaps. All project partners will be encouraged to contribute to collaboration events to integrate regional and national programs and initiatives and to allow other organizations, not directly participating in W2W, to know about its objectives, evolution and conclusions. Additional technologies under development in similar project, important for the design of the W2W framework will be identified and linked with the project. At least 2 events will be organized, targeting the core W2W approach aiming to bring together the stakeholders around the proposed value chains and allow them to further interact with the results of the project. Outputs: D19.3

T19.3 International collaboration and exchange of best practises [M01-M18] (UJOH, ICCS, NTUA): will create and provide access to an established network of universities and private organisations working on developing highly skilled built environment professionals in the field of sustainable infrastructure development, in order to exchange knowledge and best practises. Constant interaction with relevant international projects and initiatives, making use of the ongoing connections that exist among the consortium members with relevant initiatives will ensure that the project outputs as well as input data are shared and communicated regularly during the duration of the project. Liaison with decision-makers, at international level will be sought (in constant collaboration with T16-18.4), in form of meetings, invitations to attend and speak at dissemination events, as well as involvement in developing a set of practical recommendations that will give support to international policy initiatives. At the beginning of this task, UJOH will leverage its own rich networks to involve industry stakeholders, which has extensive experience with technical solutions as well as financial modelling and digitalisation, in the waste management and built environment sectors. Outputs: D19.3

T19.4 Exploitation, business models and IPR management [M01-M18] (R2M, All partners): will undertake the project's exploitation activities in a collaborative fashion, resulting into the definition of the system's Business Case and allow partners to work towards the realisation of the joint business model that will be handpicked by the consortium partners and that will define an exploitation approach where selected partners will play the role of exploitation champions, which will be used for the commercial exploitation of the overall solution. This task will also perform the IPR activities within the project including monitoring technical and scientific progress, identifying exploitable assets and their added value, facilitating constructive discussion and fostering innovative concepts within the time and budget restraints of the project, maintaining research integrity and providing consultation/review/editing in the preparation of articles, maintaining legal compliance of all development and exploitation activities, as well as aligning project activities with the revenue streams which are defined in T19.5. Outputs: D19.2, D19.4

T19.5 Market analysis, sustainable business models and go-to-market strategy [M01-M18] (QPLAN, All partners): will deliver an evidence-based go-to-market strategy for W2W's circular techs, as follows: [M01-M18]: build upon the

evidence produced within the project and employ market research to better define and understand the market targeted by the circular technologies of W2W, as well as new business opportunities at the time. Based on the results of our analysis we will a co-design sustainable business model (or models if meaningful) with new revenue streams in close cooperation with our technology providers and industry partners. The process will utilise the Triple Layered Business Model Canvas, complemented by lean value proposition design. The key elements of the model(s) will be included in D19.5. Outputs: D19.5

Work package WP20 – Impact Maximisation and Increasing Competitiveness for RP2

Work Package Number	WP20	**Lead Beneficiary**	19. R2M
Work Package Name	Impact Maximisation and Increasing Competitiveness for RP2		
Start Month	19	**End Month**	36

Objectives

● Shape a tailored dissemination and communication plan, ensuring continuous stakeholder engagement and collaboration throughout the project; ● Create synergies with relevant associations, projects and initiatives, and provide the methodological framework to serve as a continuous point of reference for the establishment of collaborations; ● Develop replication guidelines and methodology with technical and practical information to replicate a successful case internationally; ● Map the market landscape, identifying opportunities and obstacles; ● Develop an efficient exploitation and IPR management plan towards sustainability; ● Provide customised revenue streams to improve the economy scale in the upcycling of the new materials.

Description

T20.1 Dissemination and communication activities for promoting resources consciously [M19-36] (ISWA, All Partners): will build the foundation for and prepares W2W's dissemination, communication and outreach of the project achievements and research results to the priority target audience groups mapped under T4.1. At the very beginning of the project, the visual identity (logo, a set of graphic elements and images, templates for presentations and publications) based on the material of the proposal documents, and the project public website containing a portfolio of materials and contents, will be established – the project website will be constantly updated with the project's latest activities and outcomes, including the communication objectives, its main messages, the underlying communication principles, the communication channels and activities and the respective consortium partners' roles and responsibilities. This task will also coordinate the efforts of W2W partners in the dissemination of the project's achievements by: (a) submitting high quality papers at high impact, peer-reviewed journals and magazines, (b) presenting and publishing results in prestigious conferences, (c) organising workshops, trainings, conference session, and tutorials, and (d) link the dissemination and communication activities with related initiatives and ongoing projects. Apart from the aforementioned dissemination activities, the consortium will additionally be engaged in dedicated stakeholders clustering and decision makers engagement and awareness raising, target mainly core initiatives and decision-making communities that hold more potential in commercially exploiting the results and applying them in daily practice and in regional or even national extent, as well as best practices for social acceptance. Capitalising on already formulated ecosystems, and enhancing them with additional stakeholder groups (such as policy formulators, decisions makers, policy makers, organisations and initiatives focused on entrepreneurship, business support services, such as (but not limited to) incubators or accelerators programmes), the consortium will raise awareness with regards to its objectives and results, and through them try to promote the project results and bring W2W closer to market, ensuring dialogue with potential funders and customers.
Outputs: D20.1

T20.2 Cross-sectoral cooperation and link existing initiatives [M19-36] (ICCS, ISWA, CIRCE): will set up synergies between different projects ,funding sources and technology roadmaps along the innovation chain proposed by W2W. Collaboration will be established with European-wide initiatives, as well as networking activities with related SPIRE projects, in previous and future calls at HE or other relevant programmes to link the project results and conclusions with available or under development roadmaps. All project partners will be encouraged to contribute to collaboration events to integrate regional and national programs and initiatives and to allow other organizations, not directly participating in W2W, to know about its objectives, evolution and conclusions. Additional technologies under development in similar project, important for the design of the W2W framework will be identified and linked with the project. At least 2 events will be organized, targeting the core W2W approach aiming to bring together the stakeholders around the proposed value chains and allow them to further interact with the results of the project. Outputs: D20.2

T20.3 International collaboration and exchange of best practises M19-36] (UJOH, ICCS, NTUA): will create and provide access to an established network of universities and private organisations working on developing highly skilled built environment professionals in the field of sustainable infrastructure development, in order to exchange knowledge and best practises. Constant interaction with relevant international projects and initiatives, making use of the ongoing connections that exist among the consortium members with relevant initiatives will ensure that the project outputs as well as input data are shared and communicated regularly during the duration of the project. Liaison with decision-makers, at international level will be sought (in constant collaboration with T16-18.4), in form of meetings, invitations to attend and speak at dissemination events, as well as involvement in developing a set of practical recommendations that will give support to international policy initiatives. At the beginning of this task, UJOH will leverage its own rich networks to involve industry stakeholders, which has extensive experience with technical solutions as well as financial modelling and digitalisation, in the waste management and built environment sectors. Outputs: D20.2

T20.4 Exploitation, business models and IPR management [M19-36] (R2M, All partners): will undertake the project's exploitation activities in a collaborative fashion, resulting into the definition of the system's Business Case and allow partners to work towards the realisation of the joint business model that will be handpicked by the consortium partners and that will define an exploitation approach where selected partners will play the role of exploitation champions, which will be used for the commercial exploitation of the overall solution. This task will also perform the IPR activities within the project including monitoring technical and scientific progress, identifying exploitable assets and their added value, facilitating constructive discussion and fostering innovative concepts within the time and budget restraints of the project, maintaining research integrity and providing consultation/review/editing in the preparation of articles, maintaining legal compliance of all development and exploitation activities, as well as aligning project activities with the revenue streams which are defined in T20.5. Outputs: D20.3

T20.5 Market analysis, sustainable business models and go-to-market strategy [M19-36] (QPLAN, All partners): will deliver an evidence-based go-to-market strategy for W2W's circular techs, as follows: [M01-M18]: build upon the evidence produced within the project and employ market research to better define and understand the market targeted

by the circular technologies of W2W, as well as new business opportunities at the time. Based on the results of our analysis

we will a co-design sustainable business model (or models if meaningful) with new revenue streams in close cooperation with our technology providers and industry partners. The process will utilise the Triple Layered Business Model Canvas, complemented by lean value proposition design. [M19-M36]: the key assumptions of the business model(s) (costs, revenue streams, value propositions, etc.) will be explored via interviews and surveys of prospective customers and stakeholders participating in our pilot activities (wood industries, investors, value chain actors such as formulators and recyclers, public authorities, and standardisation actors etc.) within and outside of our consortium. This will enable to validate the assumptions and define the building blocks of a sustainable business model to serve as the core for our goto-market strategy. Insights collected will be reported in D20.4 with the updated business model(s). Outputs:D20.4

Work package WP21 – Impact Maximisation and Increasing Competitiveness for RP3

Work Package Number	WP21	**Lead Beneficiary**	19. R2M
Work Package Name	Impact Maximisation and Increasing Competitiveness for RP3		
Start Month	37	**End Month**	48
Objectives			
● Shape a tailored dissemination and communication plan, ensuring continuous stakeholder engagement and collaboration throughout the project; ● Create synergies with relevant associations, projects and initiatives, and provide the methodological framework to serve as a continuous point of reference for the establishment of collaborations; ● Develop replication guidelines and methodology with technical and practical information to replicate a successful case internationally; ● Map the market landscape, identifying opportunities and obstacles; ● Develop an efficient exploitation and IPR management plan towards sustainability; ● Provide customised revenue streams to improve the economy scale in the upcycling of the new materials.			
Description			

T21.1 Dissemination and communication activities for promoting resources consciously [M37-48] (ISWA, All Partners): will build the foundation for and prepares W2W's dissemination, communication and outreach of the project achievements and research results to the priority target audience groups mapped under T4.1. At the very beginning of the project, the visual identity (logo, a set of graphic elements and images, templates for presentations and publications) based on the material of the proposal documents, and the project public website containing a portfolio of materials and contents, will be established – the project website will be constantly updated with the project's latest activities and outcomes, including the communication objectives, its main messages, the underlying communication principles, the communication channels and activities and the respective consortium partners' roles and responsibilities. This task will also coordinate the efforts of W2W partners in the dissemination of the project's achievements by: (a) submitting high quality papers at high impact, peer-reviewed journals and magazines, (b) presenting and publishing results in prestigious conferences, (c) organising workshops, trainings, conference session, and tutorials, and (d) link the dissemination and communication activities with related initiatives and ongoing projects. Apart from the aforementioned dissemination activities, the consortium will additionally be engaged in dedicated stakeholders clustering and decision makers engagement and awareness raising, target mainly core initiatives and decision-making communities that hold more potential in commercially exploiting the results and applying them in daily practice and in regional or even national extent, as well as best practices for social acceptance. Capitalising on already formulated ecosystems, and enhancing them with additional stakeholder groups (such as policy formulators, decisions makers, policy makers, organisations and initiatives focused on entrepreneurship, business support services, such as (but not limited to) incubators or accelerators programmes), the consortium will raise awareness with regards to its objectives and results, and through them try to promote the project results and bring W2W closer to market, ensuring dialogue with potential funders and customers.
Outputs: D21.1

T21.2 Cross-sectoral cooperation and link existing initiatives [M37-48] (ICCS, ISWA, CIRCE): will set up synergies between different projects ,funding sources and technology roadmaps along the innovation chain proposed by W2W. Collaboration will be established with European-wide initiatives, as well as networking activities with related SPIRE projects, in previous and future calls at HE or other relevant programmes to link the project results and conclusions with available or under development roadmaps. All project partners will be encouraged to contribute to collaboration events to integrate regional and national programs and initiatives and to allow other organizations, not directly participating in W2W, to know about its objectives, evolution and conclusions. Additional technologies under development in similar

project, important for the design of the W2W framework will be identified and linked with the project. At least 2 events will be organized, targeting the core W2W approach aiming to bring together the stakeholders around the proposed value chains and allow them to further interact with the results of the project. Outputs:D21.2

T21.3 International collaboration and exchange of best practises [M37-48] (UJOH, ICCS, NTUA): will create and provide access to an established network of universities and private organisations working on developing highly skilled built environment professionals in the field of sustainable infrastructure development, in order to exchange knowledge and best practises. Constant interaction with relevant international projects and initiatives, making use of the ongoing connections that exist among the consortium members with relevant initiatives will ensure that the project outputs as well as input data are shared and communicated regularly during the duration of the project. Liaison with decision-makers, at international level will be sought (in constant collaboration with T16-18.4), in form of meetings, invitations to attend and speak at dissemination events, as well as involvement in developing a set of practical recommendations that will give support to international policy initiatives. At the beginning of this task, UJOH will leverage its own rich networks to involve industry stakeholders, which has extensive experience with technical solutions as well as financial modelling and digitalisation, in the waste management and built environment sectors. Outputs:D21.2

T21.4 Exploitation, business models and IPR management [M37-48] (R2M, All partners): will undertake the project's exploitation activities in a collaborative fashion, resulting into the definition of the system's Business Case and allow partners to work towards the realisation of the joint business model that will be handpicked by the consortium partners and that will define an exploitation approach where selected partners will play the role of exploitation champions, which will be used for the commercial exploitation of the overall solution. This task will also perform the IPR activities within the project including monitoring technical and scientific progress, identifying exploitable assets and their added value, facilitating constructive discussion and fostering innovative concepts within the time and budget restraints of the project, maintaining research integrity and providing consultation/review/editing in the preparation of articles, maintaining legal compliance of all development and exploitation activities, as well as aligning project activities with the revenue streams which are defined in T21.5. Outputs: D21.3

T21.5 Market analysis, sustainable business models and go-to-market strategy [M37-48] (QPLAN, All partners): will deliver an evidence-based go-to-market strategy for W2W's circular techs, as follows: [M01-M18]: build upon the evidence produced within the project and employ market research to better define and understand the market targeted by the circular technologies of W2W, as well as new business opportunities at the time. Based on the results of our analysis we will a co-design sustainable business model (or models if meaningful) with new revenue streams in close cooperation with our technology providers and industry partners. The process will utilise the Triple Layered Business Model Canvas, complemented by lean value proposition design. [M37-M48]: will elaborate a go-to-market strategy for W2W as a key exploitation mechanism, offering an evidence-based pathway for exploiting its circular techs. The strategy will be delivered by M48 (D21.4), taking into account: (i) the final results from the pilot activities; (ii) data gathered through secondary market research and their analysis; (iii) the regulatory and policy framework of the targeted market(s), as well as social, business and policy trends; (iv) organisational changes required for the integration of our solutions into the value chain of their process industry and the development of economies of scale; and (v) feedback from end-users and key stakeholders within the targeted market(s). Outputs: D21.4

Associated with document Ref. Ares(2023)8305270 - 05/12/2023

STAFF EFFORT

Associated with document Ref. Ares(2023)8305270 - 05/12/2023

LIST OF DELIVERABLES

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	W2W project management handbook	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	4
D1.2	Data management plan (version 1)	WP1	4 - NTUA	DMP — Data Management Plan	SEN - Sensitive	6
D2.1	Data management plan (version 2)	WP2	4 - NTUA	DMP — Data Management Plan	SEN - Sensitive	36
D3.1	Data management plan (version 3)	WP3	4 - NTUA	DMP — Data Management Plan	SEN - Sensitive	48
D4.1	Mapping of the actors and key stakeholders	WP4	16 - DRAXIS	R — Document, report	PU - Public	10
D4.2	Wood characterisation & classification schemes	WP4	21 - CEPS	R — Document, report	PU - Public	10
D4.3	Recycled wood cascade valorisation framework	WP4	4 - NTUA	R — Document, report	PU - Public	10
D4.4	Data sharing and exchange model	WP4	10 - TUB	R — Document, report	SEN - Sensitive	10

D4.5	Impact assessment methodologies and W2W KPIs	WP4	16 - DRAXIS	R — Document, report	PU - Public	10
D5.1	CDW flows valorisation and digitalisation (version 1)	WP5	12 - ELAK	R — Document, report	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.2	Wood and glass non-destructive real time techniques analysis & optimal size definition (version 1)	WP5	20 - IRIS	R — Document, report	PU - Public	18
D5.3	Adaptive glass and wood sorting systems (version 1)	WP5	1 - ICCS	OTHER	PU - Public	18
D5.4	Better separation and quality control systems (version 1)	WP5	1 - ICCS	OTHER	PU - Public	18
D6.1	CDW flows valorisation and digitalisation (version 2)	WP6	12 - ELAK	R — Document, report	PU - Public	36
D6.2	Wood and glass non-destructive real time techniques analysis & optimal size definition (version 2)	WP6	20 - IRIS	R — Document, report	PU - Public	36

D6.3	Adaptive glass and wood sorting systems (version 2)	WP6	1 - ICCS	OTHER	PU - Public	36
D6.4	Better separation and quality control systems (version 2)	WP6	1 - ICCS	OTHER	PU - Public	36
D7.1	Cascade to function, mixed wood treatment (version 1)	WP7	9 - UHE	OTHER	PU - Public	18
D7.2	New mix between wood, glass and composites for wallboards (version 1)	WP7	4 - NTUA	OTHER	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D7.3	Final products and initial validation (version 1)	WP7	15 - FOCCHI	DEM — Demonstrator, pilot, prototype	PU - Public	18
D8.1	Cascade to function, mixed wood treatment (version 2)	WP8	9 - UHE	OTHER	PU - Public	36
D8.2	New mix between wood, glass and composites for wallboards (version 2)	WP8	4 - NTUA	OTHER	PU - Public	36

D8.3	Final products and initial validation (version 2)	WP8	15 - FOCCHI	DEM — Demonstrator, pilot, prototype	PU - Public	36
D9.1	Glue separation from wood and bioremediation wood treatment (version 1)	WP9	2 - LERMAB	OTHER	PU - Public	18
D9.2	Liquefaction process of mixed wood waste (version 1)	WP9	9 - UHE	OTHER	PU - Public	18
D9.3	Final products and initial validation (version 3)	WP9	1 - ICCS	DEM — Demonstrator, pilot, prototype	PU - Public	18
D10.1	Glue separation from wood and bioremediation wood treatment (version 2)	WP10	2 - LERMAB	OTHER	PU - Public	36
D10.2	Liquefaction process of mixed wood waste (version 2)	WP10	9 - UHE	OTHER	PU - Public	36
D10.3	Final products and initial validation (version 4)	WP10	1 - ICCS	DEM — Demonstrator, pilot, prototype	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D11.1	Hydrothermal carbonisation and gasification process of wood (version 1)	WP11	3 - CIRCE	OTHER	PU - Public	18

D11.2	Valorisation of ashes from CDW and thermal gasification (version 1)	WP11	3 - CIRCE	OTHER	PU - Public	18
D11.3	Optimised engineered metabolic pathways enhancing dodecanol (version 1)	WP11	13 - KIVERDI	OTHER	SEN - Sensitive	18
D11.4	Assessment of dodecanol and validation of detergent formulations (version 1)	WP11	14 - P&G	DEM — Demonstrator, pilot, prototype	PU - Public	18
D12.1	Hydrothermal carbonisation and gasification process of wood (version 2)	WP12	3 - CIRCE	OTHER	PU - Public	36
D12.2	Valorisation of ashes from CDW and thermal gasification (version 2)	WP12	3 - CIRCE	OTHER	PU - Public	36
D12.3	Optimised engineered metabolic pathways enhancing dodecanol (version 2)	WP12	13 - KIVERDI	OTHER	SEN - Sensitive	36
D12.4	Assessment of dodecanol and validation of detergent formulations (version 2)	WP12	14 - P&G	DEM — Demonstrator, pilot, prototype	PU - Public	36
D13.1	End-to-end tracing module and DPP (version 1)	WP13	8 - CERTH	OTHER	PU - Public	18
D13.2	Material volume estimation tool (version 1)	WP13	1 - ICCS	OTHER	PU - Public	18
D13.3	Supply chain optimisation tool (version 1)	WP13	7 - UPV	OTHER	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D13.4	Life cycle analysis computation tool (version 1)	WP13	16 - DRAXIS	OTHER	PU - Public	18
D13.5	Integrated Circular Wood Upcycling Platform (version 1)	WP13	17 - eBOS	OTHER	PU - Public	18
D14.1	End-to-end tracing module and DPP (version 2)	WP14	8 - CERTH	OTHER	PU - Public	36
D14.2	Material volume estimation tool (version 2)	WP14	1 - ICCS	OTHER	PU - Public	36
D14.3	Supply chain optimisation tool (version 2)	WP14	7 - UPV	OTHER	PU - Public	36
D14.4	Life cycle analysis computation tool (version 2)	WP14	16 - DRAXIS	OTHER	PU - Public	36
D14.5	Integrated Circular Wood Upcycling Platform (version 2)	WP14	17 - eBOS	OTHER	PU - Public	36
D15.1	CDW pilot installation and validation	WP15	1 - ICCS	DEM — Demonstrator, pilot, prototype	PU - Public	48

D15.2	Assessment of the cascade valorisation scheme	WP15	4 - NTUA	DEM — Demonstrator, pilot, prototype	PU - Public	48
D16.1	Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular	WP16	6 - VTT	OTHER	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
	economy in the process industries by 2035 (version 1)					
D16.2	Skills activities, trainings and lifelong learning programmes (version 1)	WP16	5 - POLIMI	OTHER	PU - Public	18
D16.3	Standardisation activities (version 1)	WP16	24 - UNE	R — Document, report	PU - Public	18
D16.4	Policy recommendations for developing a circular and sustainable process industry (version 1)	WP16	21 - CEPS	R — Document, report	PU - Public	18
D17.1	Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 2)	WP17	6 - VTT	OTHER	PU - Public	36

D17.2	Skills activities, trainings and lifelong learning programmes (version 2)	WP17	5 - POLIMI	OTHER	PU - Public	36
D17.3	Standardisation activities (version 2)	WP17	24 - UNE	R — Document, report	PU - Public	36
D17.4	Policy recommendations for developing a circular and sustainable process industry (version 2)	WP17	21 - CEPS	R — Document, report	PU - Public	36
D18.1	Maturity Index and Scenarios on twin transition trends and occupational scenarios	WP18	6 - VTT	OTHER	PU - Public	48

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
	for European labour market in circular economy in the process industries by 2035 (version 3)					
D18.2	Skills activities, trainings and lifelong learning programmes (version 3)	WP18	5 - POLIMI	OTHER	PU - Public	48
D18.3	Standardisation activities (version 3)	WP18	24 - UNE	R — Document, report	PU - Public	48
D18.4	Policy recommendations for developing a circular and sustainable process industry (version 3)	WP18	21 - CEPS	R — Document, report	PU - Public	48

D19.1	Communication and Dissemination plan and activities (version 1)	WP19	22 - ISWA	R — Document, report	PU - Public	6
D19.2	Exploitation and Business Innovation Plan and activities (version 1)	WP19	19 - R2M	R — Document, report	PU - Public	6
D19.3	Synergies with relevant initiatives and best practises exchange (version 1)	WP19	1 - ICCS	R — Document, report	PU - Public	18
D19.4	Exploitation and Business Innovation Plan and activities (version 2)	WP19	19 - R2M	R — Document, report	PU - Public	18
D19.5	Market analysis, business models and revenue streams (version 1)	WP19	18 - QPLAN	R — Document, report	PU - Public	18
D20.1	Communication and Dissemination plan and activities (version 2)	WP20	22 - ISWA	R — Document, report	PU - Public	36

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D20.2	Synergies with relevant initiatives and best practises exchange (version 2)	WP20	1 - ICCS	R — Document, report	PU - Public	36
D20.3	Exploitation and Business Innovation Plan and activities (version 3)	WP20	19 - R2M	R — Document, report	PU - Public	36

D20.4	Market analysis, business models and revenue streams (version 2)	WP20	18 - QPLAN	R — Document, report	PU - Public	36
D21.1	Communication and Dissemination plan and activities (version 3)	WP21	22 - ISWA	R — Document, report	PU - Public	48
D21.2	Synergies with relevant initiatives and best practises exchange (version 3)	WP21	1 - ICCS	R — Document, report	PU - Public	48
D21.3	Exploitation and Business Innovation Plan and activities (version 4)	WP21	19 - R2M	R — Document, report	PU - Public	48
D21.4	Market analysis, business models and revenue streams (version 3)	WP21	18 - QPLAN	R — Document, report	PU - Public	48

Deliverable D1.1 – W2W project management handbook

Deliverable Number	D1.1	**Lead Beneficiary**	1. ICCS
Deliverable Name	W2W project management handbook		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	4	**Work Package No**	WP1
Description			
An internal document setting the basis for the project management, the communication channels and methods, and gender and ethics issues to serve as a reference point during the duration of the project.			

Deliverable D1.2 – Data management plan (version 1)

Deliverable Number	D1.2	**Lead Beneficiary**	4. NTUA
Deliverable Name	Data management plan (version 1)		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
Document including datasets related to secondary materials identification and characterisation and their descriptions; standards and metadata; data sharing, maintenance and storage procedures.			

Deliverable D2.1 – Data management plan (version 2)

Deliverable Number	D2.1	**Lead Beneficiary**	4. NTUA
Deliverable Name	Data management plan (version 2)		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP2
Description			
Document including datasets related to secondary materials identification and characterisation and their descriptions; standards and metadata; data sharing, maintenance and storage procedures.			

Deliverable D3.1 – Data management plan (version 3)

Deliverable Number	D3.1	**Lead Beneficiary**	4. NTUA
Deliverable Name	Data management plan (version 3)		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP3
Description			

Document including datasets related to secondary materials identification and characterisation and their descriptions; standards and metadata; data sharing, maintenance and storage procedures

Deliverable D4.1 – Mapping of the actors and key stakeholders

Deliverable Number	D4.1	**Lead Beneficiary**	16. DRAXIS
Deliverable Name	Mapping of the actors and key stakeholders		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	10	**Work Package No**	WP4
Description			
Stakeholder mapping and analysis, and methodological framework of the co-creation processes.			

Deliverable D4.2 – Wood characterisation & classification schemes

Deliverable Number	D4.2	**Lead Beneficiary**	21. CEPS
Deliverable Name	Wood characterisation & classification schemes		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	10	**Work Package No**	WP4
Description			
A study on parameters linked to the wood characterisation and a proposal for a common classification scheme			

Deliverable D4.3 – Recycled wood cascade valorisation framework

Deliverable Number	D4.3	**Lead Beneficiary**	4. NTUA
Deliverable Name	Recycled wood cascade valorisation framework		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	10	**Work Package No**	WP4
Description			
Analysis on available routes and valorisation schemes from the provided feedstock			

Deliverable D4.4 – Data sharing and exchange model

Deliverable Number	D4.4	**Lead Beneficiary**	10. TUB
Deliverable Name	Data sharing and exchange model		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	10	**Work Package No**	WP4
Description			
Data sharing and exchange techniques as a set of required data and related requirements, in accordance with the FAIR data management for project's partners, external stakeholders and DPP.			

Deliverable D4.5 – Impact assessment methodologies and W2W KPIs

Deliverable Number	D4.5	**Lead Beneficiary**	16. DRAXIS
Deliverable Name	Impact assessment methodologies and W2W KPIs		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	10	**Work Package No**	WP4
Description			
Report of the most suitable methodologies and KPIs required to quantitatively evaluate the results of W2W.			

Deliverable D5.1 – CDW flows valorisation and digitalisation (version 1)

Deliverable Number	D5.1	**Lead Beneficiary**	12. ELAK
Deliverable Name	CDW flows valorisation and digitalisation (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
A methodology for the CDW separation and sorting systems for the three district levels.			

Deliverable D5.2 – Wood and glass non-destructive real time techniques analysis & optimal size definition (version 1)

Deliverable Number	D5.2	**Lead Beneficiary**	20. IRIS
Deliverable Name	Wood and glass non-destructive real time techniques analysis & optimal size definition (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Optical and spectroscopic characterisation techniques to determinate which is the most efficient particle size			

Deliverable D5.3 – Adaptive glass and wood sorting systems (version 1)

Deliverable Number	D5.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Adaptive glass and wood sorting systems (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
An adaptive system for sorting glass and wood waste from CDW			

Deliverable D5.4 – Better separation and quality control systems (version 1)

Deliverable Number	D5.4	**Lead Beneficiary**	1. ICCS

Deliverable Name	Better separation and quality control systems (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Better seperation and quality control system including cobot station, application for an MR-HMD and interface			

Deliverable D6.1 – CDW flows valorisation and digitalisation (version 2)

Deliverable Number	D6.1	**Lead Beneficiary**	12. ELAK
Deliverable Name	CDW flows valorisation and digitalisation (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
A methodology for the CDW separation and sorting systems for the three district levels			

Deliverable D6.2 – Wood and glass non-destructive real time techniques analysis & optimal size definition (version 2)

Deliverable Number	D6.2	**Lead Beneficiary**	20. IRIS
Deliverable Name	Wood and glass non-destructive real time techniques analysis & optimal size definition (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Optical and spectroscopic characterisation techniques to determinate which is the most efficient particle size			

Deliverable D6.3 – Adaptive glass and wood sorting systems (version 2)

Deliverable Number	D6.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Adaptive glass and wood sorting systems (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
An adaptive system for sorting glass and wood waste from CDW			

Deliverable D6.4 – Better separation and quality control systems (version 2)

Deliverable Number	D6.4	**Lead Beneficiary**	1. ICCS
Deliverable Name	Better separation and quality control systems (version 2)		

Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Better seperation and quality control system including cobot station, application for an MR-HMD and interface			

Deliverable D7.1 – Cascade to function, mixed wood treatment (version 1)

Deliverable Number	D7.1	**Lead Beneficiary**	9. UHE
Deliverable Name	Cascade to function, mixed wood treatment (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
Isolation of cellulose nanofillers, functionalisation of cellulose, and lignin isolation and modification			

Deliverable D7.2 – New mix between wood, glass and composites for wallboards (version 1)

Deliverable Number	D7.2	**Lead Beneficiary**	4. NTUA
Deliverable Name	New mix between wood, glass and composites for wallboards (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
Composite compounds to serve as low-cost sustainable recycled material aiming at construction applications			

Deliverable D7.3 – Final products and initial validation (version 1)

Deliverable Number	D7.3	**Lead Beneficiary**	15. FOCCHI
Deliverable Name	Final products and initial validation (version 1)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7
Description			
Validation process, products' validation result and performance report			

Deliverable D8.1 – Cascade to function, mixed wood treatment (version 2)

Deliverable Number	D8.1	**Lead Beneficiary**	9. UHE
Deliverable Name	Cascade to function, mixed wood treatment (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8

Description
Isolation of cellulose nanofillers, functionalisation of cellulose, and lignin isolation and modification

Deliverable D8.2 – New mix between wood, glass and composites for wallboards (version 2)

Deliverable Number	D8.2	**Lead Beneficiary**	4. NTUA
Deliverable Name	New mix between wood, glass and composites for wallboards (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Composite compounds to serve as low-cost sustainable recycled material aiming at construction applications			

Deliverable D8.3 – Final products and initial validation (version 2)

Deliverable Number	D8.3	**Lead Beneficiary**	15. FOCCHI
Deliverable Name	Final products and initial validation (version 2)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP8
Description			
Validation process, products' validation result and performance report			

Deliverable D9.1 – Glue separation from wood and bioremediation wood treatment (version 1)

Deliverable Number	D9.1	**Lead Beneficiary**	2. LERMAB
Deliverable Name	Glue separation from wood and bioremediation wood treatment (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP9
Description			
Chemical & morphological characterisation of fibres, and mycoremediation and submerged fermentation			

Deliverable D9.2 – Liquefaction process of mixed wood waste (version 1)

Deliverable Number	D9.2	**Lead Beneficiary**	9. UHE
Deliverable Name	Liquefaction process of mixed wood waste (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP9
Description			

Optimised liquefaction and cationic ring-opening polymerisation processes

Deliverable D9.3 – Final products and initial validation (version 3)

Deliverable Number	D9.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Final products and initial validation (version 3)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP9
Description			
Validation process, products' validation result and performance report			

Deliverable D10.1 – Glue separation from wood and bioremediation wood treatment (version 2)

Deliverable Number	D10.1	**Lead Beneficiary**	2. LERMAB
Deliverable Name	Glue separation from wood and bioremediation wood treatment (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Chemical & morphological characterisation of fibres, and mycoremediation and submerged fermentation			

Deliverable D10.2 – Liquefaction process of mixed wood waste (version 2)

Deliverable Number	D10.2	**Lead Beneficiary**	9. UHE
Deliverable Name	Liquefaction process of mixed wood waste (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Optimised liquefaction and cationic ring-opening polymerisation processes			

Deliverable D10.3 – Final products and initial validation (version 4)

Deliverable Number	D10.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Final products and initial validation (version 4)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP10
Description			
Validation process, products' validation result and performance report			

Deliverable D11.1 – Hydrothermal carbonisation and gasification process of wood (version 1)

Deliverable Number	D11.1	**Lead Beneficiary**	3. CIRCE
Deliverable Name	Hydrothermal carbonisation and gasification process of wood (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP11
Description			
Experimental test of operational conditions and online characterisation of product gas streams			

Deliverable D11.2 – Valorisation of ashes from CDW and thermal gasification (version 1)

Deliverable Number	D11.2	**Lead Beneficiary**	3. CIRCE
Deliverable Name	Valorisation of ashes from CDW and thermal gasification (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP11
Description			
Silicoaluminate mineral structures of stabilised and fixed heavy inorganic compounds			

Deliverable D11.3 – Optimised engineered metabolic pathways enhancing dodecanol (version 1)

Deliverable Number	D11.3	**Lead Beneficiary**	13. KIVERDI
Deliverable Name	Optimised engineered metabolic pathways enhancing dodecanol (version 1)		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP11
Description			
Optimised engineered metabolic pathways and demo facility properly functioning with commercial CO2 and H2			

Deliverable D11.4 – Assessment of dodecanol and validation of detergent formulations (version 1)

Deliverable Number	D11.4	**Lead Beneficiary**	14. P&G
Deliverable Name	Assessment of dodecanol and validation of detergent formulations (version 1)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP11
Description			
Compatibility assessment for detergents formulation and 2 detergent formulations validated under exposure tests			

Deliverable D12.1 – Hydrothermal carbonisation and gasification process of wood (version 2)

Deliverable Number	D12.1	**Lead Beneficiary**	3. CIRCE
Deliverable Name	Hydrothermal carbonisation and gasification process of wood (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP12
Description			
Experimental test of operational conditions and online characterisation of product gas streams			

Deliverable D12.2 – Valorisation of ashes from CDW and thermal gasification (version 2)

Deliverable Number	D12.2	**Lead Beneficiary**	3. CIRCE
Deliverable Name	Valorisation of ashes from CDW and thermal gasification (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP12
Description			
Silicoaluminate mineral structures of stabilised and fixed heavy inorganic compounds			

Deliverable D12.3 – Optimised engineered metabolic pathways enhancing dodecanol (version 2)

Deliverable Number	D12.3	**Lead Beneficiary**	13. KIVERDI
Deliverable Name	Optimised engineered metabolic pathways enhancing dodecanol (version 2)		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP12

Description
Optimised engineered metabolic pathways and demo facility properly functioning with commercial CO2 and H2

Deliverable D12.4 – Assessment of dodecanol and validation of detergent formulations (version 2)

Deliverable Number	D12.4	**Lead Beneficiary**	14. P&G
Deliverable Name	Assessment of dodecanol and validation of detergent formulations (version 2)		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP12
Description			
Compatibility assessment for detergents formulation and 2 detergent formulations validated under exposure tests			

Deliverable D13.1 – End-to-end tracing module and DPP (version 1)

Deliverable Number	D13.1	**Lead Beneficiary**	8. CERTH
Deliverable Name	End-to-end tracing module and DPP (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP13
Description			
First and final releases of the end-to-end tracing module and Digital Product Passport.			

Deliverable D13.2 – Material volume estimation tool (version 1)

Deliverable Number	D13.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	Material volume estimation tool (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP13
Description			
First and final releases of the material volume estimation tool			

Deliverable D13.3 – Supply chain optimisation tool (version 1)

Deliverable Number	D13.3	**Lead Beneficiary**	7. UPV
Deliverable Name	Supply chain optimisation tool (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP13
Description			
First and final releases of the supply chain optimisation tool.			

Deliverable D13.4 – Life cycle analysis computation tool (version 1)

Deliverable Number	D13.4	**Lead Beneficiary**	16. DRAXIS
Deliverable Name	Life cycle analysis computation tool (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP13
Description			
First and final releases of the LCA computation tool			

Deliverable D13.5 – Integrated Circular Wood Upcycling Platform (version 1)

Deliverable Number	D13.5	**Lead Beneficiary**	17. eBOS

Deliverable Name	Integrated Circular Wood Upcycling Platform (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP13
Description			
First and final releases of the Integrated Circular Wood Upcycling platform.			

Deliverable D14.1 – End-to-end tracing module and DPP (version 2)

Deliverable Number	D14.1	**Lead Beneficiary**	8. CERTH
Deliverable Name	End-to-end tracing module and DPP (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP14
Description			
First and final releases of the end-to-end tracing module and Digital Product Passport			

Deliverable D14.2 – Material volume estimation tool (version 2)

Deliverable Number	D14.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	Material volume estimation tool (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP14
Description			
First and final releases of the material volume estimation tool			

Deliverable D14.3 – Supply chain optimisation tool (version 2)

Deliverable Number	D14.3	**Lead Beneficiary**	7. UPV
Deliverable Name	Supply chain optimisation tool (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP14
Description			
First and final releases of the supply chain optimisation tool			

Deliverable D14.4 – Life cycle analysis computation tool (version 2)

Deliverable Number	D14.4	**Lead Beneficiary**	16. DRAXIS
Deliverable Name	Life cycle analysis computation tool (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public

Due Date (month)	36	**Work Package No**	WP14
Description			
First and final releases of the LCA computation tool			

Deliverable D14.5 – Integrated Circular Wood Upcycling Platform (version 2)

Deliverable Number	D14.5	**Lead Beneficiary**	17. eBOS
Deliverable Name	Integrated Circular Wood Upcycling Platform (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP14
Description			
First and final releases of the Integrated Circular Wood Upcycling platform			

Deliverable D15.1 – CDW pilot installation and validation

Deliverable Number	D15.1	**Lead Beneficiary**	1. ICCS
Deliverable Name	CDW pilot installation and validation		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP15
Description			
Report on the outcomes from the installation, operation and validation activities of the CDW sorting system.			

Deliverable D15.2 – Assessment of the cascade valorisation scheme

Deliverable Number	D15.2	**Lead Beneficiary**	4. NTUA
Deliverable Name	Assessment of the cascade valorisation scheme		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP15
Description			
Optimum material exchange & logistics network, and sustainability assessment of cascade valorisation scheme			

Deliverable D16.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 1)

Deliverable Number	D16.1	**Lead Beneficiary**	6. VTT
Deliverable Name	Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 1)		

Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP16
Description			
Maturity index on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035			

Deliverable D16.2 – Skills activities, trainings and lifelong learning programmes (version 1)

Deliverable Number	D16.2	**Lead Beneficiary**	5. POLIMI
Deliverable Name	Skills activities, trainings and lifelong learning programmes (version 1)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP16
Description			
Guidelines and training templates and material, and plan for cooperation with relevant skills building initiatives			

Deliverable D16.3 – Standardisation activities (version 1)

Deliverable Number	D16.3	**Lead Beneficiary**	24. UNE
Deliverable Name	Standardisation activities (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP16
Description			
Possible contributions in secondary materials standardisation & certification and evaluation activities.			

Deliverable D16.4 – Policy recommendations for developing a circular and sustainable process industry (version 1)

Deliverable Number	D16.4	**Lead Beneficiary**	21. CEPS
Deliverable Name	Policy recommendations for developing a circular and sustainable process industry (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP16
Description			
Policy briefs targeting EU and international policy-makers and agencies			

Deliverable D17.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 2)

Deliverable Number	D17.1	**Lead Beneficiary**	6. VTT

Deliverable Name	Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP17
Description			
Maturity index on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035.			

Deliverable D17.2 – Skills activities, trainings and lifelong learning programmes (version 2)

Deliverable Number	D17.2	**Lead Beneficiary**	5. POLIMI
Deliverable Name	Skills activities, trainings and lifelong learning programmes (version 2)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP17
Description			
Guidelines and training templates and material, and plan for cooperation with relevant skills building initiatives			

Deliverable D17.3 – Standardisation activities (version 2)

Deliverable Number	D17.3	**Lead Beneficiary**	24. UNE
Deliverable Name	Standardisation activities (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP17
Description			
Possible contributions in secondary materials standardisation & certification and evaluation activities.			

Deliverable D17.4 – Policy recommendations for developing a circular and sustainable process industry (version 2)

Deliverable Number	D17.4	**Lead Beneficiary**	21. CEPS
Deliverable Name	Policy recommendations for developing a circular and sustainable process industry (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP17
Description			
Policy briefs targeting EU and international policy-makers and agencies			

Deliverable D18.1 – Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 3)

Deliverable Number	D18.1	**Lead Beneficiary**	6. VTT
Deliverable Name	Maturity Index and Scenarios on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035 (version 3)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP18
Description			
Maturity index on twin transition trends and occupational scenarios for European labour market in circular economy in the process industries by 2035.			

Deliverable D18.2 – Skills activities, trainings and lifelong learning programmes (version 3)

Deliverable Number	D18.2	**Lead Beneficiary**	5. POLIMI
Deliverable Name	Skills activities, trainings and lifelong learning programmes (version 3)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP18
Description			
Guidelines and training templates and material, and plan for cooperation with relevant skills building initiatives.			

Deliverable D18.3 – Standardisation activities (version 3)

Deliverable Number	D18.3	**Lead Beneficiary**	24. UNE
Deliverable Name	Standardisation activities (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP18
Description			
Possible contributions in secondary materials standardisation & certification and evaluation activities			

Deliverable D18.4 – Policy recommendations for developing a circular and sustainable process industry (version 3)

Deliverable Number	D18.4	**Lead Beneficiary**	21. CEPS
Deliverable Name	Policy recommendations for developing a circular and sustainable process industry (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP18
Description			
Policy briefs targeting EU and international policy-makers and agencies.			

Deliverable D19.1 – Communication and Dissemination plan and activities (version 1)

Deliverable Number	D19.1	**Lead Beneficiary**	22. ISWA
Deliverable Name	Communication and Dissemination plan and activities (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP19
Description			
Initial, updated and final versions of the D&C plan and undertaken activities.			

Deliverable D19.2 – Exploitation and Business Innovation Plan and activities (version 1)

Deliverable Number	D19.2	**Lead Beneficiary**	19. R2M
Deliverable Name	Exploitation and Business Innovation Plan and activities (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP19
Description			
Initial, updated and final versions of the exploitation and business innovation plan and activities.			

Deliverable D19.3 – Synergies with relevant initiatives and best practises exchange (version 1)

Deliverable Number	D19.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Synergies with relevant initiatives and best practises exchange (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP19
Description			
Documents all the produced knowledge by summarising the main results, lessons learnt and best practices.			

Deliverable D19.4 – Exploitation and Business Innovation Plan and activities (version 2)

Deliverable Number	D19.4	**Lead Beneficiary**	19. R2M
Deliverable Name	Exploitation and Business Innovation Plan and activities (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP19
Description			
Initial, updated and final versions of the exploitation and business innovation plan and activities.			

Deliverable D19.5 – Market analysis, business models and revenue streams (version 1)

Deliverable Number	D19.5	**Lead Beneficiary**	18. QPLAN
Deliverable Name	Market analysis, business models and revenue streams (version 1)		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	18	**Work Package No**	WP19
Description			
Market analysis, business models, possible revenue streams, and commercialisation plan and activities.			

Deliverable D20.1 – Communication and Dissemination plan and activities (version 2)

Deliverable Number	D20.1	**Lead Beneficiary**	22. ISWA
Deliverable Name	Communication and Dissemination plan and activities (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP20
Description			
Initial, updated and final versions of the D&C plan and undertaken activities			

Deliverable D20.2 – Synergies with relevant initiatives and best practises exchange (version 2)

Deliverable Number	D20.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	Synergies with relevant initiatives and best practises exchange (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP20
Description			
Documents all the produced knowledge by summarising the main results, lessons learnt and best practices.			

Deliverable D20.3 – Exploitation and Business Innovation Plan and activities (version 3)

Deliverable Number	D20.3	**Lead Beneficiary**	19. R2M
Deliverable Name	Exploitation and Business Innovation Plan and activities (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP20
Description			
Initial, updated and final versions of the exploitation and business innovation plan and activities			

Deliverable D20.4 – Market analysis, business models and revenue streams (version 2)

Deliverable Number	D20.4	**Lead Beneficiary**	18. QPLAN
Deliverable Name	Market analysis, business models and revenue streams (version 2)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP20
Description			

Market analysis, business models, possible revenue streams, and commercialisation plan and activities.

Deliverable D21.1 – Communication and Dissemination plan and activities (version 3)

Deliverable Number	D21.1	**Lead Beneficiary**	22. ISWA
Deliverable Name	Communication and Dissemination plan and activities (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP21
Description			
Initial, updated and final versions of the D&C plan and undertaken activities			

Deliverable D21.2 – Synergies with relevant initiatives and best practises exchange (version 3)

Deliverable Number	D21.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	Synergies with relevant initiatives and best practises exchange (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP21
Description			
Documents all the produced knowledge by summarising the main results, lessons learnt and best practices.			

Deliverable D21.3 – Exploitation and Business Innovation Plan and activities (version 4)

Deliverable Number	D21.3	**Lead Beneficiary**	19. R2M
Deliverable Name	Exploitation and Business Innovation Plan and activities (version 4)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP21
Description			
Initial, updated and final versions of the exploitation and business innovation plan and activities.			

Deliverable D21.4 – Market analysis, business models and revenue streams (version 3)

Deliverable Number	D21.4	**Lead Beneficiary**	18. QPLAN
Deliverable Name	Market analysis, business models and revenue streams (version 3)		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP21
Description			
Market analysis, business models, possible revenue streams, and commercialisation plan and activities			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Establishment of consortium collaboration, visual id & presence and W2W handbook	WP1	1-ICCS	Internal Collaborative Tool operates with access to all partners. Website and social media accounts go online.	4
2	Establishment of DMP, D&C and Exploitation plans	WP1, WP19	4-NTUA	D1.2, D19.1 and D19.2 submitted.	6
3	The framework definition phase is completed - Pilar 1	WP4	16-DRAXIS	D4.1-5 submitted	10
4	External Advisor Board (EAB) is set up	WP1	1-ICCS	EAB participates in a plenary session and feedback is documented.	12
5	First release of digital tools and TRL5 technologies	WP11, WP13, WP9, WP5, WP7	1-ICCS	All prototypes are up and running, relevant research outputs published	18
6	Second release of digital tools and TRL5 technologies	WP6, WP12, WP10, WP14, WP8	1-ICCS	All the technologies are up and running for all the value chains. Relevant research outputs updated accordingly	36
7	Results of successful real demonstration	WP15	4-NTUA	Standardisation contribution, training, learning programs on digital skills, replication packages, business innovation plans are available	48
8	Project ends successfully – Final event	WP3	1-ICCS	Final deliverable versions submitted & minutes of final event sent to Officer.	48

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Project milestones or deliverables are delayed, schedule is partially mismatched	WP17, WP2, WP4, WP6, WP9, WP12, WP19, WP7, WP10, WP8, WP18, WP3, WP21, WP20, WP1, WP11, WP13, WP5, WP14, WP16, WP15	In the scope of project management continuously monitoring activities, detailed analysis will be done on both global project and lower (WP/Task) project implementation levels. Thus, it will be ensured that cases that could delay any milestones or deliverables are recognised in early stages, ensuring timely and effective corrective actions in the work plan. In crucial cases, the Project Coordinator (PC) can work on the plan adaptation in close collaboration with the Project Officer (PO). (M,M)
2	Underestimated or overspending partner resource	WP17, WP2, WP4, WP6, WP9, WP12, WP19, WP7, WP10, WP8, WP18, WP3, WP21, WP20, WP1, WP11, WP13, WP5, WP14, WP16, WP15	Situations will be detected by T1.1, T2.1 and T3.1 (depending on the RP). Executive Board (ExB) will analyse the following possibilities to ensure that planned work can be completed: i) re-arranging resources among partners; ii) committing further internal resources of partners; iii) re-planning activities; iv) mobilise more skilled and efficient human resources. Ad-hoc recovery plans could be put in place.(L,M)
3	One or more partners leaving the consortium	WP17, WP2, WP4, WP6, WP9, WP12, WP19, WP7, WP10, WP8, WP18, WP3, WP21, WP20, WP1, WP11, WP13, WP5, WP14, WP16, WP15	Given the good reputation of all consortium partners, most of them with proven success records in H2020 and HE Projects, this possibility is considered to be very unlikely. The General Assembly will decide whether uncovered project activities can be carried-out by other partners or a partner with relevant experience should recruited.(L,M)
4	One of the partners (SME or start-up) goes bankrupt	WP17, WP2, WP4, WP6, WP9, WP12, WP19, WP7, WP10, WP8, WP18, WP3, WP21, WP20, WP1,	The economic uncertainty driven by the Covid-19 pandemic has pushed struggling companies over the edge into bankruptcy proceedings. To overcome a potential partner bankrupt, advanced allocation of funding will be carefully organised so as to minimise this risk, and the ExB will identify existing or new partners to fill the gap.(M,M)

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
		WP11, WP13, WP5, WP14, WP16, WP15	
5	Missing/delayed equipment	WP6, WP11, WP9, WP12, WP5, WP7, WP10, WP8, WP15	Potential delays (due to chipset/equipment unavailability or transport restrictions) may occur in the delivery/availability of hardware equipment. Partners involved in the technical development and pilot specifications will monitor the supply chain of the identified components, to mitigate delays, and if needed, provide alternative solutions.(M,H)
6	Reluctance from pilot partners to provide data due to security issues - insufficient data from value chains	WP6, WP13, WP5, WP14	The CA will include and define the terms regarding access to data and existing knowledge, with emphasis on the pilots. Secure mechanisms will be deployed for data sovereignty and sharing for data which is characterised as “sensitive” and “confidential”. T1.2, T2.2 and T3.2 will undertake the data management (depending on the reporting period). (M,M)
7	Deviations from technical expectations – invalid design specifications or requirements	WP6, WP11, WP13, WP9, WP12, WP5, WP7, WP10, WP14, WP8, WP15	Iterative development periodically verified in all project meetings will guarantee an early detection of deviations and invalid specs. The integration of mature technologies should minimise these chances. Technical Manager (TM) will monitor the technical design and development processes (T1.2, T2.2, T3.2) with WP leaders and technical partners, to avoid alleviate misalignment of technical visions, to ensure detailed specifications extraction of the software/hardware components and architecture alignment and to quickly identify ineffective designs and take proactive measures.(M,M)
8	Interoperability problems between heterogeneous components	WP6, WP11, WP13, WP9, WP12, WP5, WP7, WP10, WP14, WP8, WP15	Best practices will be used to prevent issues while assessing the design specifications of components, aligned with open and interoperability standards. Consortium partners possess deep knowledge of interoperability with the technical partners to agree on any technical decision that may impact interoperability, with the TM taking the final decisions.(L,L)

9	Challenges in the extraction of cellulose nanofibers might hinder the achievement of the targeted performance of end products	WP7, WP8	Robust testing and optimisation protocols will be implemented to ensure that the extraction and modification processes are thoroughly optimized. Also, extensive characterisation using state-of-the-art techniques will be carried out to assess the physicochemical and thermal properties of the nanofibers and modified lignin. A feedback loop will be established with the partners involved in T7-8.1 to construct correlations between cascade conditions and biocomposites processability. Also, extensive testing and optimisation will be carried out to ensure that the targeted specifications are met.(M,M)
10	The defibration and pre-treatment by steam	WP9, WP10	Extensive pilot-scale testing will optimise the processes and ensure that the results are

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
	explosion may not be optimised effectively for fibre generation, leading to lower yields and quality		reproducible. Thorough research on similar systems and technologies will be conducted before starting the lab-scale experiments to identify potential issues The experiments will be continuously monitored and evaluated, and necessary adjustments will be made, such as modifying the parameters of the process, testing new techniques, and using quality control measures to ensure the fibres and products meet the desired specifications. There will be close collaboration with experts in the field of wood waste upcycling, bioremediation, and chemical processing, involved in the work package. In case of setbacks or delays alternative methods for producing fibres and products will be considered, or the timeline will be adjusted to allow for additional testing and optimization(M,M)
11	Optimising the different processes may require several iterations to achieve the desired results, causing delays	WP11, WP12	The project will conduct thorough research and analysis to ensure a clear understanding of the processes involved and potential challenges that may arise during optimisation, and implement a robust management plan, including regular progress reviews and milestones, to closely monitor the optimization process and identify any potential delays or setbacks early. The involvement of experts with a diverse range of skills and experience to work on the optimisation process will allow for a comprehensive and collaborative approach to problemsolving.(L,M)

12	Disputes over ownership of IPR amongst partners	WP17, WP4, WP6, WP9, WP12, WP7, WP10, WP8, WP18, WP11, WP13, WP5, WP14, WP16, WP15	Standard IPR and access rights clauses will be included in the CA which will be signed before the project starts. Further support to be provided by the legal experts involved in the project for IP conflict resolution. Ownership will be decided according to CA.(M,L)
13	Competition and low end-users' acceptance	WP17, WP4, WP6, WP9, WP12, WP19, WP7, WP10, WP8, WP18, WP21, WP20, WP11, WP13, WP5, WP14, WP16, WP15	Market intelligence activities upon exploitation planning will ensure continuous monitoring of competition landscape. Both R2M and QPLAN will ensure the thoroughness and quality of the resulting reports. Well defined cost-benefit validation of the solution will be implemented. Engagement activities will ensure co-creation with end-users to enhance acceptance(L,L)
14	New superior relevant methodologies and practices arise	WP4, WP6, WP11, WP13, WP9, WP12,	Trends within research areas will be constantly monitored, and strong liaisons with relevant projects and initiatives will be established to join forces. The technical and scientific choices

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
		WP5, WP7, WP10, WP14, WP8, WP15	of the project will be reoriented, considering the latest advances in order to avoid the project's approach being obsolete.(M,M)
15	Coordination with relevant projects, initiatives and associations	WP20, WP19, WP21	T19.2-3, T20.2-3 and T21.2-3 will support cooperation with relevant activities and projects, ensure proper links with different Working Groups and the rest projects of same topic, to build on consensus future systems with different perspectives applied in various environments and beyond.(L,M)

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	**Timing** (month)	**Location**	**Comments**
RV1	18	in person/hybrid	from EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON premises)
RV2	36	in person/hybrid	from EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON premises)
RV3	48	in person/hybrid	from EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON premises)

A Wood-to-Wood Cascade Upcycling Valorisation Approach - W2W 101138789 - HORIZON-CL4-2023-TWIN-TRANSITION-01-42

Associated with document Ref. Ares(2023)8305270 - 05/12/2023_

History of Changes

Version Number	Date	Change
V1.0	25/08/2023	First DoA, PART A Deliverables break down, PART A Milestone beneficiaries, PART A&B Addressing ESR shortcoming, PART B Section 4 added Excellence: Robustness of AI : New text added about the usage of AI Quality : Cost of cameras : The total cost was corrected Quality : BE11 role: The leadership of T19-21.3 was corrected
V2.0	15/09/2023	New version including only the following updates based on ESR: Excellence: Robustness of AI : New text added about the usage of AI Impact : IPR management: Explanation added for IPR Quality : Cost of cameras : The total cost was corrected Quality : BE11 role: The leadership of T19-21.3 was corrected
V3.0	10/10/2023	Tables with 3.1g, 3.1h, 3.1i updated and table with Effort per partner. Transfer of PMs for DRAXIS among WP1,2,3
V4.0	27/10/2023	CF2P terminated, Table with Effort per partner updated, Table 3.1g, 3.1h updated.

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1 Excellence #@REL-EVA-RE@#

1.1 Objectives and Ambition #@PRJ-OBJ-PO@#

W2W Motivation and Strategic Vision: Wood is a natural, renewable, and biodegradable material with impressive mechanical and thermal properties. According to future projections, due to its increasing use for energy production, and in products such as building materials and chemicals, wood production may be insufficient to meet European demand by 2030[1]. At the same time, there is an increasing volume of wood contained in Construction & Demolition (C&D) waste (20-30% of the total building-related C&D) and furniture waste (10 million tonnes of furniture are

discarded by businesses and consumers in EU Member States each yearAssociated with document Ref. Ares(2023)8305270 - 05/12/2023). Recycling these streams can provide an

abundant and cost-effective raw material source. In addition, waste wood streams (from industrial and urban waste) contain significant amounts of fossil-based carbon from coatings, paints, plastics, and preservatives, which could be recycled using innovative processes (among others, chemical recycling), making their valorisation one of the priorities of the Processes4Planet (P4P) partnership for wood waste processing. However, recycling processes are greatly complicated by the presence of additives (glues, varnishes, and paints), pollutants (wood treatment products

[1] Alex Forrest, Mark Hilton, Ann Ballinger & Daniel Whittaker, Circular Economy Opportunities in the Furniture Sector, European Environment Bureau, (2017).

and heavy metals), and contaminating materials (glass, plastics, metals, etc.), and further research and development are needed to improve them. Other challenges are the irregular and unpredictable availability of such waste streams, the complexity of sorting and cleaning processes, and the gaps in existing regulations which may not fully support the use of upcycled wood materials. As a result, current wood waste management strategies still prioritise landfill and energy recovery, over material recovery[2]. To tackle these challenges, **W2W** proposes a comprehensive framework for multi-dimensional cascade[4] valorisation of wood from C&D and furniture waste, consisting of four core components/pillars: **1)** Advanced Separation and Sorting Technologies; **2)** Upcycling Processes and Technologies; **3)** Digital Tools for Improving Circular Flows of Secondary Materials; and **4)** Supportive Framework in Policy, Market, and Skills. W2W will demonstrate efficient and sustainable value chains for the production of: wood without pollutants; biocomposite building materials; biopolymers; polyols; chemical detergents; and the recovery of nutrients, through the use of technologies and tools that allow the selection of optimal cascade paths for further uses of wood products and their materials. The W2W holistic approach will reduce the demand for virgin materials, reduce the amount of waste that ends up in landfills or incineration facilities, allow for the creation of new value-added products from waste materials, and support the transition towards a circular economy by promoting closed-loop systems where materials are continually repurposed and reused, extending their useful life.

1.1.1 W2W Objectives and Success Criteria

O1: To produce and validate the W2W comprehensive framework in terms of stakeholder interactions, wood characterisation and classification, wood valorisation and management technologies, data sharing principles, and impact assessment methods.
W2W will produce the requirements for a comprehensive framework for Construction & Demolition Waste (CDW) cascade valorisation and recycled wood management, drawing on the principles of multi-dimensional cascade valorisation by identifying the different cascade value chain stakeholders and their interactions. A harmonised framework for recycled wood classification will be developed, considering the different schemes available in EU countries. The requirements for an integrated wood cascade valorisation framework will be developed, proposing a set of technologies (related to waste management and digitalisation), and considering nontechnological aspects (policy, market, skills, standardisation, certification). The multi-dimensional approach will be utilised to consider the time, value, and function of recycled wood throughout its lifecycle, with a focus on maximising its economic, social, and environmental benefits while minimising its negative impacts. **Link to Key Exploitable Results (KERs):** KER17 - KER24
Relation to the call: *"Increase the use of secondary resources in the process industry […] reduction of CO2 emissions; reduction of waste sent to landfill and overall positive environmental impact."***;** *"energy efficiency techno-economic and life-cycle assessment considerations of the overall process."*
KPI O1.1: To produce **a harmonised framework** for wood classification in Europe published in a policy brief. **KPI O1.[3]:** To develop **8 technologies** supporting the implantation of the developed framework. **KPI O1.3:** To develop **a data sharing model** and the respective tools among all the identified stakeholders. **KPI O1.4:** To validate the sustainability performance of the designed framework based on techno-economic and Life Cycle Assessment (LCA) considerations.
O2: To improve the separation capabilities of wood sorting mechanisms in line with environmental and technical valorisation parameters.
W2W will develop an advanced separation and sorting system for optimising the cascade valorisation of wood particles. The project will focus on determining the optimal particle size for efficient and effective valorisation, with the aim of maximising the value and utility of the material while minimising its environmental impact. An adaptive sorting system based on cascade valorisation thresholds will be developed to enable the efficient and automated separation of wood particles based on their quality and potential uses. Human-robot collaboration and mixed reality technology will be utilised to improve the accuracy and speed of the separation and sorting process, enabling more efficient and effective management of waste materials. The project will also provide a set of guidelines and recommendations for the development and implementation of advanced separation and sorting

[2] Besserer, A.; Troilo, S.; Girods, P.; Rogaume, Y.; Brosse, N., Cascading Recycling of Wood Waste: A Review. Polymers 2021, 13, 1752, https://doi.org/10.3390/polym13111752 [4] **Cascading use is defined as the efficient utilisation of resources by using residues and recycled materials for material use to extend total biomass availability within a given system.**

[3] Höglmeier, K., Weber-Blaschke, G.; Richter, K. Utilization of Recovered Wood in Cascades versus Utilization of PrimaryWood—A Comparison with Life Cycle Assessment Using System Expansion. Int. J. Life Cycle Assess. (2014), 19, 1755–1766.

systems for the cascade valorisation of wood particles, with the potential to contribute to the broader goal of promoting a circular economy and sustainable waste management practices. **Link to KERs**: KER1 - KER3

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Relation to the call: *"The proposed technologies should contribute to the matching of supply-versus-demand of feedstock at the level of quality constraints."*; *"Considering the upgrading of secondary resources which include development of better separation and sorting technologies."*

KPI O2.1: To achieve a **95%** accuracy of wood separation from mixed CDW sources.
KPI O2.2: To adoptively sort wood based on dynamically updated classification criteria with **90% accuracy**.
KPI O2.3: To allow collaboration between Robots and humans with **non-latency constraints**.

O3: To develop sustainable and innovative technologies for upcycling waste materials, including cascade refinement, chemical and bioremediation processes, and energy, gas and ashes valorisation, to promote resource efficiency, waste reduction, and sustainable waste management.

By encompassing three main themes, namely cascade refinement, chemical and bioremediation processes, and energy/gas/ashes valorisation, W2W aims to promote resource efficiency, waste reduction, and the development of a circular economy through the sustainable and scalable cascade valorisation of resources coming from C&D wood waste, the development of advanced chemical and bioremediation processes for additive removal, and the exploration of innovative approaches for energy and gas valorisation, in cases where material recovery is not possible. By developing these innovative approaches and technologies, the project seeks to contribute to a more sustainable future by reducing environmental impacts, promoting circularity, and enabling informed decisionmaking in waste management practices. **Link to KERs**: KER4 - KER11

Relation to the call: *"Ensure consistent quality and safety of recyclates and their suitability for the upcycling process itself."*; *"Detection and removal additives in the secondary resources stream."*

KPI O3.1: At least **3 value chains** are developed and validated end to end including all the relevant stakeholders.
KPI O3.2: The replication potential of the developed technologies to be performed **in at least one more sector**.

O4: To develop innovative approaches and digital tools for enabling the integration of life cycle thinking and advanced sustainability assessments throughout the cascade process.

W2W will develop tools that will enable stakeholders to perform sustainability assessments of the full wood cascade process, taking into account closed loops of materials processed by a variety of available technologies, and considering supply chain limitations from feedstock to end use. This will make it possible for industry and policy stakeholders to identify trade-offs between alternative cascade paths and allow for improved, evidencebased decision-making. W2W will develop an LCA-based tool to predict and assess the impact of different cascade paths or sequences for possible further uses of wood products and their materials. It will also develop a Supply Chain management tool for the optimisation of upcycling of secondary materials. The use of these tools will be supported by a Digital Product Passport (DPP), containing detailed information about wood-containing product's materials, components, and environmental impact throughout their lifecycle, and a multi-interface digital platform for circularity/upcycling information exchange that will enable relevant stakeholders (businesses and/or consumers) to share information about the environmental impact of products. **Link to KERs**: KER12 - KER16

Relation to the call: *"Foster data sharing, and FAIR digital assets principles, considering the application of DPP between recycling companies and the process industry to improve the economy of scale in upcycling of material streams."*; *"Successful upcycling relies on advanced monitoring and sensing in the process industries and value chains, and on improved data completeness, accuracy and interoperability between the process and the recycling companies. Upcycling may create new business opportunities and models."*; *"Proposals should include energy efficiency techno-economic and life-cycle assessment considerations of the overall process."*

KPI O4.1: LCA-based cascade process sustainability assessment tool validated in the 3 proposed value chains.
KPI O4.2: Supply chain optimisation tool validated in the 3 proposed value chains.
KPI O4.3: DPP incorporates and regularly updates a minimum of **5 material parameters**.
KPI O4.4: Multi-digital interface platform integrating several digital tools/services (i.e. dynamic LCA, dynamic DPP) developed by Month 48 with **at least 200 active users**.

O.5: To validate W2W technologies/solutions in 3 Use Cases, representing different value chains.

The project focuses on sustainable waste management practices in the construction, pulp and paper and chemicals sectors, with the implementation of three Use Cases that will concentrate on specific areas of waste management. The **first Use Case** will emphasise on resource efficiency and waste reduction of C&D wood waste through cascade biorefinement, which involves the separation of lignin and cellulose to create composite wood materials for various industries, including pulp and paper. The **second Use Case** will focus on innovative approaches to additive removal using advanced chemical and bioremediation processes for sustainable waste management, relevant to both the pulp and paper and chemicals sectors. The **third Use Case** will explore energy and gas recovery potential from CDW through hydrothermal carbonisation, gasification, ashes valorisation, and dodecanol production via H2/CO2 upcycling, also relevant to both sectors. The developed cascade valorisation approach will be assessed for replicability in other industrial sectors, such as glass waste derived from CDW. This will enable
the exploration of the potential for the transferability and adaptation of sustainable waste management approach to a broader range of industries. **Link to KERs**: ALL
Relation to the call: *"Prove the technical and economic feasibility of the use of secondary resources in the process industry leading to products with identical properties and performances as those produced using primary resources and allowing production without quality restriction."***;** *"The innovative upcycling of the secondary raw materials should be demonstrated through at least two realistic use cases [...] technology providers;"***;** *"pursue the involvement of all the actors in the value chain from the process industry."*
KPI O5.1: To demonstrate the application of all the developed technologies **at TRL 6 in 3 use cases**. **KPI O5.2:** To develop **at least 3 cross-sector business models**.
O.6: To increase competitiveness and promote sustainable practices in industries through skills development, collaborative training programs, capacity building, contributions to standards and certification schemes, policy influence, and business models in circular and sustainable processes.
W2W aims to augment competitiveness, certification, and policy for promoting circular and sustainable practices in industry, including an integrative approach to skills development that incorporates social partnerships, synergies between learning initiatives and circular economy hubs, collaborative design of training programs, promoting capacity building and inclusive growth, and driving sustainable practices through contributions to standardisation and certification. A market analysis and the co-design of business models with prospective customers and stakeholders participating in the pilot activities will offer an evidence-based pathway for further developing and commercialising W2W outputs. **Link to KERs**: KER20, KER24
Relation to the call: *"To increase the use of unused and new skills to unfold the potential of the technological solutions at the workplace for upcycling and contribution to inclusive growth."***;** *"A strategy for skills development should be presented. Particular attention should be given to the cooperation with existing initiatives."*
KPI O6.1: Fulfilment of Dissemination, Communication and Exploitation KPIs, as described in Section 2.2. **KPI O6.2: Financial feasibility** of 3 value chains to be validated and sustained post-project implementation. **KPI O6.3:** Development of **at least 2 major training activities and lifelong learning programs** co-designed with relevant stakeholders and initiatives, to promote upcycling, replication, and cooperation. **KPI O6.4:** To produce 1 policy brief for wood classification and 1 for C&D wood waste cascade valorisation. **KPI O6.5: At least 3 certification proposals** formed for secondary raw materials after analysing value chains.

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1.1.2 Ambition – Research and Innovation Maturity of the W2W Pillars

W2W aims to provide cross-sectoral – from sensing to sorting & from materials to products – digitally enabledsolutions to increase waste reduction, improve use of secondary raw materials, thus, lead to circular and low emission value chains. This is achieved through **4 main innovation pillars**, as presented below.

1.1.2.1 Pillar 1: Advanced Separation and Sorting

A) Separation Technology for Optimal Wood Particles Size

State of the Art Analysis & Challenges: Spectroscopic methods as near-infrared hyperspectral imaging (NIR-HSI) are often used as analytical techniques to determine the presence of wood by-products, to detect the wood by-products quality and classify the suitable materials for valorisation. In particular, they can be used for detecting the moisture content[5], for the classification of wood according to species[6] and to differentiate untreated waste wood[5]. In the case

[4] a, T., Morita, G., Inagaki, T. et al. Moisture transport dynamics in wood during drying studied by long-wave near-infrared hyperspectral imaging. Cellulose 29, 133–145, (2022). [6] Kanayama, H., Ma, T., Tsuchikawa, S. et al. Cognitive spectroscopy for wood species identification: near infrared hyperspectral imaging combined with convolutional neural networks. Analyst 144, 6438-6446, (2019). http://dx.doi.org/10.1039/C9AN01180C

[5] So, C. L., Lebow, S. T., Eberhardt, T., et al. Application of Near Infrared Spectroscopy to Preservative-Treated Wood, Wood and Fiber Science (2021).

of wood chips, laser scanning, machine vision and image processing can be applied to calculate and categorise them according to their size, and determine the percentage of rot in residues[6].

Ambition: In the W2W project, optical and spectroscopic characterisation techniques will be used to determine which the most efficient particle size of wood and glass coming from CDW to further process these materials. Certain process parameters, such as the size of shredding of wood-by products, will be correlated with the wood and glass sorting effectivity. Moreover, an innovative offline NIR handheld device working with advanced chemometric models will be used to analyse relevant product parameters.

B) Adaptive Sorting System based on Cascade Valorisation Thresholds

State of the Art Analysis & Challenges: Sorting systems currently developed for CDW are evaluated based on simple metrics, such as picks per time unit and rate of successful material classification[7,8]. For separating wood, commercial sorting systems are deploying robotic arms and specialised grippers[11,9], while for glass waste air jet-

[6] Grigorev, I., Shadrin, A., Katkov, S. et al. Improving the quality of sorting wood chips by scanning and machine vision technology. J Forest Sci, 67: 212–218, (2021).

[7] R. Sarc, A. Curtis, L. Kandlbauer, K. Khodier, K.E. Lorber, and R. Pomberger, "Digitalisation and intelligent robotics in value chain of circular economy-oriented waste management – A review", Waste Management, Volume 95, 476-492, (2019).

[8] R. Bogue, "Robots in recycling and disassembly", Industrial Robot: the international journal of robotics research and application, Volume 46, Pages 461-466, (2019). [11] OP Technik ab, https://www.opteknik.se/sorteringssida?lang=en

[9] Zen Robotics, https://www.terex.com/zenrobotics/

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based separators are used[10]. Although typical CDW sorters depend on several factors related to the quality of the stream and the sorting technology, these metrics can reach up to 4,000 picks per hour (per robot) and 95% accuracy respectively. These metrics are essential for the performance of the sorting system itself, but improving them does not necessarily maximise the overall impact of the sorting process towards the optimum valorisation of the wood waste. Thus, currently employed sorting systems do not attempt to monitor the output stream characteristics and do not take into consideration valorisation needs and material composition during their planning.

Ambition: W2W aims to redesign the entire sorting process to focus primarily on cascading valorisation of the input material. A multi-robot composite sorting system capable of separating wood and glass from mixed CDW with more than 95% accuracy will be deployed, while the sensor equipment will be chosen based on the optimal particle size decided above and the classification threshold will be dynamically updated based on the classification thresholds defined for each valorisation flow. The classification thresholds shall be dynamically defined based on sustainability assessment criteria and the input of the project digital tools supporting production planning and supply-vs-demand.

C) Human-robot Interaction/Collaboration Technology & MR Solution enabling Advanced Separation Capacity

State of the Art Analysis & Challenges: Apart from the chemical contamination on the surface, which has been the focus of most research work, a wide range of physical impurities are also present on CDW materials[11]. These impurities are normally removed after size reduction with suspension magnetic separators for ferrous metals or eddy current separators for non-ferrous metals[14]. For wood waste, it is quite common for nails, screws and bolts to remain in the material after passing through the separators due to the strong connection with the material, while plastic impurities due to their physical properties cannot be affected by the aforementioned separation processes. Nonetheless, a large number of novel studies has been devoted lately to robotic disassembly of known complex products[12]. In these processes, however, there is a key limitation in the capacity of the robot to understand and perform the large range of required disassembly tasks, a challenge that is magnified in waste sorting due to the variety of sizes and geometries of the waste particles. A human worker, on the other hand, is quite dexterous to handle many of the tasks and collaborate with the robot, while MR facilitates their coordination and communication. Mixed Reality (MR) offers an interface to seamlessly simplify the collaboration between operators and robots, applied to many settings of their interaction, such as gesture control and programming[13].

Ambition: A Human Robot Collaboration process will be established within W2W project with the objective of decreasing waste due to physical impurities. It will take advantage of important qualities of human workers, e.g., increased dexterity and advantages of the collaborative robots, such as endurance to remove all sorts of impurities from the contaminated wood. Specifically, an automated task coordinator will divide the workload based on vision input of the workpiece into tasks attainable by the robot and tasks that need to be performed by the human worker. Taking advantage of MR technology to positively influence operator-related aspects such as performance, task awareness, and safety feeling, W2W offers the right settings for efficient and effective guidance of humans through augmented interfaces to avoid unnecessary costs and time consumption[14]. MR challenges the conventional operation of collaborative robots (cobots), via user friendly and intuitive interface communication, resulting in an improvement of the accuracy to time ratio, and minimisation of the workload by pointing out to humans' tasks that cobots are either not programmed to perform, or do not have the advanced capabilities to do so.

Tool / Technology /Activity	Key Advantage & Progress – Start TRL ❼ Foreseen End TRL
Separation technology for optimal wood particles size	A wood processing unit based on correlation analysis of particle size and cascading valorisation value: **TRL4❼TRL6**
Adaptive sorting system	A high accuracy, high performance sorting system that separate different classes of wood and glass waste from CDW streams while adapting to dynamic cascade valorisation thresholds: **TRL4❼TRL6**
Human-robot interaction/collaboration	Human-robot interaction/collaboration technology and MR solution enabling advanced separation capacity: **TRL4-->TRL6**

1.2.1.2 Pillar 2: Novel Technologies for Upcycling

[10] Z. Yao, B. Qin, Z. Huang, J. Ruan, Z. Xu, "Green Combined Resource Recycling System for the Recycling of Waste Glass", *ACS Sustainable Chem. Eng*Pages 7361–7368, (2021).
[11] G. Faraca, A. Boldrin, and T. Astrup, "Resource quality of wood waste: The importance of physical and chemical impurities in wood waste for recycling", (2019).
[12] H. Poschmann et al. , "Disassembly 4.0: A Review on Using Robotics in Disassembly Tasks as a Way of Automation", Chemie Ingenieur Technik, Vol. 92, 4, (2020).
[13] Hörbst, J., Orsolits, H. (2022). Mixed Reality HMI for Collaborative Robots. In: Moreno-Díaz, R., Pichler, F., Quesada-Arencibia, A. (eds) Computer Aided Systems Theory – EUROCAST 2022. EUROCAST 2022. Lecture Notes in Computer Science, vol 13789. Springer, Cham. https://doi.org/10.1007/978-3-031-25312-6_63
[14] Costa, G.d.M.; Petry, M.R.; Moreira, A.P. Augmented Reality for Human–Robot Collaboration and Cooperation in Industrial Applications: A Systematic Literature Review. Sensors 2022, 22, 2725. https://doi.org/10.3390/s22072725

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A. Cascade Refinement Technologies for Wood Waste Upcycling

#Production of Lignin and Cellulose Fibers from Wood Waste

State of the Art Analysis & Challenges: Limited numbers of works have been reported in literature about the fractionation of mixed wood[15,16] into its main components that could be used for different applications. There are two major concerns regarding wood treatment for fiber production: **(1)** the lengths of fibers obtained from waste wood are often shorter than those obtained from native wood; and **(2)** the fibers produced retain debris, such as plastics or decorative elements (e.g., melamine coating). As they also still contain some of the additives originally contained in the waste wood (glues, paints, fungicides, etc.), achieving efficient and effective use of wood resources requires considering multidimensional valorisation of wood, including time, value, and function.

Ambition: The development and optimisation of the operation conditions of fractionation process able to produce lignin and cellulose fibers from the feedstock, will take place during W2W. Two main natural reinforcements will be isolated from wood waste by the use of chemical and mechanical treatments: **1)** Unmodified CNF[17] will be provided as water solution containing 6 to 9% of solid (cellulose). Solvent exchange will be done to replace portion of water by ethanol; and **2)** Modified CNF and lignin will be provided in solid state.

#Production of Bio Composites Reinforced with Wood Fibres

State of the Art Analysis & Challenges: The current state-of-the-art in sustainable composite[21,18] production involves the use of natural fillers or reinforcement agents, such as fibers from plants. However, the hydrophobic and nonpolar character of the polymeric matrix used in most research studies poses challenges to fiber-matrix compatibility and homogeneity; this incompatibility is challenging for both the processing (discontinuous feeding, flow of the melt) and the final properties of the composites (poor mechanical strength). Compatibilisation is attempted either through chemical modification of the fibres or through the addition of surfactants or compatibilizers/coupling agents. These factors limit the integration of wood fibers produced from wood waste into the industrial manufacturing cycle.

Ambition: W2W innovation will be based on the combination of PLA (poly(lactic acid)), as a biodegradable polymeric matrix, and biodegradable lignocellulose-based materials derived from wood waste, yielding an entirely sustainable biocomposite structure. Optimisation of compounding conditions in melt mixers and twin-screw extruders along with testing different types of commercially available PLA and PLA-copolymers will construct a toolbox to correlate PLA type-wood waste cascade refinement technology-compounding conditions-properties data and will permit the selection of the adapted materials per targeted application. Indicatively, the compatibility and mixing potential of the derived W2W wood fractions with PLA grades of different stereochemistry, different molecular weights and melt flow rates, etc., will be assessed based on the biocomposites morphology, thermal and mechanical characteristics, and will be improved through the fine tuning of the applied fractionation technology and/or compatibilization approaches.

B. Chemical and Bioremediation Technologies for Wood Waste Upcycling

#Physicochemical (Steam Explosion) Treatment of Wood Waste

State of the Art Analysis & Challenges: The presence of residual glue on the surface of the wood fibres or particles makes their recycling more difficult, especially to produce new wood panels. Recently, a process based on steam explosion for the removal of UF resin has been developed. This process opens the way for the recycling of fiberboard panels[19]. Comparing to other technologies like Hot Pressurised Water treatment (HPW), steam explosion (SE) technology is a very promising process for the recovery of medium-density fiberboard MDF fibers, allowing 83% of the resin removal accompanied by an efficient fragmentation in a very short time and water consumption. Other MDF hydrothermal treatments for MDF recycling reviewed in literature lead to a fiber shortening of ~30%[20] Steam refining performed in comparable conditions lead to a fiber yield of ~75%[21]. Although the SE and SEW techniques have taken place in recent studies in experimental level, a fiber yield of 80% w/w (±2%) was achieved.

[15] Maarouf Abdou Alio, et al., Microwave-assisted Organosolv pretreatment of a sawmill mixed feedstock for bioethanol production in a wood biorefinery, Bioresource Technology, Volume 276, 2019, Pages 170-176, ISSN 0960-8524, https://doi.org/10.1016/j.biortech.2018.12.078.

[16] Xuejun Pan, et al., Biorefining of softwoods using ethanol organosolv pulping: Preliminary evaluation of process streams for manufacture of fuel-grade ethanol and co-products, Biotechnology and Bioengineering, Volume 90, Issue 4, Pages 473-481, 2005, https://doi.org/10.1002/bit.20453.

[17] Baraka, F., Robles, E., Labidi, J. Microwave-assisted esterification of bleached and unbleached cellulose nanofibers (2023) Industrial Crops and Products, 191, art. no. 115970 [21] Didho M,n et al. Lignin as green filler in polymer composites: Development methods, characteristics, and potential applications (2022). Hindawi Advances in Materials Science.

[18] Singh, A., Genovese, M., Mancini, G., Marina, L., Athanasssiou A., (2020). Green processing route for polylactic acid-cellulose fiber biocomposites. ACS Sustainable Chemistry & Engineering, 8, 4128-4136

[19] Troilo, S., Besserer, A., Rose, C., Saker, S., Soufflet, L., Brosse, N., 2023. Urea-Formaldehyde Resin Removal in Medium-Density Fiberboards by Steam Explosion: Developing Nondestructive Analytical Tools. ACS Sustainable Chem. Eng. 11, 3603–3610. https://doi.org/10.1021/acssuschemeng.2c05686

[20] Lubis, M. A. R.; Manohar, S. Y.; Laksana, R. P. B.; Fatriasari, W.; Ismayati, M.; Falah, F.; Solihat, N. N.; Sari, F. P.; Hidayat, W. The Removal of Cured Urea-Formaldehyde Adhesive towards Sustainable Medium Density Fiberboard Production: A Review. J. Sci. Lett. 2021, 9, 23.

[21] Hagel, S.; Saake, B. Fractionation of Waste MDF by Steam Refining. Molecules 2020, 25, 2165.

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Ambition: Within W2W, wood containing Adhesives/Pollutants with potential of removal will be treated by SE and the process will be up scaled from TRL4 to TRL6. It is expected that a fiber yield of more than 80% w/w will be achieved, going beyond the efficiency achieved in experimental stage. The recovered fibers will meet the specifications to be utilized as fillers for polymers production with improved properties.

#Fermentation and Fungal Growth for Detoxification of Hazardous Wood

State of the Art Analysis & Challenges: In recent studies, the morphological characterisation and quantification of the mycelial growth of the Brown-Rot Fungus Postia placenta has taken place. Parameters affecting their growth were identified while the parameter values were used for calibrating the model for simulating the mycelial growth. The growth characteristics were statistically analysed in relation to the different duration of the colony development[22].The morphological characterisation and quantification of the mycelial growth paves the way for the treatment of wood containing adhesives/pollutants, through fungi fermentation. For this purpose, microorganisms with tolerance in preservatives like Postia placenta could be used.

Ambition: Wood waste belonging to BR2 and C classes will be initially subjected to mycoremediation to remove hazardous substances. Two ways will be explored for mycoremediation: **(1)** Solid state fermentation with wood preservative tolerant fungi by using both Postia placenta and Trametes versicolor strains as control strains to establish the proof of concept. Then a fungal strain screening will be done to find the best fungal specie(s) for mycoremediation. **(2)** Submerged fermentation with wood preservative tolerant microorganisms, by using fungalbacteria consortia with Postia placenta as fungus and Burkholderia strains. Optimisation of the liquid detoxification phase will be performed using first reference fungal strains known to be active in formaldehyde detoxification at lab scale. After validation of the detoxification efficiency, complementary analysis of the incubation medium will be performed to screen occurrence of molecules with potential interest in material manufacturing or biocontrol activity.

#Liquefaction of mixed Waste with Organic Catalysts for Polyol Production

State of the Art Analysis & Challenges: Polyols are industrially produced mainly by ring-opening polymerisation (ROP), radical and ionic polymerisation, and condensation polymerisation pathways. The synthesis of low-cost polyols from abundant and renewable biomass resources has created industrial interest. In line with the anionic ROP (AROP) process, biomass-derived polyols have been developed over the decades. Biomass oxypropylation has been researched intensively over the last decades with different natural polymeric substrates that contain hydroxyl groups. Nevertheless, this traditional method based on anionic ROP has several drawbacks related to the aggressive conditions employed. Therefore, the reaction requires special equipment to ensure safety against explosions and reaction conditions, increasing the final cost of the polyols considerably, hampering any industrial application. **Ambition:** To circumvent these drawbacks, a more industrially feasible alternative route should be developed. In W2W, a new process based on cationic ROP using acid catalysts will allow synthesising bio-polyols with a controlled structure under mild conditions. Polyols will be synthesised using two approaches: liquefaction process and using ROP. Polyol will be produced by ROP of oxiranes in the presence of mixed waste and in an organic solvent or mixture thereof using acidic catalysts. The process will be upscaled from TRL4 to TRL6.

C. Energy, Gas & Ashes Valorisation Technologies for Contaminated Wood Waste

#Hydrothermal Carbonisation through Microwaves

State of the Art Analysis & Challenges: Hydrothermal carbonisation (HTC) is carried out in wet and at supercritical conditions producing mainly process water rich in nutrients and organics and solid rich in carbon (hydrochar). This process is already commercialised mainly for sewage sludge treatment at the wastewater treatment plants[23]. However, there are not reports or evidence of the implementation of the HTC treatment as part of another waste management process, such as wood containing waste, in a commercial scale. In recent years, hydrothermal treatment has attracted increasing attention as one of the most promising methods for the treatment of various solid wastes. These wastes can be converted into stable aluminosilicate minerals materials, and meanwhile achieve the stabilization of heavy metals in the final hydrothermal products when operating in alkaline environments. In addition, current hydrothermal systems combined with microwaves are just tested in lab-scale, as their upscaling to industrial-scale devices needs complex designing and simulation know-how that first needs to be nurtured with intensive experimental outputs. Microwave-assisted hydrothermal processes for ash valorisation with this aim are not clearly described in literature.**Ambition:** HTC of mixed waste (contaminated wood waste and liquid waste e.g., sludge,

[22] Du H, Lv P, Ayouz M, Besserer A, Perré P (2016) Morphological Characterization and Quantification of the Mycelial Growth of the Brown-Rot Fungus *Postia placenta* for Modeling Purposes. PLoS ONE11(9): e0162469. https://doi.org/10.1371/journal.pone.0162469

[23] Klaudia Czerwińska, Maciej Śliz, Małgorzata Wilk, Hydrothermal carbonization process: Fundamentals, main parameter characteristics and possible applications including an effective method of SARS-CoV-2 mitigation in sewage sludge. A review,
Renewable and Sustainable Energy Reviews, Volume 154, 2022, https://doi.org/10.1016/j.rser.2021.111873.

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MSW, etc.) will be employed at lab scale (TRL4) and validated in a TRL6 HTC reactor. In this project, a more efficient "HTC 2.0" concept is proposed consisting in HTC assisted through microwave exhibiting the following advantages: **1)** The reaction temperature is limited to 150–300°C, which is lower than other thermal treatments (>900°C); **2)** The subcritical hydrothermal treatment of CDW-Flying Ash (FA) is carried out in a closed autoclave, which prevents the diffusion of dioxins or heavy metals into the air, posing fewer health threats to operators; **3)** Microwave heating method involves novel pathways with extremely high potential of energy efficiency and renewable energy-toelectrification strategies. The process will be upscaled from TRL4-TRL6.

#Biotech-driven Upcycling of Syngas into Dodecanol for Low Impact Detergents Application

State of the art and challenges: Dodecanol ($C_{12}H_{26}O$), or lauryl alcohol, is today one of the mostly used long chain alcohol intermediates to produce almost any kind of detergent; in Europe it is produced industrially from imported palm kernel oil/coconut oil or from fossil-based ethylene by using the Ziegler process, causing relevant impacts on the environmental sustainability[24,29]. Some attempts have been done at lab scale to obtain biobased medium and long chain molecules from sugars fermentation through engineered microorganisms. **In the past three years, KIV has patented and successfully completed a deep experimental campaign to develop and validate at TRL4, a custom synthetic biology toolkit and oxyhydrogen engineering chassis for the production of dodecanol**. This toolkit includes vectors, recombinases, CRISPR precision editing, libraries of gene expression regulatory elements, a genome-scale metabolic model and computational tools for operon design. This has led as main starting achievement to a library of recombinant oxyhydrogen strains (C.necator) specifically engineered to produce the C_{12} fatty alcohol dodecanol showing a productivity > 1g/lt h. The **KIV** proprietary organism has an incredibly rapid growth rate and produces biomass continuously at productivities of up to 3g/L/h.

Ambition: W2W will leverage **KIV**'s extensive knowledge for further engineering the dodecanol production strain to enhance productivity compared to the state-of-the-art process (from 1 to than 1.5-2g/lt h of dodecanol). The overall process with enhanced performances will be tested using syngas from waste derived from hydrochar gasification in combination with additional CO2 sources from biogas cogeneration; TRL will be progressively increased from TRL5 (20Lt gas fermenter volume) to TRL6 (100Lt gas fermenter volume). Dodecanol together with by-products from the process (biomass cells as source of amino acids) will be validated into the formulation of ionic-surfactants for specific applications in detergency sector with small scale consumer exposure tests proving both technical and safety requirements, achieving properties at least equivalent to those from primary resources.

Tool/Technology/Activity	Key Advantage & Progress – Start TRL ❼ Foreseen End TRL
Wood Cascade Refinement	Wood cascade to function, Fractionation of mixed feed stock: **TRL4 --> TRL6** Production of biocomposites reinforced with wood fibers, Wood composites for wallboards: **TRL4❼TRL6**
Chemical and Bioremediation processes	Wood glue separation, Physicochemical (Steam explosion) treatment of wood waste: **TRL4❼TRL6**; Bioremediation wood treatment: **TRL4❼TRL6**; Production of bio polyols suitable for the synthesis of NIPUS, Liquefaction of mixed Waste with organic catalysts for polyol production: **TRL4❼TRL6**
Energy and Gas Valorisation technologies	Hydrothermal Carbonization through MW, Ashes valorisation **TRL4❼TRL6** Biotech-driven upcycling of syngas into dodecanol for low impact detergents application: **TRL4❼TRL6**

1.1.2.3 Pillar 3: Digital Tools for Improving Circular Flows of Secondary Materials

(A) Life Cycle Sustainability Assessment for different Cascade Pathways

State of the Art Analysis & Challenges: Life Cycle Assessment (LCA) studies related to recycling of wood often omit the management stages, such as the separation and collection processes, don't cover details of the process in every life cycle step, and focus on the environmental perspective, rather than the economic and social perspectives[25]. For a cascade approach, there is a need to track material pathways over several use cycles. Another challenge is that the quality and consistency of data on the wood material stream in Europe is generally not satisfactory[26] (partly

[24] Shah, J., et al., Comparison of Oleo- vs Petro-Sourcing of Fatty Alcohols via Cradle-to-Gate Life Cycle Assessment. J. Surfactants Deterg. 19, 1333–1351 (2016).

[29] Anagha Krishnan, Bonnie, A. McNeil and David T. Stuart. Biosynthesis of Fatty Alcohols in Engineered Microbial Cell Factories: Advances and Limitations, Biotechnol., (2020)

[25] Jahan, I.; Zhang, G.; Bhuiyan, M.; Navaratnam, S. Circular Economy of Construction and Demolition Wood Waste—A Theoretical Framework Approach. *Sustainability* **2022**, *14*, 10478. https://doi.org/10.3390/su141710478

[26] Vis M., U. Mantau, B. Allen (Eds.) (2016) Study on the optimised cascading use of wood, No 394/PP/ENT/RCH/14/7689. Final report. Brussels 2016. 337 pages

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because it is often not recovered separately but instead collected, managed and disposed in a mix with other CDW), making it difficult to perform a detailed assessment.

Ambition: W2W will combine LCA, Life Cycle Cost Analysis (LCCA), and Social Life Cycle Assessment (S-LCA) to perform an integrated Life Cycle Sustainability Assessment (LCSA), where the three dimensions of sustainability (environment, economy, and social aspects) will be considered. LCSA will support the identification of trade-offs between the dimensions and allow for better decision-making in policy making and industry. Machine Learning methods will be used for the prediction of missing data, forecasting impact parameters both directly and indirectly.

(B) Material Volume Estimation

State of the Art Analysis & Challenges: Estimating the volume of waste materials is a challenging task and it has attracted a lot of research interest mainly because of their diversity. Challenges concern low accuracy of computer vision methods, and failure to clearly distinguish among waste materials. In addition, different settings render the development of more generic methods a conundrum. There are various approaches to waste estimation using computer vision methods and photogrammetry to tackle the aforementioned issues. Deep learning methods are used for segmenting the various materials[27,28] but they usually need a more contained environment in order to approach the task efficiently. Stereo vision is another popular approach where the materials are captured from different angles in order to generate a 3D model that will eventually serve for the volume calculation. These methods concern primarily construction and demolition waste. However, generating the correct 3D model is a complex process that might lead to inaccurate results. Furthermore, LiDAR and hyperspectral imaging approaches can also be employed as well as methods assisted by UAVs[29,30], however, such equipment increases the cost significantly.

Ambition: In W2W, computer vision methods will be applied for mixed waste volume estimation. For the efficient material volume estimation: **i)** multisource data will be used in order to accurately generate the 3D models, **ii)** images will be acquired from conventional RGB cameras, **iii)** a mixture of ground and aerial images will be used, and **iv)** semantic segmentation of the waste will be performed to characterise the waste material and complement with their volume estimation. Eventually, a tool for volume estimation and material recognition in mixed waste cases will be developed capable at operating in different settings.

(C) Supply Chain Management for the Optimisation of Upcycling of Secondary Materials

State of the Art Analysis & Challenges: A cascade approach requires additional logistics operations to realise mass and energy flows between material use, chemical use and energy use, which may be performed by different stakeholders who may be at distant locations. These processes come along with environmental effects, which must be quantified and considered with respect to the overall effect of cascading utilisation. The quantification of the positive environmental effects and of the trade-offs with other consequences of cascade utilisation, however, remains an open question both in theory and practice[31].

Ambition: W2W will develop a supply chain optimisation digital tool for secondary materials for managing the process of recycling wood along the supply chain. The tool will use advanced algorithms and data analysis techniques to simulate various scenarios to help companies identify the most efficient and cost-effective supply chain design and operation for optimising the flow of secondary materials throughout the recycling process, from collection to processing and distribution.

Tool / Technology /Activity	Key Advantage & Progress – Start TRL ❼ Foreseen End TRL
Lifecycle Sustainability Assessment for different cascade pathways	The currently available technology for LCA studies will be extended to support cascade paths. Machine Learning will be used to fill data gaps: **TRL4❼ TRL6**
Material Volume Estimation	AI-based Material Volume Estimation Tool using own equipment for use in diverse settings. The tool in its final version will be validated in a real-world setting: **TRL4❼TRL6**
Close Loop Supply Chain management tool	Close Loop Supply Chain management tool for upcycling optimisation of secondary materials: **TRL4❼TRL6**

1.1.2.4 Pillar 4: Framework for CDW Cascade Valorisation

State of the Art Analysis & Challenges: The cascading concept aims to increase the efficiency of biomass utilisation and is promoted in bioeconomy strategies as a proper practice for potential biomass limitations. There are three approaches of cascading: **(i)** cascading-in-time - conceptualised as a sequential use of biomass through reusing

[27] Chen, J., Lu, W., Yuan, L., Wu, Y., & Xue, F. (2022). Estimating construction waste truck payload volume using monocular vision. Resources, Conservation and Recycling, 177.
[28] Dong, Z., Chen, J., & Lu, W. (2022). Computer vision to recognize construction waste compositions: A novel boundary-aware transformer (BAT) model. environmental management.
[29] Filkin, T., et al., (2022). Estimation of dump and landfill waste volumes using unmanned aerial systems. Waste Management, 139, 301-308.
[30] Julge, K., Ellmann, A., & Köök, R. (2019). Unmanned aerial vehicle surveying for monitoring road construction earthworks. The baltic journal of road and bridge engineering.
[31] Aidin Parsakhoo, Sattar Ezzati, Saba Pirov et al. Operational Planning of Optimal Wood Supply-Chain Network with Focus On Inbound Logistics Constraints, 17 March 2022.

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or recycling and energy production at the end-of-life cycle, **(ii)** cascading-in-value - prioritising the highest possible value over the whole life cycle through optimisation of the time steps of the approach, and **(iii)** cascading-in-function - aiming to maximise the value of the original biomass through co-production[32,33]. A complete framework of cascade valorisation of C&D wood waste is currently lacking, and legislation has not been updated to reflect the potential of novel technologies for upcycling[39]. There is also a lack of a common European Framework for wood waste classification into categories[34]. As an inert and biodegradable material, wood is easy and relatively cheap to send to landfills, which does not favour the emergence of wood recycling. Due to the complexity of upcycling cascade process (requiring additional sorting, cleaning, chemical or mechanical treatment, transportation, etc.), energy recovery is prioritised.

Ambition: W2W will fill the regulation gap for wood waste classifying wood based on: **I)** the type of chemical additives and components, **II)** the contamination level, and **III)** the place of origin (post-production: waste generated in the technological process of wood manufacturing, or post-consumer: waste generated from used wood products). A comprehensive framework for cascade valorisation of C&D recycled wood will be designed including mechanical recycling, cascade biorefinement, chemical and bioremediation processes, and energy, gas and ashes valorisation processes.

Tool / Technology /Activity	Key Advantage & Progress – Start SRL ❼ Foreseen End SRL
Framework for recycled wood classification	A complete analysis of legislations/regulations, solutions of wood waste classification, and proposals for a common EU Framework: **SRL5❼SRL6**
Framework for cascade valorisation of recycled wood	A complete framework for optimal cascade valorisation of C&D wood waste: **SRL4❼SRL6**

1.2 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

1.2.1 Overall Concept and Approach: Wood Cascade Valorisation from CDW

The W2W approach aims to extract maximum value from CDW and furniture waste by prioritising multiple uses and transformations of wood, according to its composition, the availability of upcycling technologies, and supply chain restrictions. A description of how the W2W innovation pillars interact to continuously improve the value of the secondary resources throughout the upcycling process is shown in Figure 1.1. The approach considers different cascade paths, according to the circular economy principles, for upcycling wood available in C&D and furniture waste: **1. Sequential reuse of materials after their separation**: reuse of materials in their original form for similar or alternative applications; **2. Cleaning**: when materials can no longer be reused in their original form, the cascade approach promotes repurposing by removing additives and delivering wood of better quality; **3. Upcycling**: transformation into higher-value products; **4. Energy recovery** of materials that no longer be repurposed or upcycled.

[32] M. Jarre, A., et al., Transforming the bio-based sector towards a circular economy - What can we learn from wood cascading?, *(*2020), doi: 10.1016/j.forpol.2019.01.017.

[33] E. Robles, et al., "Valorization of heat-treated wood after service life through a cascading process for the production of lignocellulosic derivatives," *Resour. Conserv. Recycl.*, (2021).

[39] **The Waste Framework Directive set a 70% target for reuse and recycling applies to CDW in general by 2020. However, as wood represents 20-30% of CDW, this target did not serve as an incentive to set up wood collection schemes. In theory, this target could be achieved with 0% of wood being recycled. The amendment of Landfill Directive drives the shift from landfill to the EU's waste hierarchy but does not present any specific wood measures.**

[34] **Austria, Belgium, Bosnia and Herzegovina, Canada, Denmark, Finland, France, Germany, Ireland, Netherlands, Slovenia, Switzerland, Turkey, UK have set regulation for wood waste categorisation into 3 to 5 (mainly 4) categories considering the source, components, chemical additives, quality, contamination level and the treatment method of wood waste.** Source: ttps://unece.org/sites/default/files/2021-03/ece-tim-efc-wp2-2021-inf-5.pdf

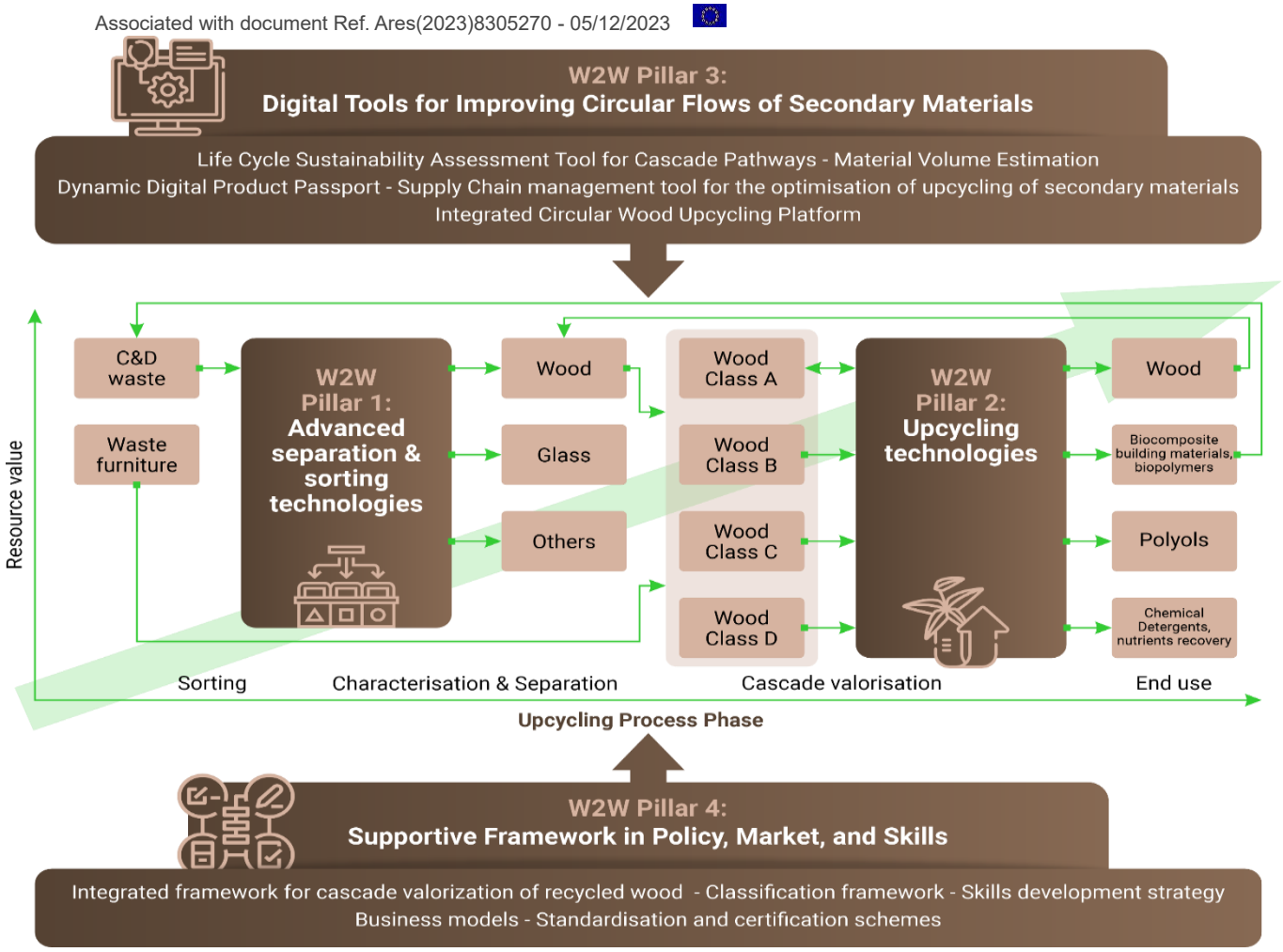

Figure 1.1: W2W overall concept

The **W2W Advanced Separation and Sorting Technologies (Pillar 1)** will enable the automated separation of wood particles from C&D waste based on their quality and potential uses. It is assumed at this stage that wood can belong to 4 generic classes (the classification framework eventually proposed by the project will be part of the policy recommendations). The **W2W Upcycling Technologies (Pillar 2)** will enable the processing of recovered wood (through mechanical recycling and cascade biorefinement, chemical and bioremediation processes, and energy/gas/ashes valorisation) to produce higher value products. The **Digital Tools for Improving Circular Flows of Secondary Materials (Pillar 3)** will facilitate transparency and collaboration among stakeholders (Digital Product Passport), support decision making (LCA cascade potential calculator, Material volume estimation), and optimise the supply chain (Supply Chain management tool for the optimisation of upcycling of secondary materials). The **Supportive Framework in Policy, Market, and Skills (Pillar 4)** will address non-technological aspects to ensure that the project's innovations are effectively implemented and supported at all levels. In particular, **a)** the **W2W skills development strategy** will provide targeted training programmes to equip workers with the necessary skills for the uptake and deployment of the project technical innovations; **b)** the **W2W policy recommendations** will help shape the regulatory environment to support the implementation of W2W principles and technologies in the EU context; **c)** the **W2W Business Models** will support the commercialisation of the W2W outputs; and **d)** the **Contribution to Standardisation and Certification** will help to promote the adoption of project results by providing a common language and framework for communication between different stakeholders. By addressing these categories in an integrated manner, the W2W project can effectively work towards developing a comprehensive, innovative, and sustainable approach to the circular management of wood-containing waste in the European Union.

The different components of the W2W approach are presented below.

Pillar 1: Advanced Separation and Sorting of Mixed Waste from CDW

An advanced multilayer sorting system of mixed waste from CDW will be developed in W2W, such as the one illustrated in Figure 1.2. Multi-sensor systems (i.e., Hyperspectral, MWIR, X-ray, RGB cameras, NIR, FTIR) will be used in critical spots of the sorter to classify the type of CDW assisted by machine- or deep- learning techniques (e.g., neural network models) while the separation of waste stream or removal of the negative compounds will be

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succeeded using robotics arms, collaborative robots and humans. In contrast to available commercial CDW sorters[35], the proposed system will dynamically adapt the separation thresholds based on LCA results and valorisation requirements, while the feature of human-machine collaboration for disassembly needs[36] will be explored and evaluated, increasing the overall sorting efficiency under the Industry5.0 human-centric pillar. In operation mode, unsorted CDW will be transported onto a conveyor belt by a feeding and vibration system to ensure an even spread across the width of the input conveyor. Utilising the multisensor system, the position of the wastes within the input conveyor will be determined and the valorised category will be selected in order to drive the robotic system. Specifically, the wastes will be separated by means of a robotic system into three distinct flows, consisting of: **(i) Wood waste** containing metallic or assembly elements (e.g., nails, staples, wheels or others) will be directed towards a human-robot collaborative (HRC) separation line, leveraging the increased capabilities of HRC units to deal with complex elements in comparison to fully-autonomous ones. After the assembly elements are removed, wood elements re-enter the sorting line and follow the **pure wood waste** pathway. **(ii) Pure wood waste** after entering a shredding system to reach optimal particle size, will go through a second characterisation stage to be separated according to the presence of foreign materials. Pure wood will then be directed according to its valorisation properties into four bins, following the developed classification. **(iii) Other C&D waste** will exit the plant and be redirected for further processing. A study will also be performed on glass waste separation based on the recycling identification thresholds, aiming to assess the replication potential of the developed system to the glass sector. In addition, a glass adaptive sorting mechanism will be developed and evaluated in terms of performance and replication in the glass sector.

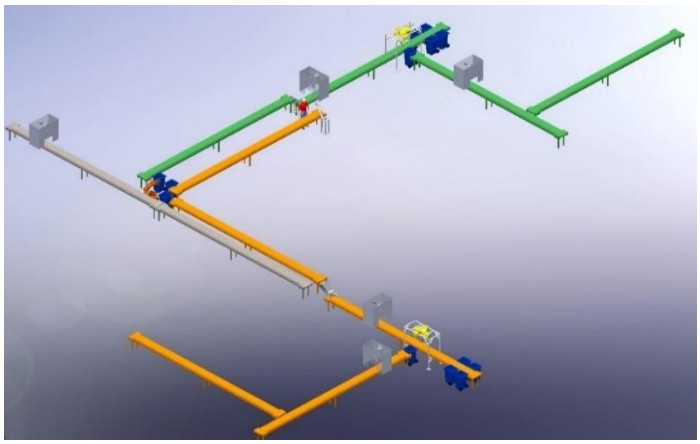

Figure 1.2: Advanced multilayer sorting system of mixed waste

Pillar 2: Upcycling Processes and Technologies

The upcycling technologies are presented as part of the 3 project Use Cases in Section 1.2.3.

Pillar 3: Digital Tools for Improving Circular Flows of Secondary Materials

(A) **Life Cycle Sustainability Assessment Tool for Cascade Pathways**: W2W will develop a tool to predict and assess the impact of different cascade paths or sequences for possible further uses of wood products and their materials. Different scenarios will be constructed based on wood type, treatment, transportation, and end-of-life management The tool will rely on three main data sources: **(i)** the Digital Product Passport for upcycled products, **(ii)** a digital product database, and **(iii)** a process database for circular actions and their impact on products. Data on how the different uses of the secondary resource occur over time will be considered to perform a dynamic LCA, making it possible to compare different paths, use a sustainable-by-design approach for the design of new products, and assist in investment decisions.

(B) **Material Volume Estimation**: This tool will enable volume estimation and material recognition in mixed waste by using computer vision methods. It will use multisource data such as images from conventional RGB cameras, ground and aerial images. Semantic segmentation of the waste will be performed to characterise the waste material.

(C) **Digital Product Passport**: The W2W Digital Product Passport (DPP) will be a digital document containing detailed information about wood-containing product's materials, components, and environmental impact throughout their lifecycle. It will provide information about the type, origin, and treatment of wood used in a product, and will help stakeholders identify the most suitable wood materials for cascading uses, as well as track the flow of wood waste in the supply chain. It will also include information on the product's environmental impact throughout its lifecycle, including resource extraction, manufacturing, transportation, use, and end-of-life management. Eventually, the W2W DPP will facilitate collaboration and information sharing among different stakeholders involved in wood waste management, such as manufacturers, waste handlers, recyclers, and researchers.

[35] Sarc, R., Curtis, A., Kandlbauer, L., Khodier, K., Lorber, K. E., & Pomberger, R. (2019). Digitalisation and intelligent robotics in value chain of circular economy-oriented waste management–A review. *Waste Management*, *95*, 476-492.

[36] Liu, Q., Liu, Z., Xu, W., Tang, Q., Zhou, Z. and Pham, D.T., 2019. Human-robot collaboration in disassembly for sustainable manufacturing. *International Journal of Production Research*, *57*(12), pp.4027-4044.

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(D) **Supply Chain Management Tool for the optimisation of upcycling of secondary materials**: The Supply Chain Optimisation tool will use advanced algorithms and data analysis techniques to simulate various scenarios to assist companies in identifying the most cost-effective and efficient supply chain design and operation for improving the flow of secondary materials throughout the recycling process. This involves determining the optimal inventory levels of secondary materials at different stages in the supply chain to meet customer demands while minimizing expenses, as well as optimising transportation by selecting the appropriate modes of transportation and carriers, managing routing and scheduling. In addition, users will be able to generate custom reports and track performance metrics over time, enabling them to identify areas for improvement and continue optimising the recycling process.

(E) **Integrated Circular Wood Upcycling Platform**: This platform will integrate the digital tools developed in the project (Life Cycle Sustainability Assessment Tool for Cascade Pathways, Supply Chain management Tool for the optimisation of upcycling of secondary materials, and Digital Product Passport) into a common multifunctional and interoperable platform. The platform will enable: **i)** Processing and analysis of data from various stages of the wood lifecycle for identifying patterns and trends that can be used to optimise cascading use strategies and minimise environmental impacts; **ii)** Predictive modelling of the environmental impacts of different wood waste cascading scenarios; **iii)** Integration with Geographic Information Systems (GIS) to analySe the spatial distribution of wood waste sources, transportation routes, and processing facilities; and **iv)** Real-time monitoring of wood waste cascade strategies and adaptive management.

Pillar 4: Supportive Framework in Policy, Market, and Skills

(I) **Framework for CDW Cascade Valorisation**: W2W will offer two different frameworks; the first framework will introduce policy recommendations for recycled wood waste classification and proposals for a common European Framework (which will aim to harmonise the current national frameworks presented in Table 1.1). The proposed framework will strengthen the wood valorisation by transforming the existing streams to improved -in terms of quality, purity, and usability- by creating new flows of secondary materials driven by the improved up-cycling technologies. The second framework will contribute to policy impacts for the optimal cascade valorisation of recycled wood through the multi-dimensional approach (time, value, function) throughout its lifecycle, with a focus on maximising its economic, social, and environmental benefits while minimising its negative impacts. The project will produce a policy brief that can provide guidance for policymakers and stakeholders seeking to promote sustainable waste management practices, the development of a circular economy and at a longer term, to pave the way toward sustainable-by-design for circular products.

Table 1.1: National schemes for wood waste classification

Class A	Class B	Class C	Class D
AT, BA, CA, FR, NL, UK	**AT, CA, FR, FI, GE, NL, SL, TR, UK**	**AT, CA, TR, NL**	**CA, FI, FR, GE, NL, Sl, SW, UK**
Unpainted & *untreated* wood, without or minor defects and a few restrictions on use	Partially contaminated (painted, varnished, coated, glued etc.) wood *without preservatives or halogenated compounds*	*Hazardous* waste wood – containing hazardous or toxic substances (e.g., paint, varnish, stain) or treated with *halogenated* organic materials	Wood waste treated with wood *preservatives* containing hazardous substances (e.g., copper and chromium or copper, chromium, arsenic, and creosote)
FI, GE, SL, TR, SW	**BA, Switzerland**	**BA, FI, FR, GE, SL, SW, UK**	**AT, BA**
Pure wood, only *mechanically treated*, insignificantly contaminated with harmful substances	Wood of average to A class quality, *without specific requirements* for pure wood	*Non-hazardous* waste wood with low-concentration chemical additives and organic halogenated compounds in the coating, with *no wood preservatives*	Wood waste that could not be assigned to other categories or components that are not wood waste (e.g., furniture with less than 50 mass -% wood)

(II) **Upskilling**: W2W will provide timely access to relevant twin transition skills data and information - providing policy makers, employers and workers with expert advice to make informed decisions about training and career paths. This evidence will form the basis for innovative skills approaches, in the form of interventions tailored to effectively support the workforce through engaging the twin transition skills ecosystem. This covers also the dimensions of capacity building for skills development providers and engagement with policy makers to support the embedding of leading practices into policies and training programmes. Upskilling impacts will be attained at the levels of practical labour market and skills information, future-proofing career pathways, and providing agile

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responses to skills needs. W2W emphasises providing the skills ecosystem actors with timely, accurate and practiceoriented data to anticipate changing skills needs and shape evidence-based interventions. It proposes novel approaches for the design of future-oriented, high-quality skills training. It deploys a certification process that is focused on equipping workers with required skills.

(III) Contribution to Standardisation: W2W will develop a strategy for effectively contributing to the standardisation and certification of secondary materials. Through interaction with relevant committees, candidate topics for contributions will be identified. W2W will contribute to the development of new standards, submit proposals for modifying existing standards, and promote the elaboration of new standards. Fast-track standards like CEN-CENELEC Workshop Agreements will be considered, as the best-suited option for such research projects.

(IV) Business Models: The project will co-design sustainable business models for the rollout of its innovative sorting, separation, and upcycling technologies, building on the evidence provided by the use cases and market research. The process will utilise the Triple Layered Business Model Canvas, complemented by lean value proposition design.

1.2.2 Scientific Methodology

The W2W concept relies on the implementation of a circular approach for the whole lifecycle of secondary resources. Circularity principles require integration of the entire value chain, implementation of detailed monitoring and sorting systems and assurance of increased use of secondary materials, as well as the integration of secondary resources in the processes. These requirements imply the execution of a comprehensive methodology that follows an iterative multistep development approach, in which user requirements and corresponding solutions evolve through multiple iterative collaboration rounds involving all the stakeholders. The core of the methodology is agile approach, adaptive development and continuous improvement based on iterative rounds including the **following 4 reciprocal phases**.

Phase 1: Data, Digital Assets and Wood Valorisation Framework (M01-M08)

Main
- To develop wood classification thresholds and a common valorisation framework
- To map the stakeholders of the defined value chains and their data exchange needs

Activities
- To set an impact assessment methodology

The first phase of the project sets the baseline for its execution. A preliminary analysis of the secondary materials quality characterisation is done aiming to develop a common framework for the characterization and classification of recycled wood. Taking inro consideration and in parallel with the developed classification thresholds A framework for CDW recycle wood will be defined setting the basis for the cascade in value valorisation that W2W is envisioning. **DRAXIS** will be responsible for the mapping of interaction between actors and stakeholders as well as for the initial definition of impact assessment methodologies and KPIs to assess the developed framework. All the needed data exchange between the stakeholders, with FAIR principles, will be defined and retrofitted for development for the next phases of the project.

This phase is performed the first 8 months of the project and covers WP1, WP4, WP16, and WP19).

Phase 2: Design and Alpha version Development phase (M09-M18)

Main
- To design and develop at TRL5 the management of waste streams
- To design and develop at TRL5 the upcycling and valorisation technologies for the 3 cases

Activities
- To design and develop at TRL5 the digital tools of W2W

The outcomes of the thresholds, framework and KPIs will be retrofitted to the partners responsible for the development of the technologies, ensuring with the participation of the end users that no quality implications to the end products may be arisen.

P2.1 Management of Waste Streams TRL5: At first, an analysis shall be performed with the participation of Public Authorities (**RCM**) and Private Recycling Center (**ELAK**) about the processing of CDW at multiple levels. Based on the results a modular CDW management system will be developed allowing the installation at various sites. **ICCS** will set up the first system on its premises allowing the processing of small quantities of CDW and being able to provide the feedstock to all the related partners developing the upcycling solutions.

P2.2 Upcycling and valorisation technologies TRL5: Use case #1: **UHE** shall receive the feedstock coming out the sorting done by **ICCS** and produce the first samples of lignin and cellulose fibers. **NTUA** will receive from **LERMAB** the samples and produce the first pellets with biopolymers. Use Case#2: **LERMAB** shall receive classified wood from **ICCS** as well as furniture feedstock coming from **ECOM** and shall perform the first steam explosion tests at their facilities. The first remediation tests will also start. **UHE** shall perform the first tests and

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validation of the liquefaction process. Use Case#3: **CIRCE** shall receive contaminated wood as feedstock and perform HTC at lab scale and then the first sample will be gasified. **KIV** shall receive the first bottles of syngas to perform a TRL5 validation. First sample of pure dodecanol will be provide to P&G for validation.

P2.3 Digital tools TRL5 validation: The development of all the digital tools will be initiated and the architecture for integration to the W2W platform will be defined.

This phase takes place from M10 to M18 and covers WP1), WP5, WP7, WP9, WP11, WP13, WP16, and WP19.

Phase 3: TRL6 Development and Implementation phase (M19-M36)

Main Activities
- To upgrade at TRL6 the management of waste streams
- To upgrade at TRL6 the upcycling and valorisation technologies for the 3 cases
- To upagrade at TRL6 the digital tools of W2W

P3.1 Management of Waste Streams TRL6: The sorting system at **ICCS** premises will process higher volumes of feedstock provided by **ELAK** and **ECOM** having all the components of the system in place.

P3.2 Upcycling and valorisation technologies TRL6: Use case #1: **UHE** will increase the production of fibers and **NTUA** will produce the first pellets that will be delivered to **FOCCHI** for validation. Use Case#2: **LERMAB** will upscale to TRL6 the equipment and provide the sample for the production of the wallboards. The first tests of the produced boards shall be performed. Use Case#3: The system will be upgraded to TRL6 and higher volumes will be achieved.

P3.3 Digital tools TRL6 validation: The W2W platform will be made available at TRL6 to all the partners allowing the first experimentation of all the developed tools.

This phase takes place from M19 to M36 and covers WP2, WP6, WP8, WP10, WP12, WP14, WP17, and WP20.

Phase 4: Pilot Activities and Validation (M37-M48)

Under this phase, the end-users of the produced products (**FOCCHI**, **P&G**) will receive the end products and evaluate their performance. The whole system will continue to operate performing a full impact assessment based on the validation results to the developed valorisation framework. Final conclusions from standardisation and certification activities will be drawn. Moreover, once the final results have been gathered together with feedback from main stakeholders, corresponding policy recommendations final business models, enhanced learning programs and further exploitation strategy will be defined.

This phase takes place from M37 to M48 of the project and covers WP3), WP15, WP18, and WP21.

Training and upskilling activities will be taking place alongside the whole duration of the project with a continuous improvement of the context based on the project results. The D&C activities will use as inputs the outcomes obtained in each WP - a detailed plan of D&C is presented in Section 2.2.

1.2.3 Overview of the Sustainable Value Chains of Secondary Materials

Use Case #1: Cascade Refinement Technologies for Wood Waste Upcycling

This Use Case involves a four-stage process (Figure 1.3) to produce composite building materials from wood waste: **i)** Production of lignin and cellulose fibers from the wood waste (**UHE**); **ii)** Production of pellets by combining the produced fibers with biopolymers (**NTUA**); **iii)** Pellets are formed into building material components (**NTUA**); and **iv)** Testing of the new building materials in mock-ups (**FOCCHI**).

Material Input: #Class B wood waste: lightly admixed products, **#Class C wood waste**: heavily admixed products

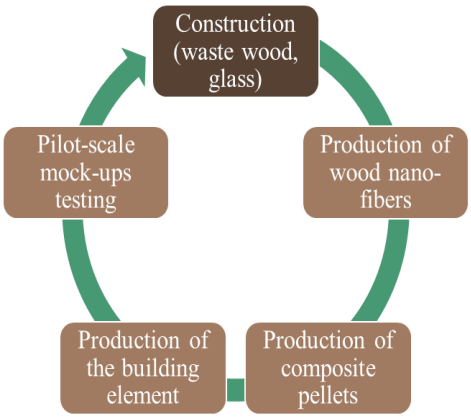

Figure 1.3: Process for cascade refinement

Challenges to be Addressed: There are significant challenges to be addressed for sustainable production of composites reinforced with natural fillers deriving from plant fibers. First, not all polymers are compatible with wood fibers; for example, polyolefins are not the most appropriate biopolymer matrices for natural fibers because of their nonpolar character. On the other hand, the use of more polar thermoplastic matrices such as PMMA (Poly(methyl methacrylate)), PVC (poly(vinyl chloride)) and biodegradable polyesters is an alternative to obtain compatibility between natural fibers and a matrix even without compatibilizers

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or special fibres treatment[37,38]Another challenge is that the fibers deriving from wood waste do not always meet the required specifications due to the fact that their length is shorter than required and additionally are contaminated with retained debris such as plastics, glues, paints, fungicides, etc. These limitations make it difficult to integrate wood fibers produced from wood waste into the industrial manufacturing cycle.

W2W Solution: #Production of lignin and cellulose fibers from wood waste: The use of reinforcement filler derived from natural waste will permit to overcome the handicaps that prevent the use of PLA from a large share of the construction market. Protocols for isolation, purification and characterization of fillers as well as standards for modification/functionalisation will be defined. Cellulose nanofibers (CNF) from cellulose pulp will be extracted by mechanical and/or chemical treatment (unmodified and modified CNF) and will be used as filler for the selected polymer. The functionalised or unfunctionalized fillers will be incorporated into PLA via melt-compounding in lab scale using internal mixing chambers, followed by a compression-molding process to produce samples for further characterization. Mechanical and physical properties of these resulting PLA-based composites will be determined. As the ultimate step, the UV-ageing of related PLA composites will be considered as a key to produce materials with improved outdoor resistances against light. #Production of biocomposites reinforced with wood fibers: Within W2W, the matrix and fiber compatibility issue will be addressed using a PLA polymer matrix in combination with fibers from wood waste that will act as biocomposites' fillers. As soon as the fibers are extracted from wood waste, they will be integrated in PLA biocomposites to improve PLA thermal and mechanical properties. Optimisation of compounding conditions in melt mixers and twin-screw extruders and testing of different types of commercially available PLA and PLA-copolymers will take place. The hydrolytic stability of PLA-based materials will be considered and -as this property is related to the COOH- two different approaches will be adopted to improve it. Accelerated thermal and hydrolytic ageing studies will be carried out by **NTUA** to assess ageing kinetics with the aim of lifetime prediction modelling.

Obtained End Products: The end products derived from the process mentioned above will be biocomposite building materials produced by wood nano-fibers, glass fibers and biopolymers.

Product Validation and KPIs: Overall, the validation process of the new composite material in the façade industry will be done through a comprehensive approach that considers different aspects such as environmental impact, performance, and market potential. By integrating the composite component in a façade prototype that will be developed, based on physical and dimensional features of a composite panel, **FOCCHI** will demonstrate the benefits of the new material and its potential to replace traditional products such as gypsum board. The comparison with "traditional" façades will be performed using specific KPIs such as thermal performance, fire resistance, and mechanical strength. Acoustic tests will be performed to provide valuable information about the sound insulation properties of the composite panel. The results of these tests will be used to compare the composite acoustic performance with other products available on the market. The tests will be conducted according to current regulations guidelines, in certified laboratories. The prototype will be installed in a real case room in which microphone sensors will monitor the acoustic performance.

Increase of secondary raw materials use	Waste reduction	GHG emissions reduction	Up-cycling, reuse and upgrade secondary raw materials technologies implemented	By-products value increase
30%	30%	20%	3	>50%

Use Case #2: Chemical and Bioremediation Technologies for Wood Waste Upcycling

This Use Case involves wood pollutants removal through steam explosion, detoxification of the aqueous effluent from the steam explosion and liquefaction of mixed waste with organic catalysts for polyols production, following the process illustrated in Figure 1.4. **LERMAB** is focusing on the valorisation of #1 Wood containing Adhesives/Pollutants with potential of removal, whereas **UHE** is focusing on the liquefaction of #2 Mixed wood waste with the usage of various catalysts. **ECOM** is the provider of feedstock regarding furniture material, whereas the CDW will be provided as a result of the W2W sorting process.

[37] Le Digabel F., Boquillon N., Dole P., Monties B., Averous L., (2004). Properties of thermoplastic composites based on wheat-straw ligonecellulosic fillers. *Journal of Applied Polymer Science* 93, 428-436

[38] Averous L., Le Digabel F., (2006). Properties of biocomposites based on lignocellulosic fillers. Carbohydrate Polymers, 66, 480-493.

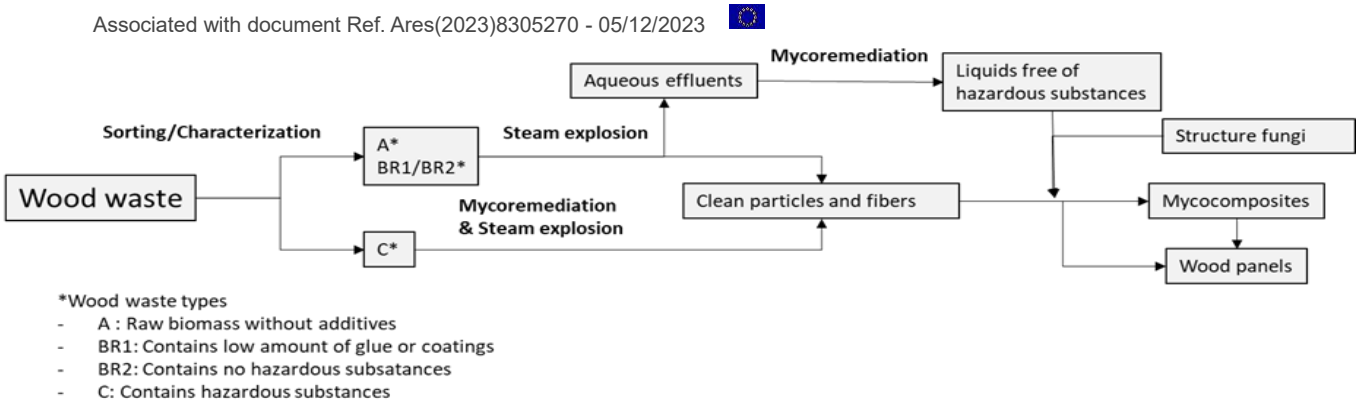

Figure 1.4: Chemical and bioremediation for wood waste upcycling process

Material Input: # **Class B** wood waste: lightly admixed products, # **Class C** wood waste: heavily admixed products

Challenges to be addressed: The main contaminant of recycled wood is the glue used in products, such as particleboards, plywoods, fiberboards, etc. Most commonly, urea-formaldehyde (UF) glue is used, alone or reinforced with small quantities of melamine. The presence of residual glue on the surface of the wood fibres or particles makes their recycling more difficult, especially for the production of new wood panels. The cross-linked glue particles do not react with a new adhesive to form a polymer network during a second cross-linking step, resulting in a significant drop in the properties of the materials formed from recycled panels. Moreover, the end of life of fiberboard panels (MDF, HDF, etc.) has become a major issue due to the occurrence of UF resin which prevents the energy recovery or the integration of fibers in particleboard, making these materials non-exploitable.

W2W Innovation: The steam explosion process (Figure 1.5) will be used in combination with mycoremediation to remove substances of special concern such as biocides used in wood preservatives. Wood waste will be screened for occurrence of preservatives and UF resins. Wood pieces containing hazardous substances (BR2 and C class) will be separated from those containing UF and coating (BR1) or free of additives (A class). Then the wood waste will be ground into 8mm particles. A differential two steps process will then be setup according to the type of wood waste. *#Fermentation and Fungal growth for Detoxification of hazardous wood:* Wood waste belonging to BR2 and C classes will be firstly subjected to mycoremediation to remove hazardous substances. Two ways will be explored for mycoremediation: (a) Solid state fermentation, (b) Submerged fermentation. *#(Physicochemical) steam explosion will be employed for the treatment of wood waste (post consumers panel board, hazardous wastes):* In a second step, the fungal pre-treated wood will be treated by steam explosion to recover previously desorbed pollutants in the liquid phase. The resulting "cleaned" wood particles will then be integrated in new panel board production. *#Detoxification of aqueous effluent (from steam explosion) and biological (fungi) processes:* Detoxification of aqueous effluent from (SE) treatment performed on wood waste belonging to A and BR1 classes will be performed by submerged fermentation. Filamentous fungi, using hazardous substances as carbon source, will be utilised for the submerged fermentation process. Process optimisation and validation will take place. *#Liquefaction of mixed Waste with organic catalysts:* For cases where mixed waste cannot be separated, the production of polyols from the produced biomass will be exploited. Polyols will be synthesised using two approaches: liquefaction process and using ring-opening polymerisation. Liquefaction is thermochemical process carried out with a solvent, in acidic or alkaline media, with or without the presence of a catalyst at high temperatures. The most commonly employed solvents are polyhydric alcohols as polyethylene glycol, ethylene glycol and glycerol. Strong acids or bases are typically used as catalysts, especially sulfuric acid, which can reduce the temperature required for reaction.

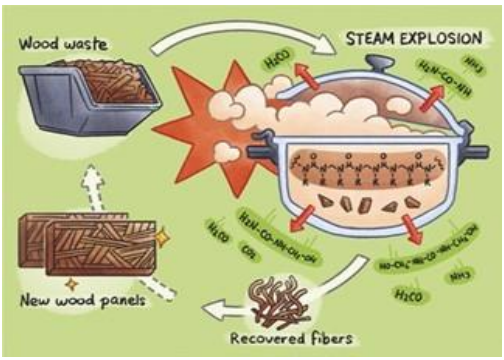

Figure 1.5: Steam explosion process

Obtained Secondary Raw Materials: As part of the pilot activities the developed system is expected to treat up to 20kg/day producing: *#1 Wood without Adhesives/Pollutants.* Classified in Category A, that could be used for any alternative flow, and *#2 Polyols from liquefaction.*

Product Validation and KPIs: The recovered wood fibers and particles will be characterised, while one-layer and three-layer particleboards will be produced and fully characterised regarding their internal bond strength, density, MOE, MOR and swelling. In addition, innovative lightweight panels with incorporated mycocomposites (with no VOC emissions) will be produced and then characterised in terms of internal bond strength, density, MOE, MOR and swelling. Finally, the obtained polyols will be used for the synthesis of non-isocyanate polyurethanes (NIPUs) adhesives through polycondensation reactions while the properties of obtained NIPUs will be also determined.

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Increase of secondary raw materials use	Waste reduction	GHG emissions reduction	Up-cycling, reuse and upgrade secondary raw materials technologies implemented	By-products value increase
30%	30%	70%	3	>50%

Use Case #3: Energy & Gas Valorisation Technologies for Contaminated Wood Waste

This Use case relies on the conversion of liquid and solid waste from different industries (construction, paper, urban, etc.) into building blocks (CO2 and H2) for further production of valuable compounds (detergents) that can be used in other industry sectors. The liquid waste will be pre-treated through a novel technology, microwave hydrothermal carbonization (HTC-MW). During the HTC-MW treatment (**CIRCE**), the carbon contented into the waste is densified producing hydrochar that has a higher heating capacity than the original feedstock. Furthermore, the hydrochar contains high dewaterability properties reducing the high energy demand steps to remove water from the liquid wastes. Hydrochar and dry solid waste will be used as feedstock for gasification (**CIRCE**) to produce H2 and CO2 (building blocks). CO2 and H2 will feed the novel Biotech reactor (**KIV**ERDI BV) in order to produce long chain carbon compounds (detergents) that are valuable in different industry sectors.

Challenges to be Addressed: Contaminated wood containing hazardous substances cannot be recycled or disposed of in landfill, making energy recovery its primary use. Moreover, handling the large volume of by-products (e.g., ash, wastewater) from wood waste thermochemical conversion processes remains a severe challenge from both environmental and economic perspective. Hydrothermal carbonisation is a well-known process for converting biomass into valuable products and is already commercialised mainly for sewage sludge treatment at the wastewater treatment plants[39]. However, there is no evidence of the HTC application as part of another waste management process in a commercial scale. Wood waste is an interesting feedstock for the hydrothermal conversion due to its high content of carbon and low content of metals. The potential advantages of using HTC as a process to treat wood waste are: **1)** The process water can be used for nutrient recovery and further biomethane production and, **2)** The hydrochar can be used as a material to produce syngas.

W2W Innovation: For contaminated wood waste, a valuable cascade process is suggested where hydrothermal carbonisation (HTC), gasification and anaerobic digestion technologies, along with up-cycling biotech process of gasification emissions (H_2 and CO_2) into dodecanol will taking place, as depicted in Figure 1.6. The hydrochar is a stable and carbon pure product that increases the quality of the syngas by reducing the production of undesired side gases (NOx, SOx, etc.) and reduces the formation of tars during the gasification process. The dodecanol production (for low impact detergents application) using as feedstock the syngas deriving from HTC can be used to close the cycle of contaminated wood valorisation. *#Hydrothermal Carbonisation through MW will be employed at lab (TRL 4) and HTC reactor (TRL6):* Mixed waste (contaminated wood waste and liquid waste e.g., sludge, MSW etc.) will undergo treatment through HTC at lab scale (TRL4) and validated in a TRL6 HTC reactor. A more advanced HTC process will be applied which will include microwave (MW) heating bringing advantages such as **(1)** Fast and uniform heating; **(2)** Selective heating; **(3)** Treatment of non-homogeneous wastes; **(4)** Energy efficiency; **(5)** Overall cost effectiveness/savings; **(6)** Improved process control; and **(7)** Ability to operate from an electrical source in comparison with the conventional HTC treatment. Hydrochar and rich in nutrients and organics process water will be produced. *#Gasification (TRL ≥ 6) of wood waste and hydrochar produced from HTC:* Then, gasification of hydrochar produced from HTC will be employed at a reactor (TRL ≥ 6). *#Anaerobic digestion from the process water from HTC:* The process water from HTC treatment is a rich in organics liquid that can undergo

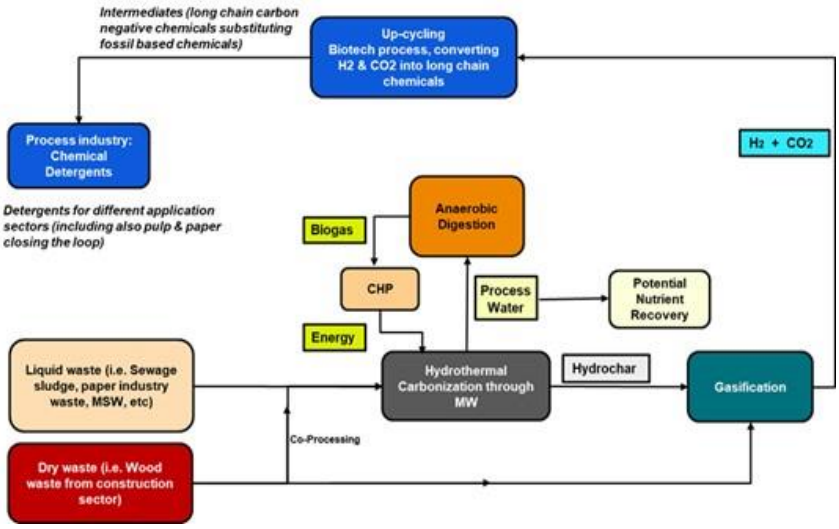

Figure 1.6: Energy & gas valorisation for contaminated wood waste process

[39] S. A. Alavi-Borazjani, I. Capela, and L. A. C. Tarelho, "Valorization of biomass ash in biogas technology: Opportunities and challenges," *Energy Reports*, vol. 6, pp. 472–476, 2020, doi: 10.1016/j.egyr.2019.09.010.

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throughout anaerobic digestion for biogas production with higher methane content. **CIRCE** will apply selected prediction models to estimate the biomethane production from the process waters generated from the HTC treatment based on their characterisation. *#Biotech-driven up-cycling of syngas into dodecanol for low impact detergents application:* **KIV** has developed a proprietary microbial conversion platform allowing the use of syngas (H2 and CO2) as feedstock production pathways. The process uses CO2 and H2, an inexpensive nitrogen source, and mineral nutrients, for dodecanol production. At first, genetic and analytical tools will be used to improve dodecanol production by reengineering fatty acid biosynthesis and fatty acid beta-oxidation. Metabolic modelling and 13ºC metabolic flux analysis will be applied to support the enhancement of the microorganism strain (*C.necator*) towards the desired metabolic pathway. Following, tests at 20Lt gas fermentation scale (TRL4) using syngas simulating detailed composition provided by **CIRCE**, will take place. The progressively improved strains will be used during fermentation trials in order to achieve targeted productivity and titer. First samples (500gr- 1 Kg) of pure dodecanol and exhausted biomass cells will be provided to **P&G** for pre-validation. Then, feedback will be provided to **CIRCE** (from pre-validation process) regarding potential contaminants in the syngas which may determine productivity reduction. **CIRCE** and **KIV** will set-up the appropriate decontamination measures. The samples from TRL5 campaign will be provided to **P&G** for preliminary feedback and characterisation (odour, colour, stability, impurities content, compatibility with other chemicals for detergents formulation, etc.) to provide feedback for fine tuning downstream process. Based on the obtained feedstock, during phase 2 the overall process (upstream and downstream) will be scaled up at 100 Lt using as feedstock bottles of syngas (H2 and CO2 the latter also deriving from CHP) from **CIRCE** process and leveraging the access from **KIV** R&D team to upstream and downstream facilities located in Spain. Samples from TRL6 tests (kg scale) will be then progressively validated into formulation of low impact detergents leveraging a micro falling film pilot reactor (TRL6) owned by **P&G** in its Brussels centre that allows to convert fatty alcohols in ionic-surfactants for specific application in detergency (e.g., laundry detergents, fabric enhancers and dishwashing liquid) including small scale consumer exposure tests for validating technical and safety performances. **KIV** will perform process adjustments through a feedback loop if necessary.

Obtained Intermediate Products: *#1 Hydrochar produced from HTC and being gasified for H_2 and CO_2 production; and #2 H_2 & CO_2 produced from hydrochar gasification process + exhausted CO_2 from CHP being converted into dodecanol + protein rich biomass via low impact biotech route.*

Obtained End Products: *#1 Chemical Detergents for different application sectors, and #2 Nutrients recovery from HTC i.e., recovery of ammonia from process water.*

Product Validation and KPIs: Productivity > 1.5g/lt h; titer of >50g/L; validation of purified syngas (CO2/H2) derived dodecanol and aminoacids from exhausted biomass into at least 2 detergents formulations proving: **i)** the fulfilment of technical and safety requirements (properties identical to those obtained from standard dodecanol; **ii)** environmental sustainability (GHG savings: 50% compared to fossil based dodecanol and 70% compared to dodecanol obtained from palm/coconut oils from outside EU equivalent to 1.5 ton of COeq saved/ton of dodecanol; capturing and utilisation of 2.85 ton CO2/ton dodecanol); **iii)** cost-competitiveness (price not higher than current price of standard dodecanol in the market).

Increase of secondary raw materials use	Waste reduction	GHG emissions reduction	Up-cycling, reuse and upgrade secondary raw materials technologies implemented	By-products value increase
50%	30%	70%	3	>50%

1.2.4 Replication in Other Sectors

W2W is setting the base of the cascade valorisation approach tackling resource usage with efficiency and higher value. W2W is mainly targeting the polluted wood from industrial and urban waste but in parallel is targeting a twofold replication strategy based on: **i) Direct Technology Transfer in other sectors:** W2W technologies shall be validated in wood originating from CDW and used furniture but can occur in the wood processing and pulp paper industries as well. In lower quantities, it may occur in all sectors due to wood packaging or similar usages. These sectors can directly apply the whole proposed technologies. In addition, W2W offers novel approaches for the valorisation of ashes from CDW which can also be applied in ashes originating from any thermal or combustion process having a direct impact in the majority of the P4P sectors. **ii) Conceptual Replication of Cascading concept:** A study of the proposed cascade framework shall be performed for the replication in other sectors, starting from the glass sector. All the developed digital tools shall have a generic approach offering the direct replication in additional sectors. The separate collection or sorting of glass in multiple classes, the automatic separation of them based on recyclability sectors shall be performed, offering the needed input for the glass replication scenario. Glass is not part of the current P4P partnership, but W2W aims to give the required incentives through a complete replicability study.

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1.2.6 Positioning of W2W with Respect to Other Projects and Innovation Activities

Relevant Objectives	Added Value in W2W
DigInTrace (Horizon Europe), Jan 2023-Dec 2026 – **ICCS** (Project Coordinator), **NTUA**, **CIRCE**, **IRIS**	
Creation of a decentralised traceability platform with innovative tracking and sorting techniques, updated quality certification schemes, AI-based decision-making mechanisms.	The outcomes will facilitate the creation of sophisticated separation and sorting systems and associated digital tools leveraging AI, resulting in efficient circular resource valorisation.
CORALIS (H2020), Oct 2020-Sep 2024 – **CIRCE** (Project Coordinator), **ICCS**	
Pathways creation for resource decarbonisation in energy intensive sector through implementation of viable IS approaches.	**CIRCE** and **ICCS** will use knowledge for circular resources valorisation, monitoring systems and IS data sharing and management.
REDOL (Horizon Europe), Dec 2022-Nov 2026 – **CIRCE** (Project Coordinator), **ICCS**, **IRIS**	
Establishment of circularity hubs in urban areas with zero waste policies by optimising value chains with digital tools.	Project findings will support the creation of effective values chains in relation to W2W developed technologies.
WoodCircus (H2020), Nov 2018-Dec 2021 – **VTT**	
To promote wood-based value chains as part of circular bioeconomy.	Results are findings from the project will be used towards the development of the cascade valorisation framework and the skills development strategy
SHAREBOX (H2020), Sep 2015-Aug 2019 – **IRIS**	
Secure Management Platform for Shared Process Resources for flexible energy use and material flow integration.	W2W will learn from the project experiences how the digital solutions can provide robust and reliable information in realtime to share resources
ACCELWATER (H2020), Nov 2020-Oct 2024 – **ICCS, NTUA**	
Offers a holistic WaterEnergyFood nexus framework in food industries for more costeffective use of resources.	**NTUA** is involved in protein recovery from waste and byproducts, and application development expanding it towards the new materials developed in W2W
REHAP (H2020), Oct 2016-Mar 2021 – **VTT**	
Transformation of agro-forestry residues into new sustainable and eco-efficient derivatives through innovative cost-effective processes to demonstrate their potential use in several	Project findings will be used towards methods development for further wood valorisation, and in building components evaluation. Additionally, the implementation of Life Cycle and Cost Assessment, and waste management strategy and

products at an industrial level and their application in sectors like Construction.	measures for future standardisation will be used for the developm of the W2W cascade valorisation framework. Associated with document F Ares(2023)8305270 - 05/12/2023
DIMAT (Horizon Europe), Jan 2023-Dec 2025 – **CERTH** (Project Coordinator), **UPV**, **DRAXIS**	
Digital Technologies for modelling, simulation and optimisation in the material value chain with data analysis services and visualisation techniques for improving quality, sustainability, effectiveness, and competitiveness of materials.	The DIMAT Cloud Materials Database structure will provide input to the W2W DPP. The DIMAT Materials Environmental and Cost Life Cycle Assessment will used as a basis for the W2W Life Cycle Sustainability Assessment Tool for Cascade Pathways.
BASAJAUN (H2020), Oct 2019-Mar 2024 – **FOCCHI**	
Aims to demonstrate how wood construction chains can be optimsed to foster both rural development and urban transformation.	The experience on optimising the façade system made of biobased composites to respond to market needs will be used in W2W.
POP-MACHINA (H2020), Jun 2019-May 2023 – **CERTH**, **QPLAN**	
To promote environmental sustainability and generate socio-economic benefits in European cities through the circular production paradigm.	The W2W business models will utilise the experience gained through the accelerator programme for entrepreneurship support and development of POP-MACHINA.

1.2.7 W2W Need for a Trans-disciplinary Approach Positioning

To promote overall sustainability, new circular solutions for secondary materials and waste management, as well as the increased use of secondary resources in the process industry, the project must consider many different aspects, thus require the joint involvement of different scientific domains. W2W works in a strong inter-disciplinary field, as it involves expertise and methods from various disciplines including natural and applied disciplines, such as chemistry and chemical engineering, physics, ICT, data science, electrical and mechanical engineering together with social sciences, such as sociology, economics, political science, ethics, law, business, communications, training and learning, governance, etc. These are interwoven within the W2W activities and will be brought together and integrated in pursuit of the project objectives forming inter- and trans-disciplinary conditions, used to develop innovations in different fields, and support other activities related to policy influencing, certification, business case and learning programs for skills development. W2W will also actively pursue the involvement of all the actors in the value chain to consider the complex requirement of the specific markets in terms of standardisation, norms constraints, and performances. W2W recognises the difficulties of a trans-disciplinary collaboration (e.g., different language, terminology among different disciplines and methods of work, different levels of scientific understanding and cognitive structures etc.) and adopts a proactive approach towards them by planning to follow the fundamentals of the Methodology in Interdisciplinary Research framework[40]. At WP levels, the consortium will employ a quality assurance process (supported in WP1-3) to ensure that all trans-disciplinary aspects are integrated in the developed solutions, thus creating a meta-knowledge about all the different disciplines, methods and epistemologies, and purposefully and reflectively integrate and synthesise different perspectives, to better understand and contribute to new processes for circularity of secondary materials.

1.2.8 Integration of Social Sciences and Humanities (SSH)

In the framework of its interdisciplinary approach, W2W will combine knowledge from SSH to explore how to better contribute to policy making, to enhance public understanding on clean, climate-neutral and resilient industrial value chains, how European governments and research institutes deal with diverse truth claims about the adverse effects of secondary materials and how to offer perspectives for innovation forms of action. There are many different opportunities on how to use technologies depending on human behaviour, needs and objectives and governance structures. W2W will study the special sectoral, social and regional needs and urgencies in the demo cases, the value chains and across different stakeholders upon creating the new solutions and the related training activities and lifelong learning programmes. To boost innovations' impact, the training and capacity building activities will be cocreated and codesigned since people can develop new ideas and innovations based on the new technologies and products; the process of co-creation will be based on the Responsible Research and Innovation (RRI) principles: **i)** anticipation increasing resilience with holistic approaches of expected social risks, gaps and potentials of recycle technologies and traceability; **ii)** reflexivity and continuous re-examination of positions, activities and assumptions; **iii)** inclusion giving space to currently unheard voices to mitigate concerns; and **iv)** responsiveness building capacities to adapt to changing technological and societal conditions.

[40] Tobi, H., Kampen, J.K. Research design: the methodology for interdisciplinary research framework. (2018). https://doi.org/10.1007/s11135-017-0513-8

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1.2.9 Gender Dimension in the R&I Content

W2W will overcome sex and gender bias by disaggregating socioeconomic and biological differences between man, woman or other gender identities, contributing to a more inclusive research and innovation by integrating related activities inspired from gendered innovations proposed by EC[29]. Project management will be engaged to gender issues including them in the reporting period's reports. To overcome gaps (considering both occupational and social

status of genders) a draft study will be performed resulting in W2W Gender Equality Analysis with potential actions, as an integrated part of project activities. The study will be based on Gender Situation Analysis and Needs Assessment (GSANA, T1-3.3) to ensure gender-sensitive project objectives, results, activities, partners assessment, management, indicators for monitoring and resource allocation. The project's motivation to perform these activities are both equality and the release of special creativity and abilities of all genders and individuals, creating added value for research, innovation and society. **Proportionality and balance in research**: equal participation of women, men and other genders at all project levels will be ensured, promoted already from the early stages of the project conceptualisation. All partners are encouraged to increase women's and other genders' participation in activities including management, research, analysis, networking etc. to assure equal opportunities, recruitment and working arrangements. In addition, there is gender balance in the core project management activities and in the principal investigator per partner. All partners provide equal opportunities on employment and participation in research multinational projects. **Scientific and linguistic bias**: addressing in texts and material (e.g., personas) written in English will follow gender-neutral language, ensuring data, algorithms, digital tools, technologies and visuals are free of gender bias and stereotypes. In all studies and surveys, gender dimension will be considered. Focus groups dedicated to diversity and gender issues will be included in co-creation activities. Collection of gender statistics employing AI and use of related metadata is also encouraged to identify challenges and include in policy-making, internal monitoring and to better understand gender-specific needs, issues and gaps. Annotating source data for sex and gender will be ensured. Special biological and other factors limiting or exposing to special dangers and genderspecific risk (e.g., specific professions and activities) will considered. Gender participation and real influence will be studied upon the operation and the decision-making in the governance structures of W2W as well as in the variety of entities and sectors involved. **Equal projects' results**: W2W ensures that new solutions usage (technologies, business models, processes, digital tools, etc.), access to resources and services and the related trainings, foster the involvement of women and other gender identities scientists and engineers in the future landscape. **Equal dissemination**: dissemination (quotations, visibility in publications and patents), communication in general public though releases and press, industry and business stakeholders, policy recommendations and institutional regulations, business and project's results should be addressed to all genders equally and adjusted to their needs. **Connection with established gender activities and expertise**: W2W will seek connection with EU organisations and institutes, academia studying gender, national gender equality bodies etc. for knowledge sharing, participation in activities and provision of data and tracking in EU gender statistic database (EIGE), policy-making or studies. Existing tools will be considered to ensure the integration of gender analysis into the project such as gender specific questionnaires, checklists (i.e., Yellow Window, IGAR) guidelines and research methodologies. **Gender equality promoting**: W2W is rooted in industry environment, so it focuses on gender equality to be oriented on the participating partners making difference within their workspaces. To assure full integration of the gender dimension, a dedicated session will be shared in each demo site, where general topic knowledge, all necessary information and guidelines (incl. gender-neutral language guidance) related to project will be held for the partners and possibly other participants at the beginning of the project. Partners will be encouraged and supported to implement their own gender equality plans, based on the knowledge gained during the project.

1.2.10 Open Science Practices applied in the Project

W2W will establish open science principles as an integral part of its methodology and its management, fully aligned with the ambition of the European Open Science Cloud as a trusted environment for hosting and processing research data to support EU science in its global leading role; aligned to Open Innovation 2.0 paradigm and supporting the principle 'as open as possible' in the goals of the Open Science Policy under Horizon Europe. Appropriate open science practices will be implemented supported by WP1-3 in terms of coordination, monitoring and management and WP19-21 included in dissemination, communication replication, cooperation activities etc. Specifically, the contributions are expected to be: **Research output management and impact to the industry**: W2W research outputs will be made available to dedicated target groups (i.e., industrial, sectoral, recycle, product and material associations, industry and businesses, SMEs, scientific and research community, European and international associations and organisations (e.g., SPIRE, P4P, DGs, SPI, Orgalim, EIT RawMaterials) CEN & CENELEC, national standardisation bodies, etc.) aiming to allow the community to leverage from the project results and ensure replicability. **Early and open sharing of research results and data**: W2W aims to provide 3 case studies

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publications in Open Research Europe, one for each of the value chains demonstrated. The first versions will be published upon completion of development phase and continuously updated till the end of project. **Measures to ensure reproducibility of research outputs**: The strong commitment for providing standardisation deliverables of secondary materials, the harmonisation of data as well as the decentralised based digital activities involving and coordinating different actors of the value chains with different information, will ensure that research outputs follow common standards and support mainstream and widespread reporting and documentation methodologies. **Providing open access to research outputs and Participation in open peer-review**: W2W consortium partners are committed to use the Open Research Europe scholarly fully open access publishing service for Horizon Europe to enable rapid publication times and publication outputs that support research integrity, reproducibility, transparency and enable open science practices. The project has dedicated part of its budget to cover the fees associated with open-access publication to facilitate dissemination and reuse of the project's results. Project participants, for various reasons, may need to submit articles to journals (or proceedings) that only offer a lower level of open access, requiring either parallel publication or an embargo period. In this case, Green Open Access and Gold Open Access will be chosen depending on each publication. For green open access, developers will deposit the final peer-reviewed manuscript in a public repository of their choice, ensuring open access to the publication within the embargo period of sixmonths maximum.

1.2.11 Research Data Management in line with the FAIR Principles -AI robustness

W2W is expected to collect data in the scope of the project activities and the value chains to assess the technologies and effectiveness of the proposed solutions in real life conditions of the supply chains. All the data made available directly by the partners involved or research results like models, algorithms, scripts, code and analysis will be aggregated and anonymised according to their desired visibility and security level defined. As for experimental measurements and datasets (raw, processed, logs etc.) regarding the materials and the products during the real demonstration of the value chains, will be anonymised, safe and secured in the Circular Wood Upcycling platform. The exact protocols and data types will be defined according to the specific needs in T4.4 and constantly monitored through T1-3.2. The necessity of publishing all the data generated in the project, if it does not affect exploitation of results or competitiveness, is aligned with ethic issues and protects the privacy of data. This, aligned with Open science, W2W will make data not only "as open as possible: but also "as closed as necessary" to encourage collaboration across disciplines for solving today's complex societal challenges. So, with respect in confidentiality, most of research data generated by the project will be made open and will be offered to the Open Research Data Pilot (Open AIRE). W2W will generate research outputs and data which will be managed in line with the **FAIR** principles. Data and research results will be **Findable**: with open science research described, proper identifiers e.g., DOIs and standards used; **Accessible**: integrated to databases -created by initiatives such as these in 1.2.10- and continuously expanding; **Interoperable** and **Reusable**: standardised, harmonised using common vocabularies, open standards and sharing principles to be easily adjusted and used in further research in several fields. Open science guidelines and exact data management plan will be created by M06 as part of D1.1, constantly updated (D2.1 and D3.1). The data within project's workspace will be maintained after the conclusion of the project for at least 2 years, so that the knowledge management will abide by the principles of discoverability, accessibility, intelligibility and availability also beyond the end of the project.

All the AI-based solutions will be of limited risk based on the EC's AI risk classification and regulatory framework[41], as they don't involve any critical infrastructure or activity such as health, employment, public services, fundamental rights, justice, law etc. Thus, they cannot cause harm of any kind to humans, they can only support decision making for the improvement of their safety (e.g., better mapping of DPPs or LCA results). Their purposes are strongly related to sustainability and they are operated by related scientists who understand their processes, thus they duly consider the context and their economic and social impact. All AI components will be continuously monitored and maintained to ensure that models achieve the expected accuracy and are regularly re-trained, while the relevant infrastructure meets the highest security standards (and is regularly updated) and access to the system is secured through a rolebased authorization protocol. W2W will take all appropriate measures to ensure security, accuracy, reliability, fallback plans and reproducibility. All security aspects and expectations regarding accuracy will be properly and regularly communicated with the end users.

1.2.12 The "Do not Significant Harm" (DNSH) Principle

W2W will ensure that it is fully compliant with the DNSH Principle since its activities as well as the expected life cycle impact of the innovations at a commercialisation stage, substantially contribute in a positive way and/or do not

[41] https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

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significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation (EU) 2020/852. W2W will apply a life-cycle oriented approach (validated through the utilisation of relevant LCA tool when possible) to ensure that proposed solutions are sustainable and consisting of 3 steps: **i)** Identification of actual and potential adverse impacts during the life cycle of the project activities; **ii)** Prevention or mitigation of adverse impacts; and **iii)** Accounting for how adverse impacts are addressed by tracking performance and communicating results. This also applies for the proposed ICT technologies which will be validated with Green ICT principles. W2W activities are expected to have a direct positive impact to the environmental objectives defined: Climate Change Mitigation by reducing GHG emissions (use Recommendation 2013/179/EU or ETSI ES 203 199, ISO 14067:2018 or ISO 14064-2:2019 to calculate lifecycle GHG and net emissions), to pollution prevention and control through waste reduction, and identifying harmless substances and contaminants in each case, to transition to Circular Economy through the establishment of symbiotic and circular flows within same or different industries for materials and wastes, whereas is not expected any negative impact to the sustainable use and protection of water and marine resources, protection and restoration of biodiversity and the Climate Change adaptation.

2 Impact #@IMP-ACT-IA@#

2.1 W2W's Pathways towards Impact

In this section, an overview is provided on how the W2W results generate Outcomes (Section 2.1.1) and Wider Impacts (Section 2.1.2). The target groups and the impact created by the W2W are presented in Table 2.1, whereas the W2W KERs per innovation pillar are listed in Table 2.2, for the reader's convenience.

Table 2.1: W2W target groups and impact created

Target Groups	Impact Created by W2W
A: Process industry companies, including those involved in the production of woodbased products and combustion processes	*New opportunities to use secondary resources, which can increase resource efficiency and reduce CO2 emissions; Development of new, more sustainable products and processes.*
B: Recycling companies involved in the collection, sorting and processing of secondary raw materials	*New business opportunities and revenue streams, particularly for SMEs; Expansion of their operations by providing a new source of feedstock.*
C: Manufacturers of new products made from secondary raw materials	*New sources of feedstock to develop new products; Diversify product portfolio and develop more sustainable products.*
D: Public authorities responsible for waste management and environmental protection	*Help to achieve their environmental and sustainability goals, such as reducing waste sent to landfill and CO2 emissions.*
E: Standardisation organisations developing standards for the use of secondary raw materials in the process industry	*Contribution to the development of new standards and guidelines for the use of secondary resources in the process industry.*
F: Research and academic institutions working on technologies and approaches for the upcycling of secondary raw materials	*Access to new knowledge and insights related to the use of secondary resources in the process industry, upcycling, and circular economy.*
G: Workers and employees in the process industry and recycling sectors	*New training and skills development opportunities which can improve their employability.*
H: End-users and consumers of products made from secondary raw materials	*More sustainable products and processes, which can help to reduce the environmental impact of production and consumption.*
I: The broader society and environment	*Reduced waste and pollution, increased resource efficiency.*

Table 2.2: W2W KERs for each innovation pillar

Pillar 1: Advanced Separation and Sorting Technologies
KER1: Optimal wood size particles for cascade valorisation; **KER2:** Cascade valorisation thresholds-led sorting system; **KER3:** Optimised separation enabled by Human-Robot Collaboration and MR technologies.

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Pillar 2: Upcycling Processes and Technologies
KER4: Wood Cascade to function; **KER5:** Wood composites for wallboards; **KER6:** Wood glue separation; **KER7:** Bioremediation wood treatment; **KER8:** Mixed wood liquefaction; **KER9:** Hydrothermal Carbonisation; **KER10**: Biotech-driven up-cycling of syngas into dodecanol; **KER11**: Ashes valorisation.
Pillar 3: Digital Tools for Improving Circular Flows of Secondary Materials
KER12: Life Cycle Sustainability Assessment Tool for Cascade Pathways; **KER13:** Material Volume Estimation; **KER14:** Dynamic DPP; **KER15:** Supply Chain management tool for the optimisation of upcycling of secondary materials; **KER16:** Integrated Circular Wood Upcycling Platform.
Pillar 4: Supportive Framework in Policy, Market, and Skills
KER17: W2W Stakeholders Network; **KER18**: Classification Framework for recycled wood; **KER19**: Wood cascade data sharing model; **KER20:** Training Programs for Upcycling, Replication, and Cooperation based on collaborative Design; **KER21:** Synergies hub for circularity; **KER22:** Circularity / Upcycling framework for cascade valorisation; **KER23:** Contribution to standardisation and certification; **KER24:** Business models.

2.1.1 Contribution of W2W towards Achieving the Expected Outcomes of the Target Topic

Expected Outcome 1: "Prove the technical and economic feasibility of the use of secondary resources in the process industry leading to products with identical properties and performances as those produced using primary resources and allowing production without quality restriction."
Pathway to impact: The application of the novel W2W technologies (W2W Pillar 1 & 2) will process woodcontaining waste to produce wood without pollutants, biocomposite building materials, biopolymers, polyols, and chemical detergents. These end products will be analysed in collaboration with the process industry (Tasks T7-

8.4, T9-10.4, T11-12.5) to determine their characteristics for meeting quality specifications. In particular, for

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composite building materials: thermal performance, fire resistance, mechanical strength, sound insulation; for particleboards: internal bond strength, MOE, MOR, swelling, density profiles; for dodecanol: stability, colour, odour, safety. The compliance with regulatory and market requirements will also be assessed, to ensure that they can effectively replace products produced using primary resources.

KERs: KER1-KER11, **Target Groups:** A, B, C, D, F, H, I

Expected Outcomes KPIs	Target
KPI1.1 Increase in efficiency of successful wood material classification	**90%**
KPI1.2 Improvement in latency of the human-robot collaboration execution	**80%**
KPI1.3 Improvement in overall efficiency and productivity of the separation process	**80%**
KPI1.4 Increase in amount of Class B and Class C wood waste recovered and used to produce composite building materials and panels	**30%**
KPI1.5 Validation of purified syngas-derived dodecanol and aminoacids	**>2 detergents**
KPI1.6 Increase in the suitability of wood fibers derived from wood waste for use in end products	**30%**
KPI1.7 Increase in the efficiency of the removal of urea-formaldehyde (UF) from waste wood	**50%**

Significance: Proportion of the process industry that adopts the use of secondary resources in their production processes: **30%**	**Scale:** Number of end product types produced through the use of secondary resources with identical properties and performance as those produced using primary resources: **5**

Expected Outcome 2: "Increase the use of secondary resources in the process industry leading to significant increase in resource efficiency across the value chain and subsequent reduction of CO2 emissions; reduction of waste sent to landfill and overall positive environmental impact."

Pathway to impact: The disposal of waste wood in landfills creates methane, a potent greenhouse gas, with each tonne of wood in landfill creating the equivalent of 800kg of CO2 in methane emissions[42]. The incineration of waste wood recovers energy from the resource. In fact, woody biomass is classified as 'renewable energy' in the EU. However, there is criticism that it is having the opposite effect to that expected of renewable energy, by increasing atmospheric levels of carbon dioxide for substantial periods of time[43]. By applying the novel W2W technologies (Pillar 1 & 2) and demonstrating the successful upcycling of wood in 3 Use Cases, the project will extend the lifetime of wood in C&D and furniture waste and achieve a reduced demand for virgin wood, thereby preventing deforestation and promoting circularity in the wood value chain. Additionally, by reducing the amount of wood waste sent to landfill or incineration, the project can help prevent the release of greenhouse gases and other harmful substances into the environment.

KERs: KER1-KER11, **Target Groups:** A, B, C, D, F, H, I

Expected Outcomes KPIs	Target
KPI2.1 Reduction of wood-containing waste sent to landfill or incineration	**30%**
KPI2.2 % increase in lignin and cellulose fibers recovered from waste wood	**50%**

Scale: increase in the use of recovered wood: **30%**	**Significance:** Reduction in CO2 emissions resulting from diversion of wood waste from landfills or incineration (20%,70%,70% for Use Case 1,2, and 3, respectively)

Expected Outcome 3: "Increase the competitiveness of the European process industry; new business opportunities and revenue flows for recycling companies, benefiting particularly SMEs, which dominate this sector of the market."

[42] https://communitywoodrecycling.org.uk/what-we-do/our-environmental-impact/
[43] Norton, M, Baldi, A, Buda, V, et al. Serious mismatches continue between science and policy in forest bioenergy. GCB Bioenergy. 2019; 11: 1256– 1263. https://doi.org/10.1111/gcbb.12643

KPI3.1 Number of companies and organisations participating in the Synergies hub for circularity Ass Ref. Ares(2023)8305270 - 05/12/2023	**100**
KPI3.2 Number of companies and organisations adopting the W2W classification framework	**25**
KPI3.3 Number of companies and organisations using the W2W data sharing model	**40**
KPI3.4 Number of collaborations established through the W2W stakeholder network	**30**
KPI3.5 Increased revenue of recycling companies which adopt the W2W framework	**15%**
KPI3.6 Increased number of jobs in the process industry and recycling sector due to the increased use of upcycling technologies	**5%**

Scale: Number of exploitable technologies/methods generated from the project: **16** (KERs 1 – 16) Number of new standards taking into consideration the results of W2W: **2**	**Significance:** New partnerships and collaborations between process industry and recycling companies: **70** Number of publications, conference presentations, and other forms of dissemination of project results: **30**

Pathway to impact: The market analysis and the co-designed business models developed in W2W with prospective customers and stakeholders participating in the Use Case activities will maximise the business opportunities and revenue streams for recycling companies, particularly SMEs, which can utilise secondary resources for manufacturing new projects in a sustainable and cost-effective manner, and reduce the cost of raw materials, thus increasing their competitiveness in the market. The development of new and innovative upcycling technologies will create a competitive advantage for the European process industry, attracting new customers and driving growth in the sector. In addition, the contribution of W2W to the development of new standards will bring the project technologies closer to the market and spread technological advances.

KERs: KER17 – KER24, **Target Groups:** A, B, C, D, G

Expected Outcomes KPIs	**Target**

Expected Outcome 4: "The proposed technologies should contribute to the matching of supply-versusdemand of feedstock at the level of quality constraints (removal of impurities or wrong matrices, concentration etc.)."

Pathway to impact: The novel W2W technologies (W2W Pillar 1 & 2) will enable the conversion of recovered wood (Grade B &C) to Grade A wood, which has an appropriate quality that can match the demand of feedstock in the process industry. In addition, the Digital Product Passport and the Supply Chain management tool will enhance the traceability and transparency of the recovered wood in the value chain, thus improving the matching of supply and demand of feedstock. The project use cases will providing a comprehensive assessment of the quality constraints and the potential of the secondary resources, which can facilitate the decision-making process for both the suppliers and the process industry.

KERs: KER1-11, KER14, KER15, **Target Groups:** A, B, C

Expected Outcomes KPIs	**Target**
KPI4.1 Number of companies implementing the DPP	**40**
KPI4.2 Reduction of transport costs	**20%**

Scale: Number of end product quality properties validated in Use Case 1: **4** Number of end product quality properties validated in Use Case 2: **5** Number of end product quality properties validated in Use Case 3: **4**	**Significance:** increase in the availability of Grade A wood feedstock in the process industry compared to the baseline prior to the project: 2**0%**

Expected Outcome 5: "Foster data sharing, and FAIR digital assets principles, considering the application of digital product passport between recycling companies and the process industry to improve the economy of scale in upcycling of material streams."

Pathway to impact: The project Digital Product Passport will enable data sharing and collaboration among stakeholders in the value chain and ensure that businesses along the value chain (manufacturers, importers and distributors, dealers, repairers, remanufacturers and recyclers) can access information that is valuable in their work to improve environmental performance, prolong product lifetime, boost efficiency and the use of secondary raw materials, thus lowering the need for primary natural resources, saving costs and reducing strategic dependencies. The improved efficiency and traceability of material streams will lead to improved economy of scale in upcycling, reduce waste and environmental impact, and increase the competitiveness of the European process industry.

KERs: KER14, KER16, KER19, **Target Groups:** A, B, C, H

Expected Outcomes KPIs	Target
KPI5.1 Number of products with DPP information available	**10**
KPI5.2 Number of product information management processes	**15**

Scale: Number of businesses using the Digital Product Passport to access information on environmental performance, product lifetime, efficiency, and secondary raw materials use: **50**	**Significance:** Improvement in the accuracy and reliability of data shared through the digital product passport system: **90%;** Increase in the efficiency and traceability of material streams within the value chain: **85%**

Expected Outcome 6: "Increase the use of unused and new skills to unfold the potential of the technological solutions at the workplace for upcycling and contribution to inclusive growth."

Pathway to impact: The project training material and capacity building activities will ensure that workers possess the necessary skillset to support the uptake and deployment of the innovative products, services, and processes.

By doing so, the project will contribute to closing skill gaps in the relevant sectors and occupational groups, Associa
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leading to improvements in employment and social levels across Europe. W2W will also promote knowledge sharing and collaboration among stakeholders in the value chain, to create a supportive and inclusive environment for the adoption of new skills and technologies. Finally, the project can contribute to inclusive growth by engaging with local communities and promoting the social and environmental benefits of upcycling, such as reduced waste, lower emissions, and improved resource efficiency.

KERs: KER20, **Target Groups:** A, B, C, G, I

Expected Outcomes KPIs	Target
KPI6.1 Number of workers trained in new skills relevant to upcycling and circular economy principles	**100**
KPI6.2 Number of companies implementing new technologies and processes related to upcycling	**20**
KPI6.3 Number of skills gaps closed	**5**

Scale: Improved productivity and efficiency in the workplace as a result of the adoption of new upcycling technologies: **30%**	**Significance:** Enhanced social and economic inclusion by providing training and employment opportunities for underrepresented groups such as youth, women, and disadvantaged populations.

Expected Outcome 7: "At a longer term, to pave the way toward sustainable-by-design for circular products."

Pathway to impact: The W2W framework life-cycle thinking approach, and its Life Cycle Sustainability Assessment Tool for Cascade Pathways will enable comparison between different options for the manufacturing of wood-containing products or the efficient use of by-products from wood refining, highlighting trade-offs. This will enable a holistic assessment of the implications of a product life cycle, including environmental, social, and economic aspects, paving the way towards sustainable-by-design circular products, which are designed from the outset with circularity in mind, so that they can be easily recycled or upcycled at the end of their life. This involves not only the design of the product itself but also the selection of materials and the development of new production processes that minimise waste and maximise resource efficiency. The project can also contribute to this outcome by the development of new business models which incentivise the reuse and recycling of products. The project will also work with policy- and decision-makers and industry stakeholders in order to create an enabling regulatory and market environment for circular and secondary products, and to promote the adoption of sustainable-by-design principles across the whole value chain.

KERs: KER12, KER17-24, **Target Groups:** All target groups (A-I)

Expected Outcomes KPIs	Target
KPI7.1 Number of companies adopting sustainable-by-design circular product principles as a result of the project	**10**
KPI7.2 % reduction in environmental impact of products compared to their linear counterparts	**70%**

Scale: Number of circular products designed and developed as a result of the project: **3**	**Significance:** Reduction in the environmental impact of products over their life cycle, as measured by the project's Life Cycle Sustainability Assessment Tool for Cascade Pathways

2.1.2 Contribution of W2W towards Achieving Long-term Expected Impacts in the Destination

Expected Impact: "Global leadership in clean, climate-neutral and resilient industrial value chains, circular economy and climate-neutral and human-centric digital systems and infrastructures."

W2W project is fully aligned with the topic and significantly contributes to the current EU policy framework and goals. Aiming towards the development of Key Enabling Technologies for the Process Industries and their validation, W2W fully supports the goal of the EU Climate Neutrality, for which digital, low-carbon and circular industry is one of the core aspects, as set by the Green Deal. W2W project is thus fully aligned with the key objectives of this destination, contributing to making Europe the first digitally led circular, climate-neutral and sustainable economy, having a direct contribution to the transformation of its construction and production systems and indirectly supporting the energy transformation through the technologies of ashes combustion valorisation. Innovative products and technologies exploitation strategies, business cases, innovative financial solutions for circular economy, educational services for employees based on co-designed resources and certification and standardization schemes of secondary raw materials, novel products and technologies, that will be integrated into digital tools will be a coherent part of W2W activities. A new paradigm for valorisation of secondary resources is introduced offering a wider replication potential in a series of sectors.

Topics Serving the Objectives of the Destination and relevant to W2W Project

Manufacturing Industry: W2W is aiming to support manufacturing of circular, carbon-neutral and regenerative

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wood-based products by increasing the recovery value of recycled wood. The advanced separation and sorting machines, in various scales can be adjusted to the needs of specific manufacturing lines contributing the Industry5.0 principles of Robot Human collaboration towards a smart value network. The developed digital tools are integrated in a common platform as a service, allowing the optimisation of value chains in line with the Manufacturing as a service concept.

A new way to build, accelerating disruptive change in construction: W2W aims to improve the resource efficiency in the construction sector by improving the separation and sorting of CDW, supporting the pressure in the post pandemic period to repair and maintenance. A complete framework for circular CDW collection, separation, vaporisation is considered with the participation of the private and public sector, in line with the needs of SMEs that are dominating the recycling of CDW market. New materials and products are offered which can be used for newly build or repairing activities.

Energy efficient and climate neutral process industries: W2W will contribute to increase the sustainability and resilience of the whole recycled wood value chain, including resource and energy intensive industries. They will be benefited by the increasing amount of waste material reintroduced into the production cycle, which will enable the creation of circular value chains for the targeted streams. Additionally, the technologies developed by W2W will be designed to contribute to a lower environmental impact, introducing a cascade in function, cascade in value and cascade in time new paradigm approach.

Circularity and Zero Pollution in process industries: The core of the project activities is the valorisation of CDW and close the production loop; That being said, the results from W2W will directly contribute towards the optimisation of resources, keeping resources in the loop and overall reducing landfilling in the whole value chain.

Hubs for Circularity (H4C): W2W is developing a Supportive Framework in Policy, Market, and Skills which will be further elaborated in line with the advancements of the H4C through the participating members in the P4P (**ICCS**, **CIRCE**) as well as the related projects. The produced valorisation framework will also be evaluated by the ongoing projects of H4C, starting with the relevant objectives of REDOL project.

Significance: To establish a secondary resource optimisation framework of recycled wood covering all the needed enabling technologies, validated at TRL6 at the end of the project and at TRL9 by 2030. The W2W comprehensive approach for multi-dimensional cascade valorisation of wood from C&D and furniture waste will deliver a set of advanced waste management and upcycling technologies (TRL6), digital tools, and a supportive framework in policy, market and skills for establishing circular, sustainable value chains of recovered wood. During the project, these technologies and methods will be demonstrated with the active participation of feedstock suppliers, technology providers, and end users in 3 use cases. The wide dissemination of these technologies and methods through appropriate networks and their exploitation will set the stage for continued activities, and replication of the results in other sectors and regions (in and outside of the EU) after the project is over. To have a long-term impact, after the end of the project it will be required to scale up the technologies developed to an industrial level (TRL9) and demonstrate their commercial viability through partnerships with key industry players, investment in infrastructure, and support from policymakers. Ultimately, W2W will accelerate the transition from the take-make-dispose linear system to a more sustainable closed-loop system that upcycles secondary resources to reduce waste and create new value.

Quantification of Long-term Impact

Baseline: ● At European level, the CDWW deriving from all 28 EU countries is approximately 50.2 million tons (MT), with 2018 as a reference year. ● The average recycling rate in the EU is 46%: 23,1Mt of wood. ● When wood rots it creates methane, a potent greenhouse gas, with each tonne of wood in landfill creating the equivalent of 800kg of CO2 in methane emissions. ● Recycled wood is up to 40% cheaper than fresh wood.

Assumptions: ○ W2W technologies result in an additional 30% of wood recycled: 6,9Mt; ○ 80% of wood waste treated by W2W technologies is estimated to be Class B, C, D. Of this, 20% is converted to Class A; ○ The remaining 20% is valorised for energy and CO2/H2.

Estimated Long-term W2W impact: ♦ Approximately 7Mt of CDWW will be diverted from landfill and will be exploited for material valorisation; ♦ W2W results in avoided emissions from landfilling: 5,542Mt of CO2eq/tonnes of wood waste/year.

Replication to other sectors potential Impact: The replication of W2W technologies in the pulp and paper industry can contribute to reducing emissions in this sector, which are estimated[50] at: **direct** (32 MtCO2e in 2018) and **indirect** (9.98 MtCO2e in 2018). The technology for the valorisation of ashes from CDW can also be applied in ashes originating from any thermal or combustion process; these amount at 118 Mt[51].

[50] https://www.cepi.org/statistics/

[51] 2016 data, using Eurostat's env_wasgen database

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2.1.3 Barriers, Obstacles and Conditions that Determine Impact's Success

Technology and Process Barriers: Managing the complexity of value chains, products, materials, substances and different technologies is a great challenge creating individual barriers related to W2W technologies and processes. Additional barriers can include lack of data on waste resources, uncertainties of novel technologies, high cost of infrastructures for the replication of solutions, and difficulty in assessing the overall environmental impact.

Mitigating measures: W2W intends to involve all the actors of supply chain, from suppliers of primary materials to producers and recyclers, to achieve a coherent and holistic solution, supported by proper managerial and coordination structures and procedures. All needs, technical capacities and knowhow of companies across supply chains will be combined, and the diversity of profiles of supply chain actors will be considered from the very beginning of the development. The high scientific and technical expertise and experience of the consortium is expected to contribute to the mitigation of the possible barriers in line with general sustainability, resilience, circular economy and DNSH principles. The developed digital tools for dynamic assessment will also allow a better and cost-efficient management of all the potential solutions decreasing the retrofitting impact of the value chain.

Societal and Public Acceptance Barrier: Potential barriers related to public acceptance of novel technologies and methodologies might be related to lack of awareness, lack of digital skills, and scepticism on using recycled materials.

Mitigating measures: Dissemination and communication activities will ensure the promotion of the benefits, results, best practice and performances obtained, providing training, educational activities and awareness and engagement raising though their co-creation. Means such as tutorials, webinars, workshops, or tools will be explored. The project will aim to engage the industrial community by interacting and networking with industry and workers associations. Mitigation of possible poor acceptance will be further supported by W2W standardisation system as a tool for dissemination of the project results and interaction with the market stakeholders.

Legal and Regulatory Barriers: W2W involves different activities related to recycled wood treatment coming from various sectors and applying novel, not yet standardized technologies. As a result, there are multiple and overlapping regulations that have to be taken into consideration. Those regulations might often be conflicting, inconsistent or unclear. Additionally, in some cases, national strategies might not be aligned with the relevant European documents, contributing with different materials classifications or waste management institutional obligations (e.g., local mandatory management). Also, the evolution of EU standards and regulations is expected to be more restrictive and to set highly demanding implementation targets. (e.g., on quality or in the presence of unknown dangerous substances that are no longer allowed). All these may cause obstacles to adopt innovations and best practices or obtain approvals for project activities.

Mitigating measures: W2W will collect, analyse, follow and plan its actions accordingly with all the related EU strategies already referred and regulations (e.g., Waste Framework Directive 2008/98/EC, Regulation No 1013/2006 of end-of-waste criteria, Ecodesign Directive 2009/125/EC, GDPR, IRP protection, Cybersecurity directives 2016/1148/EC, 2019/881/EC etc.), national laws and regional/local particular practices and institutions' regulations. The W2W consortium consists of a multi-disciplinary group of partners, including digital and industrial expertise, standardisation and certification bodies, all with wide experience and knowledge on the above practices and regulations that will ensure the required effort. Finally, the project will aim to engage member States representatives involved at local level to properly address the challenges by interacting and networking with the relevant ministries, regional and local authorities, and associations, assuring the mitigation of legal and institutional barriers.

Increased Prices of Recycled Materials: Potential barriers related to increased prices of the recycled materials may appear as the offer vs demand of materials should be fully adjusted to the new proposed paradigm. Industry acceptance might be also affected by the feedstock availability and price stability, as well as occurrence of competing materials. Potential high costs for collection, sensing, sorting, upgrade and recycling processes and infrastructure of multiple technologies compilation might affect the final price of products posing a barrier of wider implementation of project's solutions.

Mitigating measures: W2W will encourage collaboration and coordination among stakeholders in the value chain, promoting the sharing of knowledge, resources and infrastructure, and the development of common standards, helping to incrase efficiency and reduce costs. The increase in public awareness of the environmental benefits of recycling and the importance of using recycled products, can help to create a more stable market for recycled materials and reduce price volatility.

2.2 Measures to Maximise Impact #@COM-DIS-VIS-CDV@#

To achieve its stepwise methodology, the W2W consortium will utilise a concrete D&C plan together with an exploitation/commercialisation plan, taking advantage of the in-project user community and external stakeholders.

2.2.1 Dissemination, Communication and Exploitation Plan

W2W Dissemination Strategy

While focused on Awareness, Interest, Desire and Action (AIDA) activities, the project's dissemination measures will follow a multi-actor, multi-step and multi-channel approach. Table 2.3 identifies the target audiences, specific dissemination objectives and target values, tailored to the needs and profile of each target audience. **Table 2.3**: Dissemination Plan – Measures, Target Audience and Target Values

Scientific Publications	Principal Decision Influencers, Decision-Making Actors, Industry Associations, Environmental Agencies, Academia and Researchers, and Relevant Consortiums.
Scientific Conferences (examples): Eur. Conference of Waste Management (ECWM), Digital Recycling Conference for Circular Economy, Sustainability, and Waste Management (eREC), Eur. Conference on Wood Modification (ECWM), Int. Conference on Wood Biology and Wood Technology, Eur. Conference on Process Safety and Big Data, Int. Conference on Sustainable Solid Waste Management, Materials Science and Technology in Europe, International Conference on Waste Recycling and Management, etc. **Peer-Review Open-Access Journal (preselected)**: Recycling, Waste Management & Research, Polymer, Sustainability, Waste Management, Journal of Material Cycles and Waste Management, Wood Science and Technology, Resources Conservation and Recycling, etc.	
Target Values: Publishing in Peer-reviewed Journals: > **15**; Presenting in Scientific Conferences: > **20**.	
Technical Publications	European Commission Relevant Offices, European Environment Agency, Public and Private Actors, ICT SMEs and large Enterprises, Formulators, Recyclers.
W2W will publish and contribute to technical blog posts, articles, position/white papers, books, or any other reference from technology suppliers (bottom-up), in addition to references linked to the application scenario domains under consideration. The forthcoming issues of Institute of Environment and Development Blog and Spire database of the outputs from projects of SPIRE portfolio are excellent instances of this type of work.	
Target Values: Publishing Technical Publications: > **25**; Participation in Industrial Exhibitions with Booths: > **9**	
Source Code Repository	Ecosystem of Open-Source Developers, Incubators, Associations and Digital Innovation Hubs, Data Spaces for circular material use, ICT SMEs, Researchers, relevant Consortia.
Stimulating a transparent and open-source culture, W2W will make widely available its software repositories through well-known distributed source code platforms such as GitHub and Bitbucket.	
Target Values: Publicly Available Deliverables: > **15**; Source code: publish to >**2** different repositories.	
Q & A Platform	Public Authorities, Associations, Open-source Developers, NGOs, SMEs, relevant Hubs.
Discussions regarding the project repository will be centered around reference sites. The designated Q&A platform, developed as an integral part of the project website, will be Stack Overflow - the most extensive and reputable online community for developers to exchange knowledge and acquire new insights.	
Target Values: No. of Scientific Conversations to Participate/Contribute: > **20**.	
Workshops and Trainings	Industrial Organisations, SPIRE Associations, Public Authorities, Potential Investors, Funding Organisations, ICT SMEs, Academia and research Organisations.
Particular emphasis will be placed on collaboratively designing training activities and lifelong learning programs to ensure that W2W actively contributes to skills development and standardisation, as outlined in WP17-19.	
Target Values: Technical Workshops: > **10**; Online Training Tutorials: > **8**; Webinars: > **12**.	
Joint Activities with relevant Initiatives	All Stakeholders with special focus on secondary resources in process industry, Policymaking Actors, Key Environmental Influencing Organisations, relevant Public Bodies.
To further amplify and maximise the impact and outreach of the project's results and to feed them into related international work streams, joint dissemination activities will be contracted. W2W will seek out opportunities to collaborate with other relevant projects and initiatives within Europe and beyond, and will organise and participate in events that promote knowledge sharing, as outlined in T19-21.2 and T19-21.3.	
Target Values: Symposiums/Fairs to Participate: > **9**; Banners in each event: > **3**; Procures to be distributed: > **300**.	
Standards and Policy Contribution	Standards Developing Organisations, Certification Committees, Open-Source Communities, Policy Makers, Relevant Public Bodies, Local Agencies.

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W2W aims to support the standardisation of secondary material upcycling technologies. Collaborative efforts with relevant Technical Committees will be established to ensure seamless integration into current and future standardisation initiatives. The project's contributions will be incorporated into new or revised standards via one of the following channels: **i)** sharing information with Technical Committees, **ii)** facilitating mutual knowledge exchange, **iii)** submitting technical proposals, and **iv)** promoting the creation of new standard documents, based on the maturity of the specific solution, as outlined in T16-18.3 and T16-18.4.
Target Values: Standards Contribution (new or revised): ≥ **3**; Policy Briefs: > **2**.

W2W Communication Strategy

The following design elements will be used to strengthen the project uniformity and identity and to deliver clear messages to the target audience: W2W naming, logo, presentations template, templates for reports, project newsletter, project posters/leaflets, etc. A recognisable visual identity is provided at the proposal stage, to be adapted if required. W2W's communication strategy is built upon the Promotion Mix, the 4th element of the Marketing Mix, focuses on creating awareness and persuading the audience to initiate the engagement, as outlined in Table 2.4.

Table 2.4: Communication plan – Communication Channels/Target Audience/Timeline and Target Values

Project Website	ALL End-users.	From M06 to 3y after
W2W dynamic project website will be the backbone of all the D&C activities, allowing easy retrieval of main action data with few clicks, while acting as the core source of information about the project's objectives, activities and results. The website will be launched by M06, and it will be lined with all W2W's social media accounts. It will be constantly maintained and updated regularly, running also for 3years after the project end.		
Target values: Online by: **M06**; Unique Visitors from M12: > **1,500**; Unique Visitors from M48: > **6,500**		
Social Media	ALL End-users.	From M06 to M48
W2W will create and actively maintain its online presence across many social media platforms, utilising Twitter and LinkedIn when connecting with research and innovation networks and disseminating new publications, and Facebook, and Podcasts for broad community participation.		
Target Values: Size of online community (by M48): >**5,000**; No. of impressions (monthly average) :>**500**		
e-Newsletters and e-mail Campaigns	All End-users.	Bi-yearly Issues starting from M06
When project findings are presented, relevant stakeholders will get e-newsletters. Emails are a very effective engagement tool, particularly for promoting events and outcomes across domains; hence, project communications will be delivered to a selected set of contact points. Professional marketing solutions (such as Sendinblue and Mailchimp) will be utilised to automate the dissemination to contact points.		
Target Values: e-Newsletters contributed/released: **8**; Total Mailing List Contact Points: >**2,000**		
Policy Fora and Debate Hubs	Societal Actors, Debaters, Recycle Operations, Sustainable Product Organisations, Workers' Associations, Alliances.	From M18 to M48
With the Science-Policy dialogue as its primary objective, W2W will contribute to discussions and make recommendations regarding how the developed solutions could contribute to the sustainable, balanced, and inclusive development of secondary material upcycling and waste management across various industry sectors and products. W2W intends to contribute to and support the activities of relevant hubs and forums, such as the World Economic Forum, Science Policy Forum, EIT RawMaterials Expert Forum, and GS1 linked DPP forums.		
Target Values: No. of Fora and Hubs to interact: >**7**; Contributions to Policy Fora/Debate Hubs: >**15**		
Blogs	Digital Innovation Hubs, Relevant Associations, Academia.	From M18 to M48
W2W will link and execute periodic wide marketing operations in open communities, DIHs, blogs, and non-profit organisations and channels, such as Gitter Communities, Nature.com, and non-profit organisations, such as RawMaterials Innovation Centers of the EIT, etc.		
Target Values: Unique Publications/Contributions to Blogs: >**15**; Different Blogs to Post: >**8**		
Printed Material	ALL end users attending physical events	From M06 to M48

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W2W will develop a distinct visual identity and reference promotional instrument when participating in events, including brochures, catalogues, posters and any other laid out paper-based resource. **Most of the content will be in digital format. Printed material will be avoided highlighting on the overall "green" aspect of the project.**		
Target Values: No. of hard copies (i.e., brochures, flyers, posters, etc.) distributed: >**1500**; Roll-up Banners: >**6**		
Multimedia	ALL end-users focusing on Regional and Local Authorities	From M09 to M48
The project will develop multimedia content for a self-explanatory and aesthetically pleasing presentation of the project, utilizing various distribution channels for marketing (e.g., YouTube, Vimeo). In addition, in close connection with the W2W capacity building efforts, a series of video interviews will be conducted with scientific partners to collect feedback, taking use of plenary sessions and other pertinent events.		
Target Values: No. of short videos and interviews produced in total: **7**; No. of reviews: >**10,000**		

W2W Exploitation Plan

Exploitation plan and sustainability

Turing brilliant R&I into successful commercial products is a complex task with more failures than successes. Most failures occur because there is a lack of Product/Market fit or to put it simply, the product that is built is not what the market wants to buy. To orientate the outcomes from W2W towards viable commercial solutions and assets, the consortium will adopt a **focused commercialisation strategy with measurable outcomes by embedding best practice into the project's lifecycle**. The exploitation (**T19-21.4)** and commercialisation tasks (**T19-21.5)** will focus on delivering solutions that have a clear market opportunity that can be proven and validated through the pilots with key stakeholders and in close collaboration with formulators, recyclers, public authorities, and standardisation actors. The completed business models and marketing plan will aim to outline viable and commercially relevant products and assets from the project by taking a broad cross-consortium perspective, with credibility built on a foundation of embracing industry best practice for bringing new products to market. To achieve sustainability for the KERs and prepare the ground for market entry, the consortium has drafted the exploitation plan presented in Figure 2.1, which is split in 3 distinct phases: **Phase 1**: setting the necessary conditions for the exploration of the business plan; **Phase 2**: validating the developed solution, aligns with market and prepares the go to market strategy; **Phase 3**: featuring the sustainability and commercialisation roadmap.

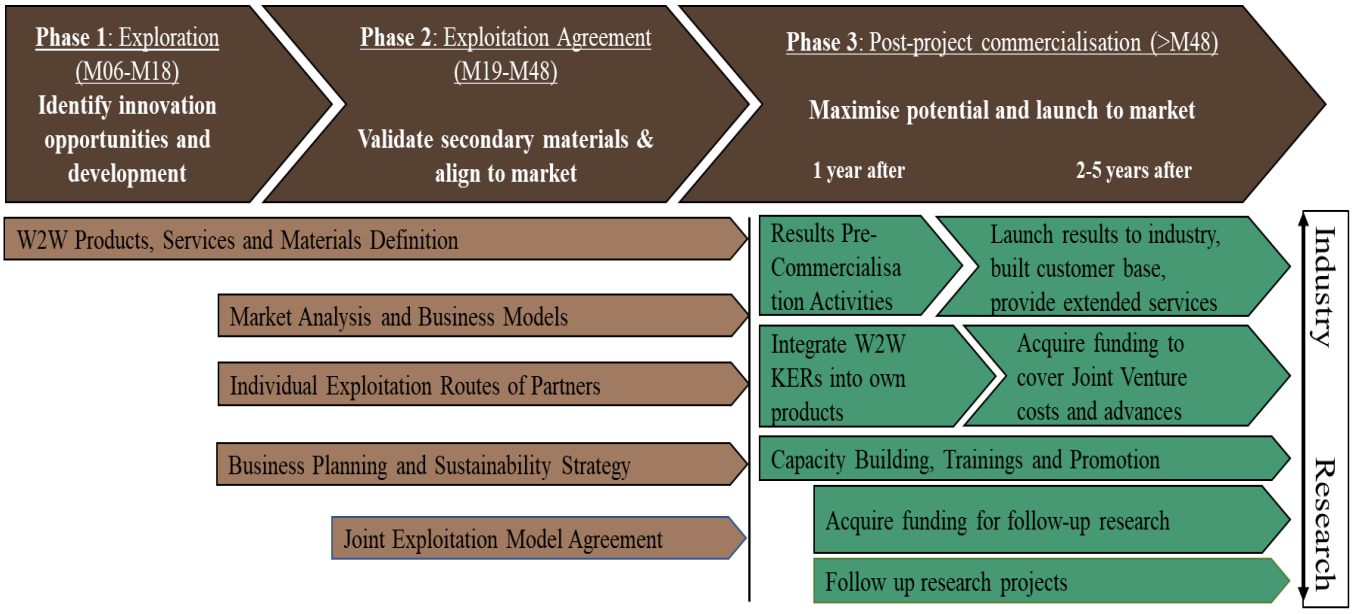

Figure 2.1: W2W exploitation plan and timeline

Key Exploitable Results (KERs) and individual exploitation plans

The consortium partners have decided to take a multi-level approach towards managing the knowledge produced, where the respective knowledge producer holds the rights to the knowledge and shares the software and material implementation with the rest of the consortium. The initial exploitation intentions towards the W2W KERs are presented in Table 2.5, which will be further refined and finalised during the project's duration.

Table 2.5: W2W KERs (Type: Product (P), Service (S), Other (O); Services: Support (Su), Training (T), Consulting (C), Extension (E))

W2W KERs	Owner	Type	Licensing	Services	Time to market
KERs resulting from W2W Pilar 1: Advanced Separation and sorting Technologies					

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Optimal wood size particles for cascade valorisation	**IRIS**	P/S	Open-source /Commercial	Su, C, E	1y after project
Cascade valorisation thresholds-led sorting system	**ICCS**	P	Open-source /Commercial	Su, E	1y after project
Optimised separation enabled by Human-Robot Collaboration and MR technologies	**ICCS**	P	Open-source /Commercial	Su, E	1y after project
KERs resulting from W2W Pilar 2: Upcycling Processes and Technologies					
Wood Cascade to function	**UHE**	P/O	Commercial	Su, E	1y after project
Wood composites for wallboards	**NTUA**	P	Commercial	Su, C, T, E	1y after project
Wood glue separation process and bioremediation wood treatment	**LERMAB**	O	Open-source /Commercial	Su, C, T, E	1y after project
Liquefaction process of mixed wood waste and green adhesives based on polyols	**UHE**	P/S	Open-source /Commercial	Su, C, T, E	1y after project
Hydrothermal carbonisation and ashes valoriasation procedures	**CIRCE**	P/O	Open-source /Commercial	Su, C, T, E	1y after project
Biotech-driven up-cycling of syngas into dodecanol	**KIV**	P/S	Open-source /Commercial	Su, C, T, E	1y after project
KERs resulting from W2W Pilar 3: Digital Tools for Improving Circular Flows of Secondary Materials					

Dynamic digital product passport	**CERTH**	P/S	Open-source /Commercial	Su, C, E	0.5y after project
Material volume estimation tool	**ICCS**	P/S	Open-source /Commercial	Su, C, E	0.5y after project
Supply chain management tool for upcycling of secondary materials	**UPV**	P/S	Open-source /Commercial	Su, C, E	0.5y after project
Life cycle sustainability assessment tool for cascade pathways	**DRAXIS**	P/S	Open-source /Commercial	Su, C, E	0.5y after project
Integrated circular wood upcycling platform	**EBOS**	P/S	Open-source /Commercial	Su, C, E	0.5y after project
KERs resulting from W2W Pilar 4: Supportive Framework in Policy, Market, and Skills					
Wood cascade data sharing model	**TUB**	S/O	Open-source	C, T, E	0.5y after project
Circularity and upcycling for cascade valorisation & Recycling classification frameworks for recycled wood	**All relevant partners**	O	Open-source	C, E	0.5y after project
Training programs for upcycling, based on collaborative design	**All relevant partners**	S/O	Open-source	T	0.5y after project

Given the low to medium TRL level of the under-development tools, KER characteristics and exploitation routes are to be constantly evaluated and updated as project progresses and engagement with stakeholders provide necessary feedback to the consortium regarding exploitation characteristics and potential. The business potential of each of the KERs is also very much related to the local regulations of upcycling/recycling and will be continuously monitored during the implementation of the project.

Preliminary market analysis with needs, value propositions, and target stakeholders for exploitation The EU's wood-based industries cover a wide range of downstream activities, including woodworking, furniture, pulp and paper, manufacturing and converting industries and printing and altogether, approximately **393,000 enterprises were active in wood-based industries** in 2020, represented the 19% of the manufacturing enterprises across Europe. The Gross Value Added (GVA) of the wood industry was evaluated to **€136 billion**[44], while there is a significant wood recycling market with at least **450 wood recycling companies serving Europe**[53]. Along these lines, the wood wastes in Europe accounted for **50.2Mt** in 2018, coming from municipal waste - 24.33Mt, wood industry – 5.6Mt and CDW – 19.17Mt[54]. Focusing on CDW, wood waste is considered its second leading element, contributing to at least 20–30% of the total CDW stream is USA and **for more than 35% of all waste generated in Europe**[45]. Adding to this percentage, according to the European Federation of Furniture Manufacturers, furniture waste in Europe accounts for more than 4% of the total municipal solid waste stream[46]. As **37% of the European wood waste is intended for landfill and incineration** (33% is used for heat production or cogeneration and 33% for material recovery)[47], the EU endorsed three legislative establishing rules for waste management and binding, targeting **a)** for recycling wood from packaging by 25% until 2025 and 30% by 2030, **b)** reuse and recycle municipal waste by 55% by 2025 and 60% by 2025, and **c)** reducing landfilling waste (maximum of 10% of municipal waste by 2030)[48]. Thus, it may come as no surprise that the **wood recycling market is estimated to grow at a CAGR of 7.32% between 2022 and 2027**, while its size is forecasted to increase by USD 10.69 billion over the same timeframe[49]. An additional reason for this shift is the fact that the recycled wood is up to **40% cheaper than fresh wood** and generally dryer, leading to **energy savings** for the particleboard manufacturing process, as well as **prolongs material circulation** and reduces the need for additional lumbering. While higher grades of wood chips can be used for material recycling, lower-grade waste wood can replace conventional fossil energy sources[50].

[44] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Wood_products_-_production_and_trade#Wood-based_industries
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[49] https://eur04.safelinks.protection.outlook.com/GetUrlReputation
[50] https://www.tomra.com/en/discover/waste-metal-recycling/news/2022/wood-recycling-from-waste-to-resource

Based on this preliminary market analysis, the use of innovative solutions like those to be developed by W2W targeting the more resource efficient use of waste wood, are vital for the waste management sector in Europe[51]. **W2W separation technologies will increase by 30% the total wood collected for recycling** instead of landfilling: by taking into account the fact that the 80% of waste wood – approximately 40.1Mt - is estimated to belong to B, C and D categories, it is estimated that the use of **W2W separation technologies will enable the upcycling of 20%** of this quantity (8.02 Mt) **towards category A**, the non-processed wood and the cleanest among the wood classes that can be recycled and reused. The remaining non-recyclable wood waste–31.9Mt of B, C and D categories (60% of the initial quantity) after the application of the W2W upcycling methods - will be valorised as feedstock for bioenergy, that will further reduce landfilling. As the pre-treatment of the biomass increases the bioenergy's competitiveness with fossil fuels, and given that 0.056t CO_2/GJ are emissions during heat production, at a carbon price of €25/tCO_2, bioenergy is competitive at prices up to €10.8/GJ, while at a carbon price of €150/tCO_2, bioenergy is competitive at prices up to €17.8/GJ, including the costs of biomass and the costs of pre-treatment[52]. Since, recycling wood waste can create an inexpensive **renewable material source** to enhance economic growth and create employment opportunities while minimise landfill costs and space costs[53], we believe that **the status of the market offers favourable framework conditions to position the innovative circular technologies of W2W across a diverse array of customer groups**, as outlined in Table 2.6.

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Table 2.6: Preliminary outline of W2W customers' needs and value propositions

Customers	Key Needs	Main Value Propositions
Wood waste Suppliers	• Valorisation of by-products streams; • Increased profitability; • Reduced carbon footprint; • Reduce disposal costs.	- Sustainable pathways for utilising by-product/waste; - Alternative and increased revenue streams along with reduced taxes due to decreased waste management; - Compliance with the EU regulation frameworks regarding waste, packaging, emissions and circular economy.
Producers / Processors	• Scale-up production to meet increased demand; • Return on investment on sustainable technologies; • Compliance with regulations.	- Increased efficiency in production processes with costeffective technologies; - Improved sustainability and marketable products with lower carbon footprint; - New business models serving as concrete demand, improving economic performance.
End-users	• Environmental-friendly products; • Reduced waste; • Preservation of natural environment.	- Increased quality of the produced products; - Affordability of the products; - Improved sustainability with lower carbon footprint.

For the value propositions to effectively address the needs of the potential customer groups, W2W solutions and products need to be introduced to market, and keep its value chains sustainably operational. To this end, an initial set of requirements and potential obstacles that the exploitation strategy needs address**,** as well as key relevant actions and levers that will be employed were identified, as presented in Table 2.7.

Table 2.7: Initial set of requirements, potential obstacles, actions and supporting factors for exploitation

Requirements & Obstacles	Relevant Actions and Supporting Factors for W2W Exploitation

[51] Besserer, A.; Troilo, S.; Girods, P.; Rogaume, Y.; Brosse, N. Cascading Recycling of WoodWaste: A Review. Polymers 2021, 13, 1752. https://doi.org/10.3390/ polym13111752
[52] A.SPIRE, (2020) Processes4Planet 2050 Strategic Research and Innovation Agenda (SRIA)
[53] Jahan, I.; Zhang, G.; Bhuiyan, M.; Navaratnam, S. Circular Economy of Construction and Demolition Wood Waste—A Theoretical Framework Approach. Sustainability 2022, 14, 10478. https://doi.org/10.3390/su141710478

Lack of interest in the innovative services and digital tools produced	The benefits of W2W innovative services and digital tools will be demonstrated and validated, alongside their pilot implementation. W2W solutions will be designed with actual end-users in complete value chains, safeguarding, that they will cover real user needs. User acceptance will be validated at several stages to build confidence, that solutions and products can respond to market demands.
Further advancement of techs in terms of TRL after the end of the grant, to reach the commercialisation stage	Partners (researchers, technology developers and value chain players) are highly motivated by the potential of the W2W solutions and are committed to exploit the project's results, investing the resources required to develop them further, to reach the commercialisation stage. Go-to-market strategies (with identified options for acquiring external financing / funding if needed) to this end will be delivered, before the end of the project.
IPR management	An IPR management strategy will be developed during the project, describing the baseline for managing the IP produced within W2W and setting the basis for the exploitation of project results after the end of the grant. Specifically, for IP protection, the proper protection measures will be identified and all the necessary actions to apply them will be carried out (starting within the project's duration), to facilitate exploitation at least within the target market W2W.
Needs for standardisation	As any new technology, compliance and refinement (if needed) to existing standards and certification are key requirements for its release to the market. As the further development of related standards can affect market growth, W2W will consider standardisation activities, to support easy-to-market solutions.
Sustainability of financing for exploitation after the project end	Aside from investing own resources, partners are willing to continue the development and exploitation of the W2W technologies and products, sustainably operating the value chains set up and demonstrated during the project. Further financing sources will be investigated as well, either from national and EU funds, and/or by approaching private investors, building on the go-to-market strategies developed during the project.
Availability of initial demand for driving product sales and establishing market shares	The consortium's pilot partners will provide the initial demand needed to successfully market the solutions of W2W beyond the end of the grant. As early adopters, they are also committed to further promote the technologies/products to their ecosystem for identifying additional customers.

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2.2.2 Strategy for the Management of Intellectual Property

IPR issues have been discussed by the consortium during the proposal preparation phase for the commitment of the offered technical solutions and by consulting their respective partners' legal offices. In conformance with the rules for participation of current European Programs, a minimum set of basic principles concerning ownership of knowledge and access rights have been fixed and will be detailed with the signature of a consortium agreement (CA), before the start of the project, which will regulate rights and obligations by all partners and will be further updated this during the project. The IPR provisions distinguish between 2 basic types: **i) <u>Background Knowledge</u>**: Access rights necessary for the execution of the project will be granted on a royalty free basis. **ii) <u>Foreground Knowledge</u>**: Intellectual property protection of foreground, i.e., the new technologies and knowledge developed in the project and its applications will be filled, owned and funded by the partner(s) that developed the particular results. Each partner shall inform the Project Coordinator and the project partners 4 weeks in advance of the filing of a patent application. Subject to the provisions of the EU Grant Agreement and the CA any owner of any result shall be free to use such results and to move it to the market. In case of joint ownership each owner shall be free to use such results in its own organisation but adhere to the CA and exploitation strategy with regards to commercialisation.

Relevant to knowledge management W2W will employ the following strategy: • **Intra-Consortium Knowledge Management**: **1)** Technical and scientific progress: create documents and place them in a document repository to ensure that every partner has access to them. **2)** Exploitation perspectives: the WP exploitation leader will structure, and map required knowledge and extra-knowledge needed to enhance exploitation (i.e., patents, technology transfer agreements, potential users, other projects) and propose an exploitation strategy. • **Extra-Consortium Knowledge Management**: Knowledge management will take place through dissemination to maximise the impact of the work done within the project. The exploitation leader will be in charge of collecting and proposing to the consortium all

matters referring to the results for dissemination and will monitor the productivity of the project in terms of knowledge publications (scientific papers, posters, etc.). The IPR owner of each of W2W KERs is already identified in T2.5 as well as the licensing possibilities of each of the KERs. The exact strategy of IPR management is to be set the exploitation deliverables of the project already provided by M06 in D19.2.

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2.3 Summary Associated with document Ref. Ares(2023)8305270 - 05/12/2023

Specific Needs	Expected Results	D & E & C Measures
• Waste wood recycling processes are hampered by additives, pollutants and contaminating materials. • Proper wood sorting requires costly analyses and technologies. • Limited information about the composition of wood products and components presents in waste. • Industries using waste wood as feedstock have limited control on its supply. • Wood classification not harmonised at EU level. Wood suitable for material use may be sent to incineration because of outdated characterisation schemes and legislation. • Compared to energy use, wood recycling requires additional sorting, cleaning, chemical or mechanical treatment, transportation to the processing facility and additional administrative work, making it less profitable.	**Pillar 1: Advanced separation & sorting technologies** **KER1**: Optimal wood size particles for cascade valorisation; **KER2**: Cascade valorisation thresholds-led sorting system; **KER3**: Optimised separation enabled by Human-Robot Collaboration and MR technology **Pillar 2**: **Upcycling Processes and Technologies** **KER4**: Wood Cascade to function; **KER5**: Wood composites for wallboards; **KER6**: Wood glue separation; **KER7**: Bioremediation wood treatment; **KER8**: Mixed wood liquefaction; **KER9**: Hydrothermal Carbonisation; **KER10**: Biotech-driven up-cycling of syngas into dodecanol; **KER11**: Ashes valorisation. **Pillar 3**: **Digital Tools for Improving Circular Flows KER12**: LCA Sustainability Assessment Tool for Cascade Pathways; **KER13**: Material Volume Estimation; **KER14**: DPP; **KER15**: Supply Chain management tool for the optimisation of upcycling of secondary materials; **KER16**: Integrated Circular Wood Upcycling Platform. **Pillar 4: Supportive Framework in Policy, Market, and Skills** **KER17**: W2W Stakeholders Network; **KER18**: Classification framework for recycled wood; **KER19**: Wood cascade data sharing model; **KER20**: Trainings for Upcycling, Replication & Cooperation; **KER21**: Synergies hub for circularity; **KER22**: Circularity/Upcycling framework for cascade valorisation; **KER23**: Contribution to standardisation & certification; **KER24**: Business models.	**Dissemination (D):** • Peer-review Journals **>15**; Scientific Conferences **>20** • Technical Publications **>25**; Industrial Booths **>9** • Relevant initiatives: **>15**; Scientific Conversations: **20** • Workshops **>10**; Training Tutorials: **>8**; Webinars **>12**; • Fairs to Participate: >**9** • Standards Contribution: **≥3**; Policy Briefs: **>2** **Communication (C):** • Website Online by **M06** and running for **3years after the project end**; **>6,500** unique visitors by M48 • Size of online community **>5,000**; Impressions **>500** • Newsletters: **8**; Policy Fora/Debate Hubs: **>15**; Publications to Blogs: >**15** • Hard Copies Distributed **>1,500**; Roll-up Banners **>6** • Videos produced: **7** **Exploitation (E):** Open version of the W2W Frameworks, Procedures,

		Models, Tools and Data Repository, and Practical Services / Guidelines under a fee (post-project) to ensure sustainability.
Target Groups	**Outcomes**	**Impacts**
A: Process industry companies, including involved production of wood-based products and combustion processes; **B:** Recycling companies involved in the collection, sorting and processing of secondary raw materials; **C:** Manufacturers of new products made from secondary raw materials; **D:** Public Authorities responsible for waste management & environmental protection; **E:** Standardisation organisations developing standards for the use of secondary raw materials in the process industry; **F:** Research and academic institutions; **G:** Workers and employees in the process industry and recycling sectors; **H:** End-users and consumers of products made from secondary raw materials; **I:** The broader society and environment.	• Three complete value chains demonstrating the successful upcycling of wood-containing waste; • Improved production performance, GHG savings, and reduced costs for the upcycling or wood from C&D and furniture waste; • Uptake of W2W methods, scientific and technological results by at least 25 companies and organisations; • High use of the scientific knowledge published regarding elements of the W2W technologies and tools.	**Scientific impact:** Advanced state-of-the-art in wood recycling. **Economic impact:** Cost savings for companies, new job opportunities; Enhanced competitiveness of European companies. **Societal impact:** Increased resource efficiency in the wood value chain, improvement in employment and social levels. **Environmental impact:** Reduced demand for virgin wood, prevention of the release of greenhouse gases and other harmful substances into the environment, prevention of deforestation.

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3 Quality and Efficiency of the Implementation

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3.1 Work Plan and Resources

3.1.1 Work Packages and their Interrelation

The project is organised in 21 Work Packages (WPs): eighteen correspond to the project's objectives (WP4-21), and three (WP1-3) that coordinate the project activities during the three reporting periods (M01-M18, M19-M36, and M37-M48). Figure 3.1 illustrates the relationships between the WPs and summarises the project structure, which is organised around the 4 W2W innovation pillars.

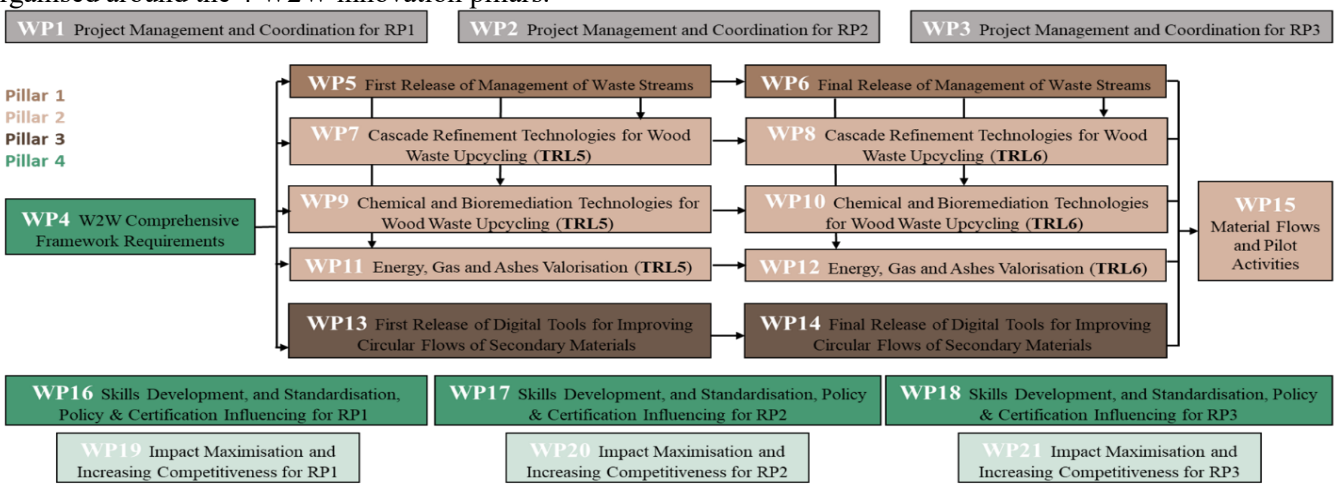

Figure 3.1: W2W PERT chart and WP dependencies

3.1.2 Timing of the Work Packages and Tasks

Work Packages and Tasks	Lead	Start Month	End Month	Phase 1	Phase 2	Phase 3	Phase 4
WP1 & WP2 & WP3 - Project Management and Coordination for RP1 & RP2 & RP3	**ICCS**	**M01/M18/M36**	**M19/M37/M48**				
T1.1 & T2.1 & T3.1 Project coordination and management (including QA and risk management)	ICCS	M01/M18/M36	M19/M37/M48				
T1.2 & T2.2 & T3.2 Technical management and scientific connection (including data management)	NTUA	M01/M18/M36	M19/M37/M48	D D		D	D
T1.3 & T2.3 & T3.3 GDPR, gender balance, legal and ethical issues management	EBOS	M01/M18/M36	M19/M37/M48				
WP4 - W2W Comprehensive Framework Requirements	**DRAXIS**	**M01**	**M10**				
T4.1 Actors mapping and key stakeholders' interactions	DRAXIS	M01	M10		D		
T4.2 Wood characterisation and classification schemes	CEPS	M01	M10		D		
T4.3 CDW recycled wood cascade valorisation framework for pure and mixed treatment	NTUA	M01	M10		D		
T4.4 Data sharing model, FAIR digital assets and digital passports principles	TUB	M01	M10		D		
T4.5 Definition of impact assessment methodologies and KPIs	DRAXIS	M01	M10		D		
WP5 & WP6 - Management of Waste Streams (first & final release)	**ICCS**	**M08/M19**	**M18/M36**				
T5.1 & T6.1 CDW flows valorisation and digitalisation	ELLAKT	M08/M19	M18/M36		D	D	
T5.2 & T6.2 Wood and glass non-destructive real time techniques analysis and definition of optimal size	IRIS	M08/M19	M18/M36		D	D	
T5.3 & T6.3 Adaptive glass sorting system based on recycling thresholds	ICCS	M08/M19	M18/M36		D	D	
T5.4 & T6.4 Adaptive wood sorting system based on cascade valorisation thresholds	ICCS	M08/M19	M18/M36				
T5.5 & T6.5 Robotics and Human Robot Collaboration for better separation	ICCS	M08/M19	M18/M36		D	D	
T5.6 & T6.6 MR system for better separation and quality control	ICCS	M08/M19	M18/M36				
WP7 & WP8 - Cascade Refinement Technologies for Wood Waste Upcycling (TRL5 & TRL6)	**NTUA**	**M08/M19**	**M18/M36**				
T7.1 & T8.1 Cascade to function, mixed wood treatment	UBC	M08/M19	M18/M36		D	D	
T7.2 & T8.2 New mix between wood, glass and composites for wallboards	NTUA	M08/M19	M18/M36		D	D	
T7.3 & T8.3 Final products and initial validation	FOCCHI	M08/M19	M18/M36		D	D	
WP9 & WP10 - Chemical and Bioremediation Technologies for Wood Waste Upcycling (TRL5 & TRL6)	**LERMAB**	**M08/M19**	**M18/M36**				
T9.1 & T10.1 Glue separation from wood	LERMAB	M08/M19	M18/M36				
T9.2 & T10.2 Bioremediation wood treatment	LERMAB	M08/M19	M18/M36		D	D	
T9.3 & T10.3 Liquefaction Process of mixed wood waste and green adhesives based on polyols	UHE	M08/M19	M18/M36		D	D	
T9.4 & T10.4 Final products and initial validation	CF2P	M08/M19	M18/M36		D	D	
WP11 & WP12 - Energy, Gas and Ashes Valorisation (TRL5 & TRL6)	**CIRCE**	**M08/M19**	**M18/M36**				
T11.1 & T12.1 Hydrothermal carbonisation	CIRCE	M08/M19	M18/M36				
T11.2 & T12.2 Fluidised bed gasification process of wood	CIRCE	M08/M19	M18/M36		D	D	
T11.3 & T12.3 Valorisation of ashes from CDW and thermal gasification	CIRCE	M08/M19	M18/M36		D	D	
T11.4 & T12.4 Upcycling of H2 and CO2	KIV	M08/M19	M18/M36		D	D	
T11.5 & T12.5 Final products and initial validation	P&G	M08/M19	M18/M36		D	D	
WP13 & WP14 - Digital Tools for Improving Circular Flows of Secondary Materials (first & final release)	**EBOS**	**M08/M19**	**M18/M36**				
T13.1 & T14.1 End-to-end tracing module and digital product passport	CERTH	M08/M19	M18/M36		D	D	
T13.2 & T14.2 Material volume estimation	ICCS	M08/M19	M18/M36		D	D	
T13.3 & T14.3 Supply chain optimisation (including primary materials & zero-defect management)	UPV	M08/M19	M18/M36		D	D	
T13.4 & T14.4 Life cycle sustainability assessment for cascade pathways	DRAXIS	M08/M19	M18/M36		D	D	
T13.5 & T14.5 Integrated Circular Wood Upcycling Platform	EBOS	M08/M19	M18/M36		D	D	
WP15 - Material Flows and Pilot Activities	**NTUA**	**M37**	**M48**				
T15.1 CDW pilot installation and validation	ICCS	M37	M48				D
T15.2 Material exchanges and logistics optimisation between the pilot sites	NTUA	M37	M48				D
T15.3 Final validation and sustainability assessment of the overall cascade valorization scheme	NTUA	M37	M48				
WP16 & WP17 & WP18 - Skills Development, and Standardisation, Policy & Certification Influencing	**VTT**	**M01/M18/M36**	**M19/M37/M48**				
T16.1 & T17.1 & T18.1 Identification of trends and scenario development	VTT	M01/M18/M36	M19/M37/M48		D	D	D
T16.2 & T17.2 & T18.2 Navigating the future of skills development	POLIMI	M01/M18/M36	M19/M37/M48		D	D	D
T16.3 & T17.3 & T18.3 Standardisation activities	UNE	M01/M18/M36	M19/M37/M48		D	D	D
T16.4 & T17.4 & T18.4 Policy influencing towards a truly circular and sustainable process industry	CEPS	M01/M18/M36	M19/M37/M48		D	D	D
WP19 & WP20 & WP21 - Impact Maximisation and Increasing Competitiveness for RP1 & RP2 & RP3	**R2M**	**M01/M18/M36**	**M19/M37/M48**				
T19.1 & T20.1 & T21.1 Dissemination and communication activities for promoting resources consciously	ISWA	M01/M18/M36	M19/M37/M48	D		D	D
T19.2 & T20.2 & T21.2 Cross-sectoral cooperation with existing initiatives	ICCS	M01/M18/M36	M19/M37/M48		D	D	D
T19.3 & T20.3 & T21.3 International collaboration and exchange of best practises	ACEN	M01/M18/M36	M19/M37/M48				
T19.4 & T20.4 & T21.4 Exploitation, business models and IPR management	R2M	M01/M18/M36	M19/M37/M48	D	D	D	D
T19.5 & T20.5 & T21.5 Market analysis, sustainable business models and go-to-market strategy	QPLAN	M01/M18/M36	M19/M37/M48		D	D	D

3.1.3 Allocation of Resources Summary of Consortium Effort

Table with PMs per WP and per Partner

Partner	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	WP11	WP12	WP13	WP14	WP15	WP16	WP17	WP18	WP19	WP20	WP21	TOTAL
1/ICCS	10.00	10.00	6.00	7.00	20.50	36.00	3.00	3.00	3.00	3.00	3.00	3.00	14.00	11.00	13.50	2.00	2.00	1.00	6.00	6.00	3.00	166.00
2/LERMAB	0.50	0.50	0.50	0.00	0.00	0.00	0.00	0.00	26.00	41.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.50
3/CIRCE	0.50	0.50	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	73.00	37.50	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.50	0.50	113.50
4/NTUA	5.00	5.00	5.00	10.00	1.00	1.00	14.00	20.00	0.00	0.00	0.00	0.00	0.00	0.00	16.00	0.00	0.00	0.00	2.00	2.00	2.00	83.00
5/POLIMI	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	2.00	0.00	11.00	11.00	9.00	2.00	2.00	2.00	44.00
6/VTT	0.50	0.50	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9.00	9.00	7.50	0.50	0.50	0.50	28.50
7/UPV	0.50	0.50	0.50	4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.00	16.00	0.00	2.00	2.00	1.40	0.70	0.70	0.70	40.00
8/CERTH	3.00	1.00	3.00	5.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.00	33.00	0.00	0.00	0.00	0.00	0.50	0.50	2.00	61.00
9/UHE	1.00	1.00	0.50	0.00	0.00	0.00	20.00	22.00	14.00	18.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	76.50
10/TUB	1.00	1.00	1.00	31.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	34.20
11/UJOH	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.00	8.00	9.00	25.00
12/ELAK	1.00	1.00	1.00	4.00	8.00	8.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.00	0.00	0.00	0.00	1.00	1.00	1.00	32.00
13/KIV	1.00	1.00	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	18.00	18.00	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	42.90
14/P&G	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	14.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	29.00
15/FOCCHI	2.25	2.25	2.25	0.00	0.00	0.00	11.00	11.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	1.75	1.75	1.75	36.00
16/DRAXIS	0.40	0.40	0.20	20.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.00	43.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	89.00
17/EBOS	5.00	5.00	5.00	6.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	16.00	20.00	0.00	0.00	0.00	0.00	5.00	2.00	2.00	66.00
18/QPLAN	1.45	1.20	0.60	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.75	0.00	0.00	0.00	19.75	9.85	2.60	39.20
19/R2M	1.00	1.00	1.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	8.00	14.00	14.00	43.00
20/IRIS	0.50	0.50	0.50	0.00	19.00	19.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.50	0.50	41.00
21/CEPS	0.70	0.65	0.65	8.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00	5.00	4.00	0.00	0.00	0.00	24.00
22/ISWA	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.00	8.00	5.00	24.00
23/RCM	1.00	1.00	1.00	2.00	3.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	1.00	1.00	1.00	16.00
24/UNE	0.50	0.50	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00	4.00	5.00	0.00	0.00	0.00	15.50
TOTAL	38.80	36.50	32.60	101.20	51.50	67.00	48.00	56.00	43.00	62.00	108.00	73.50	80.50	125.50	41.8	34.50	33.50	28.40	66.70	59.80	49.1	1237.80

'Subcontracting costs' items

Table 3.1.g Subcontracting cost items

Partner	Cost (€)	Justification
2/LERMAB	25,000	Technical work needed for the upgrade of the steam explosion equipment. The needed budget covers the technical work to be undertaken to cover all the upgrades of the system to cover the needed volumes of W2S
14/P&G	45,000	Lab synthesis to convert a few kgs of the Alanine derived from Fermentation into MGDA in order to be able to formulate, and test it.

Purchase costs' (travel and subsistence, equipment and other goods, works and services)

Table 3.1.h Purchase costs' (travel and subsistence, equipment and other goods, works and services)

1/ICCS	Cost (€)	Justification	13/KIVERDI	Cost (€)	Justification
Equipment	535,000	Hyperspectral Camera covering up to 2500nm (150K) MWIR Camera (120K) XR sensing devices (60K) 2 Delta robots (100K) 1 cobot (60K) 2 vision Cameras(20K) PLC and controls units(25K)	**Other Goods and services**	218,000	External service for the renting of open demo facility for preparatory set-up trials in WP11 and full demo trials optimization in WP12 by Kiverdi Staff . In detail, during the project, Kiverdi will leverage an external manufacturing pilot plant, Trials will be carried out and supervised by Kiverdi staff in synergy with the pilot plant owner using KIVERDI's proprietary microorganism, technology, know-how, intellectual property,

					ment Ref. Ares(2023)8305270 - tial informati specifications and protocols.
Other Goods and services	78,500	particleboards validation and characterization			
Rem.(<15%)	93,400		**Rem.(<15%)**	37,400	
Total	706,900		**Total**	255,400	
2/LERMAB	**Cost (€)**	**Justification**	**14/FOCCHI**	**Cost (€)**	**Justification**
Other Goods and services	45,000	Consumables used for the steam explosion (20K) and consumables used for bioremediation(25K)	**Other Goods and services**	35,000	Testing activities of the materials as part of th validation (35K)
Rem.(<15%)	36,000		**Rem.(<15%)**	18,400	
Total	81,000		**Total**	53,400	
4/NTUA	**Cost (€)**	**Justification**			
Equipment	90,000	Pilot scale extruder upgrade: Purchase of a twin-screw extrusion system to be used for compounding a wide range of biodegradable/biobased polymers and producing compounds for biocomposites manufacture. (70K) Thermoforming equipment for the pellets: Purchase of a heatedplates press equipment that will allow to process the biocomposite materials and transform them into building blocks. (20K)			
Other Goods and services	40,000	Consumables for the implementation of the pilot scale tests in WP7, WP8			
Rem.(<15%)	24,000				
Total	154,000				
7/UPV	**Cost (€)**	**Justification**	**20/IRIS**	**Cost (€)**	**Justification**
Other Goods and services	25,920	Project Meetings, meetings with other partners, conferences etc. 24trips*1080 = 25920	**Other Goods and services**	77,000	46k€ consumables an for building 2 nonitoring prototypes; 4k€ of cos protection, material a emination registrations fees, p fees for open acces software licenses transportation and ins prototypes 10 afety tests 15k€

Rem.(<15%)	12,600		**Rem.(<15%)**	9,600	
Total	38,520		**Total**	86,600	
9/UHE	**Cost (€)**	**Justification**	**22/ISWA**	**Cost (€)**	**Justification**
Other Goods and services	41,000	Chemicals needed for the operation of the pilot activities 31.5K. Publication fees 9.5K	**Other Goods and services**	21,000	(branding, D&C material website, newsletter, soc media, videos)
Rem. (<15%)	49,500		**Rem. (<15%)**	14,400	
Total	90,500		**Total**	35,400	ed with document Ref. Ares(2023)83052 05/12/2
11/UJOH	**Cost (€)**	**Justification**	**23/RCM**	**Cost (€)**	**Justification**
Travel and subsistence	10,060	Project meetings and travels for conferences (10,060)	Travelling Costs	10,200	Project meetings and travels for conferences (10,200)
Rem. (<15%)	19,036		**Rem. (<15%)**	0	
Total	29,096		**Total**	10,200	
12/ELAK	**Cost (€)**	**Justification**	**24/UNE**	**Cost (€)**	**Justification**
Travel and subsistence	14,400	Project meetings and travels for conferences (14,400)	**Travel and subsistence**	8,800	Project meetings and travels for conferences (8,800)
Rem. (<15%)	10,000		**Rem. (<15%)**	4,000	
Total	24,400		**Total**	12,800	

Internally invoiced goods and services

Table 3.1i: Other costs categories/ internally invoiced goods and services

Partner	Cost(€)	Justification
6/VTT	7,679.07	Costs related to the usage of internal equipment and specifically of computing platform technology platform/research infrastructure which is a distributed computational cluster for high-performance computing. With the cluster, it is possible to conduct technical analyses, calculations and simulations of different scientific phenomena requiring heavyduty parallel calculation, where dozens of computing units are needed at the same time. This item relates to WP16, 17 & 18.
9/UHE	4,000	Costs related to the usage of lab equipment essential for the activities of fraction cascading.
14/P&G	45,000	Costs related to the equipment used to convert fatty alcohols in ionic-surfactants.

3.2 Capacity of Participants and Consortium as a Whole

#@CON-SOR-CS@# #@PRJ-MGT-PM@# W2W will be executed by a well-balanced and multi-disciplinary consortium composed by 25 partners from 9 EU and 1 nonassociated non-EU countries (Figure 3.2); particularly 11 academic and research organisations, 5 large companies, 5 SMEs, a leading think tank, 2 associations and a public organisation. **ICCS** will act as coordinator; its **wide experience in EU projects of relevance of W2W** ensures the correct execution of the project. The consortium is **a high-quality partnership fully capable to attain the targeted objectives** and has been selected to include key actors essential to create synergies, ensuring that the project results will maximise the use of secondary resources in the process industry across Europe and internationally. The partners were carefully selected in order to fulfil the W2W Innovation Pillars mandatory for the execution of the project, and a series of transversal activities needed to maximise the outreach and the impact of the developed results. More specifically the role of the partners in relation to the pillars of the project are (depicted in Figure 3.3):

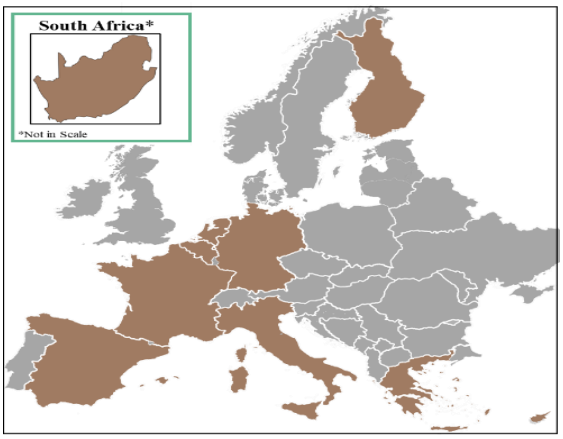

Figure 3.2: W2W geographical distribution

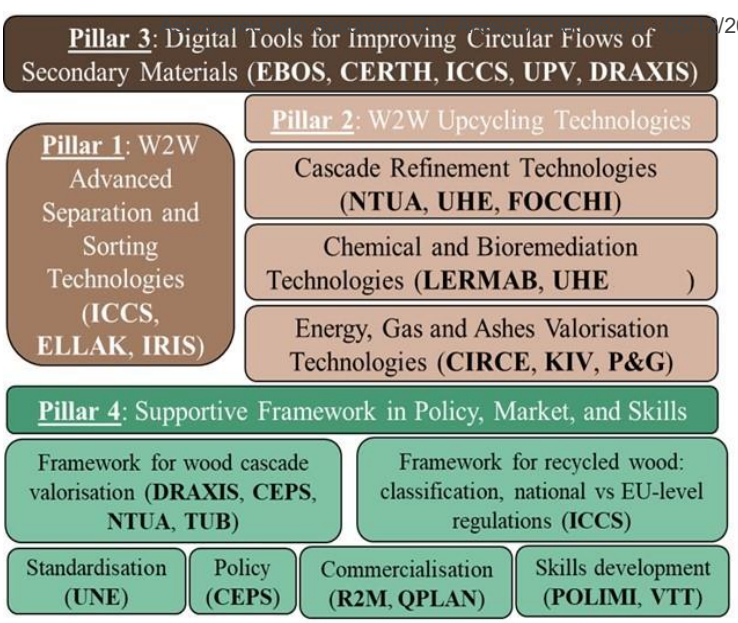

Figure 3.3: Role of the W2W partners in each innovation pillar

Pillar 1: Advanced Separation and Sorting **ELAK** is operating in a series of construction sites whereas some of the companies of the organisation are operating their own waste management plans. The related experience will support the design of a sorting mechanism ready to be deployed both at construction sites, but with an upscaling too towards a centralised CDW processing center. **IRIS** with a long experience in sensing will support the solutions related to the optimal shredding of the objects to be separated, an issue that is not commonly currently addressed neither in terms of optimal size for separation nor in terms of optimization of the used sensing devices. **IRIS** will use as a base various spectroscopic technique which are available in their labs and already demonstrated in relevant projects. **ICCS** will use its premises to perform the first tests of the sorting system, taking advantage of the multiple sensing and robotics solutions available in the lab. The experience of developing similar systems shall be used towards the design of the envisaged modular system. Existing MR devices will be initially tested as part of the system and the optimal hardware for the operation of the final system is to be decided. The first tests of Cobots with MR assisted workers shall also take place at the premises of **ICCS**. The final system, based on the final will be installed either at one of the available construction sites of **ELAK**, the premises of one of their subsidiaries or at the premises of **ICCS**. In any of the above cases the consortium members ensure access to the infrastructure needed for the design as well as the final system installation.

Pillar 2: W2W Upcycling Technologies Cascade Refinement Technologies: **UHE** has long experience in wood cascade to function technologies which are demonstrated in a series of publications and be the basis for many other studies. This will be the basis for W2W technologies ensuring access to the labs of **UHE** for the performance of the pilot activities and knowledge at a State-of-the-art level. **NTUA** already owns an extruder and can produce pellets at a small size compared to the ones needed for the project. The existing equipment will be used for the initial tests and will be further upscaled during the duration of the project. The **NTUA** labs shall be used for the installation of the equipment. **FOCCHI** owns their own testing facilities whereas with the support of **R2M**, additional certification tests shall be performed ensuring the quality of the final products. Chemical and Bioremediation Technologies: **LERMAB** has a long experience and has demonstrated in a series of publications the wood cascade valorisation approach with the usage of various upcycling techniques. As part of the project the existing steam explosion equipment will be used and further upgraded to process higher quantities of input material. The labs of **LERMAB** shall also be used for the bioremediation processes, offering the possibility to start the work on samples already seasoned for a period of time. **UHE** will collaborate with **LERMAB** on the production of green adhesives based on polyols, part of the work that both partners had individually done in the part aiming to share best practices as part of W2W. Finally, **LERMAB shall work** on the development of new products and will ensure the smooth validation and testing of the produced materials. Energy Gas and Ashes Valorisation: **CIRCE** has a long experience in energy related projects having demonstrated a series of energy optimization projects. The MW assisted HTC process will initially take place at the existing premises whereas further improvements during the project shall be performed. The needed gasification equipment is also placed at the premises of **CIRCE** and the needed processes will be implemented there. **KIV** owns pilot facilities (1Lt, 10Lt and 20Lt) to carry out pressurized gas fermentation equipped with control system for parameters optimisation, downstream units (centrifugation, separation units, distillation) and analytical equipment for characterisation of intermediates and products. During the project it will be upscaled to reach the expected volumes and showcase the TRL6 level. **P&G** has a demonstrated collaboration with **KIV** and shall ensure the validation of the developed samples both in their premises but also with the support of external labs.

Pillar 3: W2W Digital Tools for Improving Circular Flows of Secondary Materials Life Cycle Sustainability Assessment Tool for Cascade Pathways: **DRAXIS** has significant experience and scientific expertise on LCA of waste management systems, policy analysis regarding waste management, integrated plans for the management of municipal solid waste, planning of waste prevention strategies, development of monitoring indicators and procedures, and delivering effective tools for environmental monitoring and policy design, as it has participated in >10 European and national circular economy projects related to waste valorisation, industrial symbiosis, LCA and digital transition. Material Volume Estimation: **ICCS** has significant experience, demonstrated by numerous projects

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in which it acts as a coordinator or partner, in the integration of applications of automation and advanced technologies - e.g., virtual reality, artificial reality, data management etc.- in highly reliable multi-functional smart integrated systems that provide innovative solutions for addressing societal needs. Digital Product Passport: **CERTH** is one of the largest research centres in Greece. The department participating in the project will develop the W2W DPP building on its significant experience in the fields of machine learning, deep learning, multimodal retrieval, knowledge engineering, semantic integration of heterogeneous resources, semantic reasoning, discovery and mining of heterogeneous multilingual and multimedia Web resources, social media monitoring, as well as on the processing and analysis of the multimodal data extracted from them. Supply Chain Management Tool for the optimisation of upcycling of secondary materials: **UPV** is one of the top-ranked technical universities of Spain, and since 2010, it is a recognised European International Excellence Campus. The Research Centre on Production Management and Engineering (CIGIP) is at the leading edge on collaboration and interoperability in the supply chain, and in particular: Modelling and Simulation of Supply Chains, Production Planning Under Uncertainty Modelling, Enterprise Resilience, Enterprise Interoperability, Internationalisation of Operations. **UPV** will utilise its wide experience from its participation (since 1994) in several European and Spanish Research Projects and several research & development contracts with regional industrial firms from different industrial sectors. Integrated Circular Wood Upcycling Platform: **EBOS** technical role in the project, given its 18 years+ of expertise as a software powerhouse, is to design, develop and integrate the Integrated Circular Wood Upcycling Platform including its visualisation. **EBOS** will lead the digitalisation activities, leveraging its rich experience in the ICT domain, while, from the exploitation side, **EBOS** expects to advance knowledge and reinforce understanding in W2W advances, extending the company's activities.

Pillar 4: W2W Upcycling technologies: Supportive Framework in Policy, Market, and Skills Framework for CDW Cascade Valorisation and Classification Regulations: Through its wide and recognised experience in the field, **CEPS** will create the initial step towards the developed frameworks, as it will perform a study on parameters linked to wood characterisation and on the existing classification schemes at country level. **NTUA** will leverage its involvement to relevant projects and initiatives, by leading the development of the CDW recycled wood cascade valorisation framework. **DRAXIS**, **TUB** and **ICCS** will actively support the development of the W2W framework, leveraging their long involvement in the recycled material and circular economy sectors. Skills Development: **VTT** and **POLIMI** will closely collaborate, leveraging their wide experience in the training and skills development, as well as their existing learning tools and techniques, to form the basis for innovative skills approaches, in the form of interventions tailored to effectively support the workforce through engaging the twin transition skills ecosystem. Contribution to Standardisation and Certification: **UNE** will lead the W2W standardisation activities and promote the project results to its wide network of relevant experts, bodies and organisations, while **CEPS** will push the developed policy report that summarising the actions needed for a European Scale implementation of the W2W proposed cascade valorisation scheme, to its extensive network of policy- and decision-makers. Business Models and Commercialisation: **R2M** will lead the exploitation activities of the project, leveraging its rich experience in R&D and commercial projects, aiming at the alignment of the project activities with the defined revenue streams. **QPLAN** will focus on pushing the W2W KERs into the market, through its experience, existing techniques, pathways, developed revenue streams and network of stakeholders on the relevant fields.

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4. Ethics

The consortium confirms compliance with the ethical principles and the applicable EU, international, and national laws for the ethics issues identified in the Ethics Summary Report and any additional ethics issues that may emerge in the course of the grant, will be ensured. The Consortium also confirms that for any applicable ethics issue, the guidance provided in the European Commission Ethics Self-Assessment Guidelines will be rigorously followed.

4.1. Ethical dimension of the objectives, methodology and likely impact

Each member of the W2W who is leading or participating in research task that involves human participants will ensure the respect for people and for human dignity, fair distribution of the benefits and burden of research, and the protection of the values, rights and interests of the research participants. Ethical standards and guidelines will be rigorously followed and applied, irrespectively of the country in which the research is carried out. The necessary ethics approvals must be obtained, if required, as well as free and fully informed consent of the research participants. It is the responsibility of each member of the consortium leading an activity to obtain the relevant approvals and to ensure that their research complies with the laws and regulations in all the EU Member States in which the research takes place.
Human participants will be required to provide information e.g. via interviews, online forms and questionnaires that will be used for user engagement, stakeholder's identification, definition of scenarios, and gathering inputs and feedback for the project and the new solutions and technologies to be developed . Participation is voluntary and before any relevant research activity that involves humans, each individual will signify their agreement to participate with a clear affirmative act, W2W will apply clear informed consent procedures with a dedicated consent forms. Although no adverse side effects are expected, a plan for collecting, reporting, and managing solicited and spontaneously reported adverse events and other unintended effects of projects' interventions will be undertaken. All partners are fully aware of this Charter and Directive and have abided by these regulations in past and on-going research projects and any individual partners belonging to professional bodies will abide by the ethics rules and regulations of these bodies. W2W consortium will keep a record of all potential ethical issues that arise as project is developed and evaluated, and through discussion with the stakeholders. Ethics approval for the project will be sought from all necessary and applicable review panels and relevant committees in all partner countries undertaking any work.
Consortium will examine the following aspects in every related activity:
•Composition, roles, and responsibilities of the partners in charge;
•Description of research questions, specific objectives and justification for undertaking the study or analysis; •Description of study design, including type, settings and sites where data will be collected;
•Inclusion and exclusion criteria for participants;
•Time schedule of enrollment and assessments;
•Estimated number of participants needed to achieve study objectives and how these were determined;
•Plans for assessment and collection of outcome, baseline, and other trial data and study instruments;
•Who will obtain informed consent or assent from potential trial participants or authorized surrogates, and how;
•How personal information about potential and enrolled participants will be collected, shared, erased/anonymized if necessary and maintained;

Besides the high potential of AI, there are also risks, such as privacy, malicious use and jobs opportunity decrement, among others[1].

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Hence is crucial to assure its proper use having in mind its trustworthiness, as recently stated by the European Commission in the Ethics Guidelines for Trustworthy AI[2]. W2W will develop AI algorithms for waste management and sorting and for some of the developed digital tools, always following the strictest rules regarding AI defined by the EC. The usage of AI is of limited risk in the developed applications based on the EC's AI risk classification and regulatory framework [3]. It is not expected to raise any ethical concerns about human rights and values, or to have any negative impact such as environmental, social, political or financial.

[1] Jhon Zysman et. al. (2020); Governing AI: Understanding the Limits, Possibility, and Risks of AI in an Era of Intelligent Tools and Systems - Berkley University
[2] European Commission: High-Level expert group on artificial intelligence, "Ethics Guidelines for Trustworthy AI"
[3] European Commission: Regulatory framework proposal on artificial intelligence

4.2. Compliance with ethical principles and

All the local legislations for virgin raw or secondary materials, waste treatment and digitization will be analyzed and respected in collaboration with the local authorities and Environmental or other relevant Ministries, in harmony with the respective European directives and regulations. W2W will ensure that it is fully compliant with the Do No Significant Harm (DNSH) Principle since its activities as well as the expected life cycle impact of the innovations at a commercialization stage, substantially contribute in a positive way and/or do not significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation (EU) 2020/852. W2W will apply a life-cycle oriented approach (validated through the utilization of relevant LCA tool when possible) to ensure that proposed solutions are sustainable and consisting of 3 steps: i)Identification of actual and potential adverse impacts during the life cycle of the project activities ii)Prevention or mitigation of adverse impacts iii)Accounting for how adverse impacts are addressed by tracking performance and communicating results. W2W activities are expected to have a direct positive impact to the environmental objectives defined: Climate Change Mitigation by reducing GHG emissions (use Recommendation 2013/179/EU or ETSI ES 203 199,ISO 14067:2018 or ISO 14064-2:2019 to calculate lifecycle GHG and net emissions), to Pollution prevention & control through waste reduction and identifying harmless substances and contaminants in each pilot, to Transition to Circular Economy through the establishment of symbiotic and circular flows within same or different industries for materials and wastes whereas is not expected any negative impact to the sustainable use and protection of water and marine resources, protection and restoration of biodiversity & ecosystems and the Climate Change adaptation. Specifically, W2W will make sure that the related DNSH criteria (on Climate Adaption; climate adaptation; water; circular economy; pollution prevention and biodiversity) from the EU Taxonomy Compass will be applied for the below project related activities:

· -Data processing, hosting and related activities
· -Data-driven ICT solutions for GHG emissions reduction
· -Computer programming, consultancy and related activities
· -Close to market research, development and innovation And

if applicable in the below as well:

· -Manufacture of aluminium
· -Manufacture of iron and steel
· -Manufacture of organic basic chemicals
· -Manufacture of plastics in primary form
· -Engineering activities and related technical consultancy dedicated to adaptation to climate change ·

Any other activity that may arise during the project